Emma's Atlas of Numbers and Shapes

- A not-so-brief guide to all the math I know -

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An Introduction to Mathematical Reasoning

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Sets

Almost all of modern mathematics can be put in the context of the study of ***sets*** and their ***elements***. Generally, given a collection of objects $a, b, c, d, \ldots$, they can be taken together as a set, generally written in curly brackets and represented by a capital letter:

$$M := \{a, b, c, d, e, \ldots\}$$

However, an approach where any collection of mathematical objects forms a set quickly runs into trouble - famously, ***Russel's paradox*** asks:

> *"Let R be the set of all sets that don't contain themselves. Does R contain itself?"*

If R doesn't contain itself, then by definition it must contain itself, but if it contains itself, then it cannot contain itself! If $R \in R$ is true, then it must be false! We arrive at a classic paradox. The conclusion, then, is that the set of all sets that don't contain themselves cannot exist - we cannot define sets to be any collection of things defined by any property, and must instead establish ground rules about which sets exist and exactly how we can use sets we already know to construct new ones.

These ground rules are given in the form of ***axioms***, which are statements that mathematicians take for granted, which can then be used as a starting point to then derive "the rest of mathematics" from. The following chapters contain a brief and hopefully easily understandable introduction to the ZFC axiom system, which was developed in the early 20th century by Ernst Zermelo and Abraham Fraenkel and is now the most widely accepted axiomatization of mathematics.

2.1 Infinity: $\emptyset$ and $\mathbb{N}$

2.2 Extensionality

To say "b is an element of M", or equivalently "b contains M", we simply write:

$$b \in M$$

One of the most basic axioms of set theory is the ***Axiom of Extensionality***, which states:

Axiom 2.1: Axiom of Extensionality

Two sets A and B are equal if and only if they contain the same elements.

The Axiom of Extensionality establishes that sets are uniquely defined by the objects they contain and have no additional structure. In particular:

1. The order of elements in a set doesn't matter. $\{1, 2\}$ and $\{2, 1\}$ are the same set.

2. The number of times an element appears within the same set doesn't matter. $\{1\}, \{1, 1\}$, and $\{1, 1, 1\}$ are the same set.

Definition 2.2: Subsets

We call a set A a ***subset*** of a set B if and only if every element of A is contained in B. We write "A is a subset of B" as $A \subseteq B$.

The demand that "every element of A is contained in B" can be stated more formally as "$x \in A$ implies $x \in B$". Importantly, this means that every set is a subset of itself - we always have $A \subseteq A$.

This definition gives us another way of phrasing the Axiom of Extensionality - two sets are equal if and only if each is a subset of the other:

$$A = B \Leftrightarrow A \subset B \text{ and } B \subset A$$

This idea is incredibly important for writing out proofs in practice - if a problem asks you to establish that two sets A and B are the same, then the easiest approach is almost always to show that any element of A must be contained in B, and to then show that any arbitrary element of B must be contained in A.

The subset relation is frequently written as $A \subset B$. However, other authors reserve $A \subset B$ to mean "A is a subset of B that isn't equal to B" - we call such a set a ***proper subset of B***. I will therefore attempt to only use the symbols $\subseteq$ and $\subsetneq$, which clear up this ambiguity.

2.3 Specification

2.4 Pairing, Union, and Power Sets

2.5 Relations

2.6 Functions

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Part II

Algebraic Structures

Chapter 1

$(\mathbb{N}, +)$ is a Monoid

Chapter 2

$(\mathbb{Z}, +, -)$ is a Group

2.1 Group Homomorphisms

Theorem 2.1. *Let $\varphi : (G, \circ_G) \to (H, \circ_H)$ be a map that respects group operations, i.e:*

$$\varphi(g_1 \circ_G h_2) = \varphi(g_1) \circ_H \varphi(g_2)$$

Then h also respects the neutral element and inverses, i.e:

$$\begin{aligned} \varphi(e_G) &= e_H \\ \varphi(g^{-1}) &= \varphi(g)^{-1} \end{aligned}$$

Proof.

1. For any $h \in H$, we have:

$$\begin{aligned} e_H &= e_H \circ_H e_H \\ &= \left(\varphi(e_G)^{-1} \circ_H \varphi(e_G)\right) \circ_H e_H \\ &= \varphi(e_G)^{-1} \circ_H \varphi(e_G) \\ &= \varphi(e_G)^{-1} \circ_H \varphi(e_G \circ_G e_G) \\ &= \varphi(e_G)^{-1} \circ_H \varphi(e_G) \circ_H \varphi(e_G) \\ &= e_H \circ_H \varphi(e_G) \\ &= \varphi(e_G) \end{aligned}$$

   So φ respects the neutral element.

2. Since φ respects the neutral element, we get:

$$\begin{aligned} \varphi(g) \circ_H \varphi(g^{-1}) &= \varphi(g \circ_G g^{-1}) \\ &= \varphi(e_G) \\ &= e_H, \end{aligned}$$

   i.e. $\varphi(g^{-1}) = \varphi(g)^{-1}$.

□

2.2 Group Actions

Definition 2.2

Let $(G, \circ, e)$ be a group. Let X be a set. Then a map $. : G \times X \to X$ is called a ***group action*** of G on X iff:

1. It respects the neutral element:

$$\forall x \in X :$$
$$e.x = x$$

2. It associates with the group operation:

$$\forall g, h \in G, x \in X :$$
$$(g \circ h).x = g.(h.x)$$

Exercise 2.3. *Show that a map $(g, x) \to g.x$ is a group operation if and only if the map $g \to (x \to g.x)$ is a group homomorphism from G into the symmetric group over X.*

Definition 2.4. Let $. : G \times X \to X$ be an action of a group G on a set X. Let $x \in X$. Then we call the set

$$Gx := \{g.x \mid g \in G\}$$

of all points that x can be taken to by an element of G the ***orbit*** of x under G.

Exercise 2.5. *Let $X = \mathbb{R}^2$ be the plane. Let G be the group of rotations around the origin. Sketch the orbits of different points $x \in X$.*

Definition 2.6. Let $. : G \times X \to X$ be an action of a group G on a set X. Then we call the set

$$X/G := \{Gx \mid x \in X\}$$

of all orbits under G the ***orbit space*** of G in X.

Exercise 2.7. *Show that the relation*

$$x \sim y \Leftrightarrow \exists g \in G : g.x = y$$

is an equivalence relation. Show that for any $x \in X$, we the equivalence class of x is exactly the orbit of x.

Definition 2.8. Let $. : G \times X \to X$ be an action of a group G on a set X. Then we call the action . ***transitive*** if any point $x \in X$ can be taken to any point $y \in X$ by an element of G:

$$\forall x, y \in X :$$
$$\exists g \in G :$$
$$g.x = y$$

Exercise 2.9. *Show that a group action is transitive if and only if $Gx = X$ for all $x \in X$.*

Definition 2.10. Let $. : G \times X \to X$ be an action of a group G on a set X. Let $x \in X$. Then we call the set

$$G_x := \{g \in G \mid g.x = x\}$$

of all elements of G leaving x unchanged the ***isotropy group*** of x.

Exercise 2.11. *Show that the isotropy group G_x is always a subgroup of G.*

Exercise 2.12. *Show that if the action $(g, x) \mapsto g.x$ is transitive, the map*

$$\begin{aligned} \varphi_x : G/G_x &\to X \\ [g] &\mapsto g.x, \end{aligned}$$

where G/G_x is the quotient group of G by G_x, is well-defined and bijective.

We can also define the isotropy groups of points:

Definition 2.13. Let $. : G \times X \to X$ be an action of a group G on a set X. Let $A \subseteq X$. Then we define

$$g.A := \{g.a \mid a \in A\}$$

and call the set

$$G_A := \{g \in G \mid g.A = A\}$$

of all elements of G mapping elements of A to elements of A the ***isotropy group*** of A.

Chapter 3

$(\mathbb{Z}, +, -, \cdot)$ is a Ring

Chapter 4

$(\mathbb{Q}, +, -, \cdot, ^{-1})$ is a Field

Part III

Foundations of Analysis

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The Real Numbers $\mathbb{R}$

1.1 Ordered Fields

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Chapter 4

Euclidean Geometry

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Part IV

Linear Algebra

Chapter 1

Analytic Geometry

1.1 Arithmetic in n dimensions

We now focus our attention on n-dimensional euclidean space, which is modelled by the cartesian product $\mathbb{R}^n$:

$$\mathbb{R}^n = \left\{ \vec{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \;\middle|\; x_1, \ldots, x_n \in \mathbb{R} \right\} = \left\{ \vec{x} : \{1, \ldots, n\} \to \mathbb{R} \right\}$$

Elements of $\mathbb{R}^n$ are known as ***vectors***[1]. In the context of linear algebra, they are usually written vertically, as ***column vectors***.

[1] We will see in the next chapter that $\mathbb{R}^n$ is an example of a ***vector space***, including the case $n = 1$ (and $n = 0$, if you want to get a bit pathological). Therefore, formally speaking and only in certain contexts, regular old real (or even rational) numbers can be vectors too. However, when encountering the term "vector" in the wild, it is generally implied that $n \geq 2$.

Vectors simultaneously represent both ***positions*** and ***directions*** in space - in the same way that the number 5 is both the point that lies 5 units away from 0 on the real number line *and* the distance between the numbers 3 and 8 on that same number line, the vector $\binom{1}{2}$ represents both

- The point on the cartesian plane that can be reached by going one unit in the direction of one coordinate axis and then two units in the direction of the other axis, *and*
- the distance between two points with coordinates $\binom{2}{3}$ and $\binom{3}{5}$, or any other pair of points that are one unit apart in one coordinate direction and two in the other.

As is expected for algebraic structures defined on cartesian products, addition is defined **componentwise**:

$$\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{pmatrix}$$

Given a real number c, we can scale elements of $\mathbb{R}^n$ by that number:

$$c \cdot \vec{x} = c \cdot \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} c \cdot x_1 \\ \vdots \\ c \cdot x_n \end{pmatrix}$$

The factor c is known as a ***scalar***, and the operator $\cdot : \mathbb{R} \times \mathbb{R}^n \to \mathbb{R}^n$ as ***scalar multiplication***. For the sake of readability, we differentiate scalars from vectors by writing a little arrow on top of a vector.

Arguably, it is these two operations - vector addition and scalar multiplication - that define linear algebra - next chapter, we will abstractly define ***vector spaces*** to be those structures where addition and scalar multiplication are well-behaved and the scalars come from a field, and ***modules*** to be their cousins whose scalars come from a ring.

In general, there is no well-behaved[2] vector multiplication that works for any n. Later on, we will prove this observation in the form of ***Frobenius' Theorem***, which roughly states:

Theorem 1.1: Frobenius' Theorem

The only dimensions in which an associative multiplication exists on $\mathbb{R}^n$ are $\mathbb{R}^1$, i.e. $\mathbb{R}$ itself, $\mathbb{R}^2$, where multiplication is given by treating two-dimensional vectors as complex numbers, and $\mathbb{R}^4$, where multiplication is given treating four-dimensional vectors as ***quaternions***.

We will take a closer look at the complex numbers $\mathbb{C} \cong \mathbb{R}^2$ and the quaternions $\mathbb{H} \cong \mathbb{R}^4$ in the following sections. In particular, we will see that quaternion multiplication isn't commutative, so if we require multiplication to be commutative, our options shrink even further, leaving just $\mathbb{R}$ and $\mathbb{C} \cong \mathbb{R}^2$.

[2] In this case, our criterion for multiplication being well-behaved is that multiplicative inverses should exist, i.e. that division should be possible.

1.2 The Inner Product: Angles and Lengths in $\mathbb{R}^n$

Another operation that is of central importance to euclidean geometry is the ***inner product***, also known as the ***dot product*** or ***scalar product***.

Definition 1.2: Euclidean Inner Product

Given two vectors $\vec{x}, \vec{y} \in \mathbb{R}^n$, their ***Euclidean inner product***, written $\langle \vec{x}, \vec{y} \rangle$ or simply $\vec{x} \cdot \vec{y}$, is the sum of the products of the components of $\vec{x}$ with the components of $\vec{y}$:

$$\langle \vec{x}, \vec{y} \rangle = \sum_{i=1}^{n} x_i \cdot y_i$$

Theorem 1.3: Properties of Inner Products

The Euclidean inner product has the following properties:

1. ***(Positive definiteness):*** For every $\vec{x} \in \mathbb{R}^n$, we have $\langle \vec{x}, \vec{x} \rangle = 0$ if and only if $\vec{x} = \vec{0}$.
2. ***(Linearity in the first argument):*** For every $\vec{x}, \vec{y}, \vec{z} \in \mathbb{R}^n$ and $a, b \in \mathbb{R}$, we have:

$$\langle ax + by, z \rangle = a \langle x, z \rangle + b \langle y, z \rangle$$

3. ***(Symmetry):*** For every $\vec{x}, \vec{y} \in \mathbb{R}^n$, we have:

$$\langle x, y \rangle = \langle y, x \rangle$$

Exercise 1.4. *Show that the Euclidean inner product fulfills the following version of the binomial formula:*

$$\langle \vec{x} + \vec{y}, \vec{x} + \vec{y} \rangle = \langle \vec{x}, \vec{x} \rangle + 2 \langle \vec{x}, \vec{y} \rangle + \langle \vec{y}, \vec{y} \rangle$$
$$\langle \vec{x} - \vec{y}, \vec{x} - \vec{y} \rangle = \langle \vec{x}, \vec{x} \rangle - 2 \langle \vec{x}, \vec{y} \rangle + \langle \vec{y}, \vec{y} \rangle$$

The importance of inner products comes from the fact they let us define ***lengths*** and ***angles***.

Definition 1.5: Euclidean Norm

The length of a vector $\vec{x} \in \mathbb{R}^n$, also known as its ***Euclidean norm***, written as $\|\vec{x}\|_2$, $\|\vec{x}\|$, or just $|x|$, is given by:

$$\|\vec{x}\|_2 = \sqrt{\langle \vec{x}, \vec{x} \rangle} = \sqrt{x_1^2 + \ldots + x_n^2}$$

Note that for $n = 1$, we simply get $\|x\|_2 = \sqrt{x^2} = |x|$, so any result concerning the Euclidean norm on $\mathbb{R}^n$ also holds for the absolute value function.

Theorem 1.6: Some properties of Norms

The Euclidean norm has the following properties:

1. ***(Non-negativity):*** For every $\vec{x} \in \mathbb{R}^n$, $\|\vec{x}\|_2 \geq 0$.
2. ***(Positive definiteness):*** For every $\vec{x} \in \mathbb{R}^n$, $\|\vec{x}\|_2 = 0$ if and only if $\vec{x} = \vec{0}$.
3. ***(Absolute homogeneity):*** For every $c \in \mathbb{R}$ and $\vec{x} \in \mathbb{R}^n$, we have

$$\|c \cdot \vec{x}\|_2 = |c| \cdot \|\vec{x}\|_2$$

The little subscript 2 in $\|\vec{x}\|_2$ differentiates the Euclidean norm from the many other norms we will meet over time - in general, for any $p \in \mathbb{N}_1$, $\|\vec{x}\|_p$ denotes the function

$$\|\vec{x}\|_p = \sqrt[p]{|x_1|^p + \ldots + |x_n|^p} \qquad \left(= \sqrt[p]{\sum_{i=1}^{n} |x_i|^p} \right),$$

known as the ***p-norm***. For now, we only really care about the Euclidean norm, but other values of p will start coming in handy once we start applying tools from analysis to vector spaces. The limiting case $p \to \infty$ is particularly useful: We will find that

$$\|\vec{x}\|_\infty = \max |x_i|.$$

Theorem 1.7: Cauchy-Schwartz Inequality

Let $\vec{x}, \vec{y} \in \mathbb{R}^n$. Then we have

$$|\langle \vec{x}, \vec{y} \rangle| \leq \|\vec{x}\|_2 \cdot \|\vec{y}\|_2$$

Proof.

1. Assume $\|\vec{x}\|_2 = 0$. Then we immediately get

$$\langle \vec{x}, \vec{y} \rangle = 0 = \|\vec{x}\|_2 \cdot \|\vec{y}\|_2.$$

2. Assume $\|\vec{x}\|_2 = \|\vec{y}\|_2 = 1$. Then we have $\|\vec{x}\|_2^2 + \|\vec{y}\|_2^2 = 2$, and thus:

$$\begin{aligned} 1 + \langle \vec{x}, \vec{y} \rangle &= \frac{1}{2}(\|\vec{x}\|_2^2 + \|\vec{y}\|_2^2) + \langle \vec{x}, \vec{y} \rangle \\ &= \frac{1}{2}(\|\vec{x}\|_2^2 + 2\langle \vec{x}, \vec{y} \rangle + \|\vec{y}\|_2^2) \\ &= \frac{1}{2}\|\vec{x} + \vec{y}\|_2^2 \\ &\geq 0 \end{aligned}$$

Thus we get $\langle \vec{x}, \vec{y} \rangle \geq -1$. Applying the same reasoning to $1 - \langle \vec{x}, \vec{y} \rangle$, we get

$$1 - \langle \vec{x}, \vec{y} \rangle = \frac{1}{2}\|\vec{x} - \vec{y}\|_2^2 \geq 0,$$

i.e. $\langle \vec{x}, \vec{y} \rangle \leq 1$. We thus have

$$|\langle \vec{x}, \vec{y} \rangle| \leq 1 = \|\vec{x}\|_2 \cdot \|\vec{y}\|_2,$$

as desired.

3. Now, let $\vec{x}, \vec{y} \in \mathbb{R}^n \setminus \left\{\vec{0}\right\}$ be arbitrary. Then $\frac{\vec{x}}{\|\vec{x}\|_2}$ and $\frac{\vec{y}}{\|\vec{y}\|_2}$ both have norm 1, letting us simply use our previous result:

$$\begin{aligned}
\left|\langle \vec{x}, \vec{y} \rangle\right| &= \left|\|\vec{x}\| \cdot \|\vec{y}\| \cdot \left\langle \frac{\vec{x}}{\|\vec{x}\|}, \frac{\vec{y}}{\|\vec{y}\|} \right\rangle\right| \\
&\leq \left|\|\vec{x}\| \cdot \|\vec{y}\| \cdot \left\|\frac{\vec{x}}{\|\vec{x}\|}\right\| \cdot \left\|\frac{\vec{y}}{\|\vec{y}\|}\right\|\right| \\
&= \|\vec{x}\| \cdot \|\vec{y}\|
\end{aligned}$$

□

Corollary 1.8: Triangle Inequality

For every $\vec{x}, \vec{y} \in \mathbb{R}^n$, we have

$$\|\vec{x} + \vec{y}\|_2 \leq \|\vec{x}\|_2 + \|\vec{y}\|_2$$

We had previously formalized the concept of an 'angle between two lines' via the complex exponential function, and then defined the trigonometric functions $\sin x$ and $\cos x$ as the complex and real parts of e^{ix}, respectively. If we take the cosine function as a given, then scalar products provide a convenient alternate way of defining angles, which works in any dimension:

Definition 1.9: Inner angle between two vectors

Given two vectors $\vec{x}, \vec{y} \in \mathbb{R}^n \setminus \left\{\vec{0}\right\}$, we define the ***inner angle*** between them to be:

$$\measuredangle(\vec{x}, \vec{y}) = \arccos\left(\frac{\langle \vec{x}, \vec{y} \rangle}{\|\vec{x}\|_2 \cdot \|\vec{y}\|_2}\right)$$

Note that Cauchy-Schwartz gives us $\frac{\langle \vec{x}, \vec{y} \rangle}{\|\vec{x}\|_2 \cdot \|\vec{y}\|_2} \in [-1, 1]$, showing that this expression actually is well-defined.

Exercise 1.10. *Verify that we have:*

1. $\measuredangle(\vec{x}, \vec{y}) = 0$ *if and only if there exists* $c \in \mathbb{R}_{>0}$ *such that* $\vec{x} = c\vec{y}$.
2. $\measuredangle(\vec{x}, \vec{y}) = \frac{\pi}{2}$ *if and only if* $\langle x, y \rangle = 0$.
3. $\measuredangle(\vec{x}, \vec{y}) = \pi$ *if and only if there exists* $c \in \mathbb{R}_{>0}$ *such that* $\vec{x} = -c\vec{y}$.

Notably, this suggests that our definition is sensible even in the case $n = 1$, since any pair of non-zero vectors fulfills either condition 1 or 2, implying that the angle between two vectors on the real number line is either 0 or π.

These inner angles match our earlier definition of angles in terms of complex numbers, as long as we are content with not being able to meaningfully talk about negative angles or angles greater than π anymore:

Exercise 1.11. *Earlier in this book, we defined angles in terms of complex numbers. Given a complex number $y = re^{i\varphi}$ we saw that the corresponding real vector is given by $\vec{y} = \binom{r\cos\varphi}{r\sin\varphi}$.*

Show that our two definitions match as long as $\varphi \in [0, \pi]$, i.e. that given a vector $\vec{x} = \binom{x}{0}$ lying on the x-axis, we have

$$\measuredangle(\vec{x}, \vec{y}) = \varphi$$

(Hint: Remember $\cos^2\varphi + \sin^2\varphi = 1$)

We have to be content with this restricted interval $[0, \pi]$, since the cosine function isn't injective on larger intervals, meaning the arccosine would no longer be well-defined.

More generally, any definition of an 'angle between two arbitrary vectors' will suffer from this - if we only specify two vectors, it's left ambiguous which side we should measure the angle on.

This problem can be solved by finding a way to take a tuple $(\vec{x}_1, \ldots, \vec{x}_n)$ of vectors and recognize whether it is 'positively ordered' or 'negatively ordered' - in $\mathbb{R}^2$, this would be asking whether the second vector lies 'clockwise' or 'counterclockwise' from the first, while in $\mathbb{R}^3$ the question is whether the vectors are 'left-handed' or 'right-handed'. A map that gives us this information is known as an ***orientation***. We will formalize this idea later, when we have introduced the necessary tools, those being vector spaces, their ***bases***, and ***determinant functions***.

In the complex case, we could dodge this issue because one of our vectors, call it $\vec{x}$, was always part of the x-axis, and thus we could obtain a canonical orientation by declaring that the angle between $\vec{x}$ and $\vec{y}$ was positive if $y_2 > 0$ and negative if $y_2 < 0$.

Exercise 1.12. *Prove the following theorems of classical Euclidean geometry:*

1. *The* **Law of Cosines***:*

$$\|\vec{x} - \vec{y}\|_2^2 = \|\vec{x}\|_2^2 + \|\vec{y}\|^2 - 2\|\vec{x}\|_2\|\vec{y}\|_2 \cdot \cos\big(\measuredangle(\vec{x}, \vec{y})\big)$$

2. *The* **Pythagorean Theorem***:*

$$\measuredangle(\vec{x}, \vec{y}) = \frac{\pi}{2} \implies \|\vec{x} - \vec{y}\|_2^2 = \|\vec{x}\|_2^2 + \|\vec{y}\|^2$$

1.3 The Cross Product

The ***cross product*** is an operation $\times : \mathbb{R}^3 \times \mathbb{R}^3 \to \mathbb{R}^3$, such that:

1. $\vec{u} \times \vec{v}$ is orthogonal to both $\vec{u}$ and $\vec{v}$,
2. the length of $\vec{u} \times \vec{v}$ is the area of the parallelogram spanned by $\vec{u}$ and $\vec{v}$:

$$\left\|\vec{u} \times \vec{v}\right\|_2 = \left\|\vec{u}\right\|_2 \left\|\vec{v}\right\|_2 \sin\left(\measuredangle(\vec{u}, \vec{v})\right)$$

We also declare that our cross product should be ***anticommutative:*** given two vectors, we want to be able to change the sign of the cross product by switching the order in which we multiply:

$$\vec{v} \times \vec{w} := -\vec{w} \times \vec{v}$$

Notice how this immediately implies

$$\begin{aligned} &\vec{v} \times \vec{v} = -(\vec{v} \times \vec{v}) \\ \implies &\vec{v} \times \vec{v} = 0 \end{aligned}$$

Finally, notice that our geometric intuition immediately tells us that our product should be ***linear*** and ***distributive over vector addition***.

We can now derive an explicit formula for this operation on our own. First, note that the standard basis

$$\vec{e}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \vec{e}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \vec{e}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix},$$

consists of orthogonal vectors of length one, so we can start by declaring

$$\begin{aligned} \vec{e}_1 \times \vec{e}_2 &:= \vec{e}_3 \\ \vec{e}_2 \times \vec{e}_3 &:= \vec{e}_1 \\ \vec{e}_3 \times \vec{e}_1 &:= \vec{e}_2 \end{aligned}$$

Note that once again, the direction in which these vectors point depends on a somewhat arbitrary choice of orientation - we could invert all of the directions and still end up with a valid definition.

These observations are already enough to calculate the cross product of two arbitrary vectors: Using distributivity, we get:

$$\begin{aligned}
\begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} \times \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix} &= (v_1\vec{e}_1 + v_2\vec{e}_2 + v_3\vec{e}_3) \times (w_1\vec{e}_1 + w_2\vec{e}_2 + w_3\vec{e}_3) \\
&= v_1 w_1 (\vec{e}_1 \times \vec{e}_1) \\
&+ v_1 w_2 (\vec{e}_1 \times \vec{e}_2) \\
&+ v_1 w_3 (\vec{e}_1 \times \vec{e}_3) \\
\\
&+ v_2 w_1 (\vec{e}_2 \times \vec{e}_1) \\
&+ v_2 w_2 (\vec{e}_2 \times \vec{e}_2) \\
&+ v_2 w_3 (\vec{e}_2 \times \vec{e}_3) \\
\\
&+ v_3 w_1 (\vec{e}_3 \times \vec{e}_1) \\
&+ v_3 w_2 (\vec{e}_3 \times \vec{e}_2) \\
&+ v_3 w_3 (\vec{e}_3 \times \vec{e}_3)
\end{aligned}$$

To which we can apply anticommutativity and our formulas for multiplying the basis vectors to get:

$$\begin{aligned}
&\begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} \times \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix} \\
\\
&= v_1 w_1 \cdot \vec{0} \\
&+ v_1 w_2 \cdot \vec{e}_3 \\
&+ v_1 w_3 \cdot (-\vec{e}_2) \\
\\
&+ v_2 w_1 \cdot (-\vec{e}_3) \\
&+ v_2 w_2 \cdot \vec{0} \\
&+ v_2 w_3 \cdot \vec{e}_1 \\
\\
&+ v_3 w_1 \cdot \vec{e}_2 \\
&+ v_3 w_2 \cdot (-\vec{e}_1) \\
&+ v_3 w_3 \cdot \vec{0} \\
\\
&= \begin{pmatrix} v_2 w_3 - v_3 w_2 \\ v_3 w_1 - w_1 v_3 \\ v_1 w_2 - v_2 w_1 \end{pmatrix}
\end{aligned}$$

Corollary 1.13: Cross Product

The cross product $\vec{v} \times \vec{w}$ of two vectors $\vec{v}, \vec{w} \in \mathbb{R}^3$ returns a vector whose direction is orthogonal to both $\vec{v}$ and $\vec{w}$ and whose length equals the area of the parallelogral spanned between them. Given coordinate representations of $\vec{v}$ and $\vec{w}$, the cross product can be calculated via the following formula:

$$\begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} \times \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix} = \begin{pmatrix} v_2 w_3 - v_3 w_2 \\ v_3 w_1 - w_1 v_3 \\ v_1 w_2 - v_2 w_1 \end{pmatrix}$$

Proposition 1.14. *We have*

$$\left\langle \vec{u} \times \vec{v}, \vec{w} \times \vec{t} \right\rangle = \langle \vec{u}, \vec{w} \rangle \left\langle \vec{v}, \vec{t} \right\rangle - \left\langle \vec{u}, \vec{t} \right\rangle \langle \vec{v}, \vec{w} \rangle .$$

Proposition 1.15. *We have*

$$\left\langle \vec{u} \times \vec{v}, \vec{w} \right\rangle = \left\langle \vec{v} \times \vec{w}, \vec{u} \right\rangle = \left\langle \vec{w} \times \vec{u}, \vec{v} \right\rangle$$

This expression is known as the **box product** *of $\vec{u}, \vec{v}, \vec{w}$. Its absolute value corresponds to the volume of the parallelepiped spanned by the three vectors.*

Proposition 1.16. *We have*

$$(\vec{u} \times \vec{v}) \times \vec{w} = \langle u, w \rangle \cdot v - \langle v, w \rangle \cdot u = w \times (v \times u).$$

This result is known as **Grassmann's Identity**.

Proposition 1.17. *We have*

$$(\vec{u} \times \vec{v}) \times \vec{w} + (\vec{v} \times \vec{w}) \times \vec{u} + (\vec{w} \times \vec{u}) \times \vec{v} = 0$$

This result is known as **Jacobi's Identity.**

1.4 The Quaternions $\mathbb{H}$

Definition 1.18: Quaternions

The Quaternions $\mathbb{H}$ consist of the set $\mathbb{R}^4$, with the following special notation:

$$\begin{aligned} \vec{1} &:= (1,0,0,0) \\ i &:= (0,1,0,0) \\ j &:= (0,0,1,0) \\ k &:= (0,0,0,1) \end{aligned}$$

Such that in addition to scalar multiplication and vector addition, which work as usual, we have an additional vector multiplication operation $\cdot : \mathbb{R}^4 \times \mathbb{R}^4 \to \mathbb{R}^4$ that behaves as follows:

1. $\vec{1}$ is a neutral element of vector multiplication:

$$\begin{aligned} &\forall x \in \mathbb{H} : \\ &\vec{1} \cdot x = x \cdot \vec{1} = x \end{aligned}$$

2. Vector multiplication is associative:

$$\vec{x}(\vec{y}\vec{z}) = (\vec{x}\vec{y})\vec{z}$$

3. Multiplicative inverses exist:

$$\begin{aligned} &\forall \vec{x} \in \mathbb{H}_{\neq 0} : \\ &\exists \vec{x}^{-1} \in \mathbb{H} : \\ &\vec{x} \cdot \vec{x}^{-1} = \vec{x}^{-1} \cdot \vec{x} = \vec{1} \end{aligned}$$

4. Vector multiplication distributes over vector addition, from both sides:

$$\begin{aligned} &\forall \vec{x}, \vec{y}, \vec{z} \in \mathbb{H} : \\ &\vec{x} \cdot (\vec{y} + \vec{z}) = \vec{x}\vec{y} + \vec{x}\vec{z} \\ &(\vec{x} + \vec{y}) \cdot \vec{z} = \vec{x}\vec{z} + \vec{y}\vec{z}\mathfrak{t} \end{aligned}$$

5. Vector multiplication is compatible with scalar multiplication:

$$\begin{aligned} &\forall a, b \in \mathbb{R}, \vec{x}, \vec{y} \in \mathbb{H} : \\ &a\vec{x} \cdot b\vec{y} = (ab)(\vec{x}\vec{y}) \end{aligned}$$

6. The unit vectors fulfill the ***Quaternion multiplication formula***:

$$i^2 = j^2 = k^2 = ijk = -\vec{1}$$

Apart from the Quaternion multiplication formula, which is the defining feature of the Quaternions, all of our other demands together state that $\mathbb{H}$ should be an ***associative real division algebra over*** $\mathbb{R}^4$.

One very important consequence of this definition is that vectors of the form $\vec{a} := (a, 0, 0, 0)$ behave exactly like real numbers - let $\vec{x} \in \mathbb{H}$, then we have:

$$\vec{a}\vec{x} = (a \cdot \vec{1}) \cdot \vec{x} = a \cdot (\vec{1} \cdot \vec{x}) = a\vec{x}$$

For this reason, it is common to write $\vec{1}$ simply as 1, and to then proceed as we did for the complex numbers and write:

$$(a, b, c, d) = a + bi + cj + dk$$

We can now already get most of the way to deriving multiplication formula for quaternions, by simply applying distributivity:

$$\begin{aligned}
(v_1 + v_2 i + v_3 j + v_4 k) \cdot (u_1 + u_2 i + u_3 j + u_4 k)& \\
= v_1u_1 + (v_1u_2)i + (v_1u_3)j + (v_1u_4)k& \\
+ (v_2u_1)i + (v_2u_2)i^2 + (v_2u_3)ij + (v_2u_4)ik& \\
+ (v_3u_1)j + (v_3u_2)ji + (v_3u_3)j^2 + (v_3u_4)jk& \\
+ (v_4u_1)k + (v_4u_2)ki + (v_4u_3)kj + (v_4u_4)k^2&
\end{aligned}$$

Now, using $i^2 + j^2 + k^2 = ijk = -1$, we can simplify slightly:

$$\begin{aligned}
(v_1 + v_2 i + v_3 j + v_4 k) \cdot (u_1 + u_2 i + u_3 j + u_4 k)& \\
= v_1u_1 + (v_1u_2)i + (v_1u_3)j + (v_1u_4)k& \\
+ (v_2u_1)i + (v_2u_2)i^2 + (v_2u_3)ij + (v_2u_4)ik& \\
+ (v_3u_1)j + (v_3u_2)ji + (v_3u_3)j^2 + (v_3u_4)jk& \\
+ (v_4u_1)k + (v_4u_2)ki + (v_4u_3)kj + (v_4u_4)k^2& \\
= v_1u_1 + (v_1u_2)i + (v_1u_3)j + (v_1u_4)k& \\
+ (v_2u_1)i - (v_2u_2) + (v_2u_3)ij + (v_2u_4)ik& \\
+ (v_3u_1)j + (v_3u_2)ji - (v_3u_3) + (v_3u_4)jk& \\
+ (v_4u_1)k + (v_4u_2)ki + (v_4u_3)kj - (v_4u_4)&
\end{aligned}$$

All that's left is to figure out what the products of i, j and k should be.

We can derive ij and ji as follows:

1. Start with $ijk = -1$.
2. Multiply k from the right to get $ijk^2 = -k$, i.e. $-ij = -k$, i.e. $ij = k$.
3. Multiply ij from the left to get $ijij = ijk = -1$.
4. Multiply i from the left and j from the right to get $i(ijij)j = -ij$.
5. Simplify the left side to get $i(ijij)j = (ii)(ji)(jj) = ji$, i.e:

$$ij = -ji = k$$

Exercise 1.19. *Show that we also have $jk = -kj = i$ and $ki = -ik = j$.*

This means that Quaternion multiplication isn't commutative!

Our final multiplication table for $1, i, j$ and k is thus:

$\cdot$	1	i	j	k
1	1	i	j	k
i	i	-1	k	$-j$
j	j	$-k$	-1	i
k	k	j	$-i$	-1

Exercise 1.20. *Verify that this multiplication table doesn't violate associativity.*

Entering these new results into our partial multiplication formula, we get:

$$\begin{aligned}
&(v_1 + v_2 i + v_3 j + v_4 k) \cdot (u_1 + u_2 i + u_3 j + u_4 k) \\
&\qquad = v_1u_1 + (v_1u_2)i + (v_1u_3)j + (v_1u_4)k \\
&\qquad + (v_2u_1)i - (v_2u_2) + (v_2u_3)ij + (v_2u_4)ik \\
&\qquad + (v_3u_1)j + (v_3u_2)ji - (v_3u_3) + (v_3u_4)jk \\
&\qquad + (v_4u_1)k + (v_4u_2)ki + (v_4u_3)kj - (v_4u_4) \\
&\qquad = v_1u_1 + (v_1u_2)i + (v_1u_3)j + (v_1u_4)k \\
&\qquad + (v_2u_1)i - (v_2u_2) + (v_2u_3)k - (v_2u_4)j \\
&\qquad + (v_3u_1)j - (v_3u_2)k - (v_3u_3) + (v_3u_4)i \\
&\qquad + (v_4u_1)k + (v_4u_2)j - (v_4u_3)i - (v_4u_4)
\end{aligned}$$

which we can finally simplify to get:

$$\begin{aligned}
(v_1 + v_2 i + v_3 j + v_4 k) \cdot (u_1 + u_2 i + u_3 j + u_4 k)& \\
= \ (v_1u_1 - v_2u_2 - v_3u_3 - v_4u_4)& \\
+ (v_1u_2 + v_2u_1 + v_3u_4 - v_4u_3)i& \\
+ (v_1u_3 - v_2u_4 + v_3u_1 + v_4u_2)j& \\
+ (v_1u_4 + v_2u_3 - v_3u_2 + v_4u_1)k&
\end{aligned}$$

Now, this is obviously a long and unreasonably complicated formula - is there any way to think of it differently? The complex numbers had a very nice and intuitive geometric definition, maybe the quaternions do too?

Well, the key trick here is to split the underlying set $\mathbb{R}^4$ into $\mathbb{R} \times \mathbb{R}^3$, thinking of each Quaternion as consisting of a ***real part*** $a \in \mathbb{R}$ and an ***imaginary part*** $\vec{v} \in \mathbb{R}^3$. Our multiplication formula now reads:

$$\begin{aligned}
(a, \vec{v}) \cdot (b, \vec{u})& \\
= \ (ab - v_1u_1 - v_2u_2 - v_3u_3)& \\
+ (au_1 + bv_1 + v_2u_3 - v_3u_2)i& \\
+ (au_2 - v_1u_3 + bv_2 + v_3u_1)j& \\
+ (au_3 + v_1u_2 - v_2u_1 + bv_3)k&
\end{aligned}$$

which you might notice consists exclusively of familiar vector operations:

1. $ab - v_1u_1 - v_2u_2 - v_3u_3 = ab - \langle \vec{v}, \vec{u} \rangle$
2. $\begin{pmatrix} au_1 \\ au_2 \\ au_3 \end{pmatrix} = a\vec{u}, \begin{pmatrix} bv_1 \\ bv_2 \\ bv_3 \end{pmatrix} = b\vec{v}$
3. $\begin{pmatrix} v_2u_3 - v_3u_2 \\ v_1u_3 - v_3u_1 \\ v_1u_2 - v_2u_1 \end{pmatrix} = \vec{v} \times \vec{u}$

Putting everything together, we can now write out our multiplication formula in a much nicer form:

$$(a, \vec{v}) \cdot (b, \vec{u}) = (ab - \langle \vec{v}, \vec{u} \rangle, a\vec{u} + b\vec{v} + \vec{v} \times \vec{u}),$$

Exercise 1.21. *Verify that quaternion multiplication is associative and distributes over vector addition.*

Exercise 1.22. *Let $p = (a, \vec{v}) \in \mathbb{H}$. Then $pq = qp$ holds for all $q \in \mathbb{H}$ if and only if $\vec{v} \neq 0$.*

Similar to complex numbers, we can define ***quaternion conjugation***:

Definition 1.23: Quaternion Conjugation

Just like for $\mathbb{C}$, we define the ***conjugate*** of a quaternion $(a, \vec{v})$ to be the quaternion

$$\overline{(a, \vec{v})} = (a, -\vec{v})$$

An immediate application of quaternion conjugation is finding a formula for the inverse of a quaternion, which ones again ends up working out very similarly to the complex case - We have

$$\begin{aligned} \overline{(a, \vec{v})} \cdot (a, \vec{v}) &= (a, -\vec{v}) \cdot (a, \vec{v}) \\ &= (a^2 - \langle -\vec{v}, \vec{v} \rangle, a\vec{v} - a\vec{v} + (-\vec{v} \times \vec{v})) \\ &= \left(a^2 + \|\vec{v}\|^2, \vec{0}\right) \in \mathbb{R} \end{aligned}$$

And thus we can simply divide by this value to get our inverse formula:

Corollary 1.24: Inverse Quaternion

Let $(a, \vec{v}) \in \mathbb{H}_{\neq 0}$. Then $(a, \vec{v})$ has a multiplicative inverse, which is given by:

$$(a, \vec{v})^{-1} = \frac{(a, -\vec{v})}{a^2 + \|v\|^2}.$$

Definition 1.25. Let $q = (a, \vec{v}) \in \mathbb{H}$. Then

$$|q| := \sqrt{\overline{q} \cdot q} = a^2 + v_1^2 + v_2^2 + v_3^2 = \|q\|_2$$

Note how much this once again resembles both the real case $|x| = \sqrt{x^2} = \|x\|_2$ and the complex case $|z| = \sqrt{\overline{z}z} = \|z\|_2$.

1.5 Quaternions and $\mathbb{R}^3$

Theorem 1.26: Quaternions encode rotations

Let $\|\vec{v}\| = 1$ and

$$q = (\cos\varphi, \vec{v}\sin\varphi) \in \mathbb{H}$$

Then $qw\overline{q} \in \mathbb{R}^3$ is the vector obtained by rotating w around the axis $\{\lambda\vec{v} : \lambda \in \mathbb{R}\}$ by the angle 2φ.

Since 2π corresponds to a full rotation, φ and $\varphi + \pi$ describe the same rotation. Therefore, there are two different quaternions q and $-q$ associated with each rotation - in topological terms, we say that the quaternions form a ***double cover*** of the space of all rotations.

This phenomenon is actually closely related to the theory of ***spin*** in quantum physics. All known quantum particles that constitute ordinary matter, including protons, neutrons, electrons, neutrinos, and quarks, are ***spin-$\frac{1}{2}$***, meaning that they exhibit a curious behavior where a rotation of 360° leaves them in a different state than they were before, from which they return to their original state only after being rotated by another 360° around the same axis. This lets us mathematically model particles as quaternions, such that a 360° rotation transforms a particle from a state $q \in \mathbb{H}$ to the "rotationally equivalent" state $-q \in \mathbb{H}$.

Corollary 1.27. *For every isometry $F : \mathbb{R}^3 \to \mathbb{R}^3$, there exist $u \in \mathbb{R}^3$ and $q \in \mathbb{H}$ such that we have either*

1. *$F(w) = u + qw\overline{q}$, if F preserves orientation, or*
2. *$F(w) = u + q\overline{w}\overline{q}$, if F doesn't preserve orientation.*

This representation turns out to be a lot more computationally efficient than other representations of the same rotation. Other, more intuitively "trivial" representations of rotation also run into issues such as ***gimbal lock***, where they may get "stuck" in certain orientations.

This makes quaternions surprisingly practical in many applied contexts, including game development and animation[1], robotics, and three-dimensional image processing.

[1]For an introduction to quaternions in game development, I highly recommend this excellent talk by Freya Holmer.

Chapter 2

Linear Maps

2.1 Modules and Vector Spaces

Definition 2.1: Modules

Let $(R, +, \cdot)$ be a Ring. Then a (left) ***R-Module*** consists of an abelian Group $(M, +)$ equipped with a ***scalar multiplication*** $. : R \times M \to M$, i.e. scalars are elements of R and "vectors" are elements of M, such that:

1. Ring multiplication associates with scalar multiplication:

$$(r \cdot s).m = r.(s.m)$$

2. Scalar multiplication distributes over module addition:

$$r.(m + n) = r.m + r.n$$

3. Scalar multiplication distributes over scalar addition:

$$(r + s).m = r.m + s.m$$

If R is unital, then we say that M is unital if the identity of ring multiplication is the identity of scalar multiplication:

$$1.m = m$$

A ***right R-module*** is defined similarly, but with a scalar multiplication $. : M \times R \to M$, which is applied from the right instead of from the left.

Exercise 2.2. *Show that the distinction between right R-modules and left R-modules is only relevant if R is noncommutative, i.e. that any left R-module M over a commutative ring can be turned into a right R-module by defining $m.r = r.m$ and vice versa.*

To keep this text in line with familiar conventions of vector multiplication in $\mathbb{R}^n$, whenever we talk about a module without explicitly specifiying whether scalar multiplication is applied from the right or from the left, it is implied that it is applied from the left.

Exercise 2.3. *Let M be a module over R. Let $m \in M$ and $r \in R$. show that:*

1. $0_R.m = 0_M$
2. $r.0_M = 0_M$
3. $(-r).m = r.(-m) = -(r.m)$

Exercise 2.4. *Verify that:*

1. *Every ring R forms a module over itself, where scalar multiplication is given by ring multiplication.*
2. *$\mathbb{R}^n$ forms a module over $\mathbb{R}$, with scalar multiplication defined the usual way. More generally, given any ring R, the cartesian product R^n forms an R-module where scalar multiplication is given by ring multiplication applied component-wise.*
3. *Any abelian group A can be turned into a non-unital module over any ring R by equipping it with the trivial scalar multiplication*

$$r.a = 0_A$$

Definition 2.5: Vector Spaces

A module over a field (or skew field) $\Bbbk$ is known as a ***$\Bbbk$-vector space***.

Definition 2.6: Linear Maps

A ***module homomorphism*** is a structure-preserving map $F : M \to N$ between two modules over the same ring R, i.e. given $m_1, m_2 \in M$ and $r \in R$, we have that:

1. F is ***additive***, i.e. it is compatible with module addition:

$$F(m_1 +_M m_2) = F(m_1) +_N F(m_2)$$

2. F is ***homogenous***, i.e. it is compatible with scalar multiplication:

$$F(r._M m) = r._N F(m)$$

Module homomorphisms are also known as ***linear maps***.

As the name suggests, the main goal of linear algebra is to generalize the concept of a ***linear map***, and to figure out properties and applications of these maps.

Exercise 2.7. *Show that any linear map $F : M \to N$ fulfills*

$$F(0_M) = 0_N$$

Exercise 2.8. *Show that any linear map $F : R \to R$ from a unital ring into itself can be represented as multiplication by a constant, i.e. $F(r) = a.r = a \cdot_R r$ for some $a \in R$.*

Exercise 2.9.

1. *Show that complex conjugation is a linear map $\mathbb{C} \to \mathbb{C}$ if we view $\mathbb{C}$ as a module over $\mathbb{R}$.*
2. *Show that complex conjugation is not linear if we view $\mathbb{C}$ as a module over $\mathbb{C}$.*

2.2 Subspaces and Quotient Spaces

2.3 Bases and Dimension

Definition 2.10: Linear Combinations

Let M be an R-module, let $(m_i)_{i\in I} \subseteq M$ and $(r_i)_{i\in I} \subseteq R$. Then a ***linear combination*** is a sum of the form

$$\sum_{i\in I} r_i.m_i$$

Definition 2.11: Vector Span

Let M be an R-module, let $S = (m_i)_{i\in I} \subseteq M$. Then the span of S is the set $\langle S\rangle$ of all linear combinations over elements of S, i.e.

$$\langle S\rangle = \left\{ \sum_{i\in J\subseteq I} r_i.m_i \;\middle|\; (r_i)_{i\in I} \subseteq R, m_i \in S \right\}$$

For example, the span of one vector $v \in \mathbb{R}^3$ is the line through the origin on which v lies. The span of two orthogonal vectors $v, w \in \mathbb{R}^3$ is the plane through the origin on which v and w lie.

2.4 Matrices

2.5 Systems of Linear Equations

2.6 The Method of Least Squares

Chapter 3

Affine Spaces

Affine spaces are structures similar to vector spaces, but where the notions of ″points″ and ″directions″ are separate.

Definition 3.1: Affine Space

A $\mathbb{k}$-***affine space*** is a tuple $(A, \vec{A})$, consisting of a set A of ***points*** and a $\mathbb{k}$-vector space $\vec{A}$ of ***translation vectors***, alongside a translation map $+ : A \times \vec{A} \to A$, such that:

1. Translation is associative:

$$\forall a \in A, \vec{v}, \vec{w} \in \vec{A} :$$
$$(a + \vec{v}) + \vec{w} = a + (\vec{v} + \vec{w})$$

2. For any pair of points, there exists a unique vector translating one point onto the other:

$$\forall a, b \in P_A :$$
$$\exists! \vec{v} \in \vec{A} : a + \vec{v} = b$$

The dimension of an affine space is simply the dimension of its associated vector space.

Any vector space V can be canonically turned into an affine space by taking V as both the set of points and the set of vectors. On the flipside, we can turn any affine space $(A, \vec{A})$ into a vector space by restricting our attention to translations of one ″base point″ $b \in A$, and then identifying any point $p \in A$ with the vector $\vec{v} \in \vec{A}$ translating b onto p.

For most practical purposes, it is therefore sufficient to work with a vector space as your base space for whatever math you are doing, but the non-canonicity of having to choose an arbitrary base point $b \in A$ to turn an affine space into a vector space means that some people prefer affine spaces for philosophical reasons - after all, what is the ″base point″ of our physical universe?

The distinction becomes a lot more important when considering affine subspaces, which are essentialy vector subspaces translated by arbitrary points:

Definition 3.2: Affine subspaces

Let $(A, \vec{A})$ be an affine space. Then an affine subspace of $(A, \vec{A})$ is a tuple $(B, \vec{B})$ such that:

1. $B \subseteq A$.
2. $\vec{B}$ is a vector subspace of $\vec{A}$.
3. $\vec{B}$ is the set of translations between points in B, i.e. for any $b \in B$, $\vec{v} \in \vec{A}$, we have $b + \vec{v} \in B$ if and only if $\vec{v} \in \vec{B}$.

Thus, an affine subspace can always be written as

$$B = b + \vec{B} := \left\{ b + \vec{v} \,\middle|\, \vec{v} \in \vec{B} \subseteq \vec{A} \right\}$$

for an arbitrary base point $b \in B$.

An affine subspace of dimension 1 is known as a ***line***. An affine subspace of dimension 2 is known as a ***plane***. An affine subspace whose dimension is one lower than the space it resides in is known as an ***hyperplane***.

Definition 3.3. Two affine subspaces $(A, \vec{A})$, $(B, \vec{B})$ are called ***parallel*** if $\vec{A} = \vec{B}$.

Definition 3.4: Affine Maps

Let A, B be $\Bbbk$-affine spaces. Then a map $F : A \to B$ is called ***affine*** if there exists a linear map $\vec{F} : \vec{A} \to \vec{B}$ such that

$$\forall a \in P_a, \vec{v} \in \vec{A} : F(a + \vec{v}) = F(a) + \vec{F}(\vec{v})$$

As previously mentioned, if our affine space is a vector space, we can identify any point in P_A with the vector translating the origin onto that point, which lets us write:

$$F(\vec{v}) := F(0 + \vec{v}) = F(0) + \vec{F}(\vec{v}) := \vec{F}(\vec{v}) + \vec{w}$$

thus, an affine map between vector spaces is simply a linear map composed with a translation.

Theorem 3.5: Intercept Theorem

Let $v_1, v_2 \in \mathbb{R}$ be linearly independent. Let L_1 be an affine line through v_2 and v_1. Then if we have $\alpha, \beta \in \mathbb{R}_{\neq 0}$, and L_2 is a parallel line through αv_2 and βv_1, we must have $\alpha = \beta$.

Proof. Since L_1 and L_2 are parallel, their direction vectors must point in the same direction, which means there must be a $\mu \in \mathbb{R}_{\neq 0}$ such that

$$\begin{aligned} \mu(v_2 - v_1) &= \mu(\alpha v_2 - \beta v_1) \\ \implies \mu v_2 - \mu v_1 &= \alpha v_2 - \beta v_1 \\ \implies \mu v_2 - \alpha v_2 &= \mu v_1 - \beta v_1 \\ \implies (\mu - \alpha) v_2 &= (\mu - \beta) v_1 \end{aligned}$$

Since v_2 and v_1 are linearly independent, this implies $\mu - \alpha = \mu - \beta = 0$, i.e. $\alpha = \beta = \mu$. □

Theorem 3.6. *An injective map $\Phi : \mathbb{R}^2 \to \mathbb{R}^2$ is linear if and only if it maps $\vec{0}$ to $\vec{0}$ and lines to lines.*

Proof. $\Rightarrow$ Let Φ be linear. Let $g := \left\{\vec{p} + \lambda \cdot \vec{v}\right\}$ be a line. Then we have

$$\begin{aligned} \Phi(g) &= \left\{\Phi(\vec{p} + \lambda \cdot \vec{v})\right\} \\ &= \left\{\Phi(\vec{p}) + \lambda \cdot \Phi(\vec{v})\right\}, \end{aligned}$$

i.e. $\Phi(g)$ is also a line.

$\Leftarrow$ Since Φ maps $\vec{0}$ to $\vec{0}$ and lines to lines, it must map lines through the origin to lines through the origin. Let

$$\begin{aligned} \Phi(e_1) &:= v_1 \\ \Phi[\lambda e_1] &:= L_1 = \{\lambda v_1\} \\ \Phi(e_2) &:= v_2 \\ \Phi[\lambda e_2] &:= L_2 = \{\lambda v_2\} \end{aligned}$$

Since Φ is injective, L_1 and L_2 can only intersect at $\{0\}$, and thus v_1 and v_2 must be linearly independent.

Now, let's take a closer look at images of individual points. Since $\Phi[\lambda e_1] = \{\lambda v_1\}$ and $\Phi[\lambda e_2] = \{\lambda v_2\}$, we must have

$$\begin{aligned} \Phi[\lambda e_1] &= \varphi_1(\lambda) v_1 \\ \Phi[\lambda e_2] &= \varphi_2(\lambda) v_2 \end{aligned}$$

for some $\varphi_1, \varphi_2 : \mathbb{R} \to \mathbb{R}$.

Now, notice that the line through λe_1 and λe_2 is parallel to the line through e_1 and e_2. Since Φ is injective, it must preserve parallelism, and so the line through v_1 and v_2 must be parallel to the line through $\varphi_1(\lambda) v_1$ and $\varphi_2(\lambda) v_2$. The intercept theorem now gives us $\varphi_1(\lambda) = \varphi_2(\lambda) = \mu$. We thus write $\varphi(\lambda) := \varphi_1(\lambda) = \varphi_2(\lambda)$.

Since Φ is injective, φ must be as well, since otherwise we would have some v with $\Phi(\lambda_1 v) = \Phi(\lambda_2 v)$ despite $\lambda_1 v \neq \lambda_2 v$. We have also have $\varphi(0) = 0$ and $\varphi(1) = 1$, since $\Phi(0) = 0$ and $\Phi(e_1) = v_1$.

Now, we want to show that $\Phi(v+w) = \Phi(v)+\Phi(w)$ holds for linearly independent v, w. Since Φ preserves parallelism, it must map parallelograms to parallelograms. Therefore, the parallogram spanned by v and w must be mapped to the parallogram spanned by $\Phi(v)$ and $\Phi(w)$. This means that the upper corner $v + w$ gets mapped to the upper corner $\Phi(v)+\Phi(w)$ which gives us $\Phi(v+w) = \Phi(v)+\Phi(w)$ as desired.

Putting our insights together, we now know that

$$\begin{aligned}\Phi(\lambda_1 e_1 + \lambda_2 e_2) &= \Phi(\lambda_1 e_1) + \Phi(\lambda_2 e_2) \\ &= \varphi(\lambda_1)\Phi(e_1) + \varphi(\lambda_2)\Phi(e_2)\end{aligned}$$

it only remains to be shown that φ is the identity. For this, consider that we have:

$$\begin{aligned}(\varphi(a) + \varphi(b))\Phi(e_1) &= \varphi(a)\Phi(e_1) + \varphi(b)\Phi(e_1) \\ &= \Phi(ae_1 + be_1 \\ &= \Phi((a + b)e_1) \\ &= \varphi(a + b)\Phi(e_1),\end{aligned}$$

which gives us $\varphi(a)+\varphi(b) = \varphi(a+b)$. For $a \neq 0$, the line through e_1 and ae_2 must point in the same direction as the line through be_1 and bae_2, so the line through $\Phi(e_1)$ and $\varphi(a)\Phi(e_2)$ must be parallel to the line through $\varphi(b)\Phi(e_1)$ and $\varphi(ab)\Phi(e_2)$, from which the intercept theorem gives us

$$\begin{aligned}&\frac{\varphi(ab)}{\varphi(a)} = \varphi(b) \\ \implies\ &\varphi(ab) = \varphi(a)\varphi(b)\end{aligned}$$

Since we have $\varphi(0) = 0$, $\varphi(1) = 1$, $\varphi(a + b) = \varphi(a) + \varphi(b)$, $\varphi(ab) = \varphi(a)\varphi(b)$, our map $\varphi : \mathbb{R} \to \mathbb{R}$ must be a field homomorphism, of which the identity is the only one. Thus, Φ is linear.

□

Exercise 3.7. *An injective map between real affine spaces of dimension ≥ 2 is affine if and only if it maps lines to lines.*

Chapter 4

Determinants

4.1 Defining Volume

4.2 Determinants of Endomorphisms

4.3 Orientation

Chapter 5

Eigenvectors and Eigenvalues

Definition 5.1: Eigenvalues and Eigenvectors

Let k be a field, $F : V \to V$ an Endomorphism, and $\lambda \in k$. Then the ***λ-eigenspace*** of F is the set

$$\begin{aligned}\mathrm{Eig}(F,\lambda) &:= \left\{\vec{v} \in V : F(\vec{v}) = \lambda\vec{v}\right\} \\ &= \left\{\vec{v} \in V : (F - \lambda\mathrm{id}_V)(\vec{v}) = 0\right\} \\ &= \ker(F - \lambda\mathrm{id}_V)\end{aligned}$$

If $F(\vec{v}) = \lambda\vec{v}$, with $\vec{v} \neq \vec{0}$, we say that $\vec{v}$ is an ***eigenvector*** of F with ***eigenvalue*** λ.

Note that we exclude $\vec{0}$ from being an eigenvector, since $F(\vec{0}) = \lambda\vec{0}$ trivially holds for every endomorphism. Thus every λ would be an eigenvalue, which would make the definition pointless. We do, however, allow 0 as an eigenvalue - the 0-eigenspace of F is simply its kernel. Sometimes it ends up being useful to rephrase our definition of eigenspaces in terms of kernels:

$$\begin{aligned}\mathrm{Eig}(F,\lambda) &= \left\{\vec{v} \in V : F(\vec{v}) = \lambda\vec{v}\right\} \\ &= \left\{\vec{v} \in V : F(\vec{v}) - \lambda\vec{v} = 0\right\} \\ &= \left\{\vec{v} \in V : (F - \lambda\mathrm{id}_V)(\vec{v}) = 0\right\} \\ &= \ker(F - \lambda\mathrm{id}_V)\end{aligned}$$

This shows that $\mathrm{Eig}(F,\lambda)$ is a subspace of V - by definition, these subspaces must then be the maximal subspaces that are closed under application of F, i.e. $F(\mathrm{Eig}(F,\lambda)) = \mathrm{Eig}(F,\lambda)$.

Exercise 5.2. *In an informal, intuition-based way:*

1. *Find the real eigenvalues and eigenspaces of a reflection of $\mathbb{R}^2$ along an axis $\left\{\lambda\vec{v} : \lambda \in \mathbb{R}\right\}$*

2. *Find a rotation of $\mathbb{R}^2$ that has no real eigenvalues. Find two different rotations of $\mathbb{R}^2$ that do have real eigenvalues. Show that any rotation of $\mathbb{R}^2$ has complex eigenvalues.*

3. *Find the eigenvectors of the derivative map $\frac{d}{dx}$*

Our characterisation $\mathrm{Eig}(F, \lambda) = \ker(F - \lambda \mathrm{id}_V)$ means that, if F is finite-dimensional, we can find all eigenvectors of F with a given eigenvalue λ by solving $(F - \lambda \mathrm{id}_V)\vec{v} = 0$, which is simply a systen of linear equations. Finding the correct eigenvalues ends up being more tricky - in the section after this one, we will work towards proving the following key result:

Proposition 5.3. *The eigenvalues of F are given by the zeroes of the* **characteristic polynomial** $\chi_F(\lambda) := \det(F - \lambda \mathrm{id}_V)$.

However, before we do so, we want to familiarize ourselves with some of the more basic properties of eigenvectors and eigenvalues.

One important application is that finding eigenvectors lets us diagonalize endomorphisms, greatly easing computation afterwards:

Theorem 5.4. *Let V be a vector space over a field k. Let $B = (b_1, \dots, b_n)$ be a basis of V, and let F be an endomorphism represented in B by a matrix A. Then:*

1. *A basis vector b_i is an eigenvector of F iff. the j-th column of A has the form λe_j.*
2. *F is diagonalizable iff. there exists a basis of V consisting of eigenvectors of F.*

Theorem 5.5. *Let V be a vector space over a field k. Let $F : V \longrightarrow V$ be an endomorphism. Then eigenvectors of F with different eigenvalues are linearly independent.*

Corollary 5.6. *Let V be an n-dimensional vector space over a field k. Let $F : V \longrightarrow V$ be an endomorphism. Then:*

1. *F has no more than n distinct eigenvalues,*
2. *If $(\lambda_i)_{i \in I}$ are mutually distinct eigenvalues of F, the sum*
$$\sum_{i \in I} \mathrm{Eig}(F, \lambda_i) \subseteq V$$
is direct.
3. *F is diagonalizable if and only if V is a direct sum of eigenspaces of F.*
4. *If F has n distinct eigenvalues, it is diagonalizable.*

Corollary 5.7. *Let V, W be n-dimensional vector spaces over a field k. Let $F : V \longrightarrow V$, $G : W \longrightarrow W$ be endomorphisms. Then there exists a unique endomorphism $P : V \longrightarrow W$ such that $P \circ F = G \circ P$ if and only if F and G have the same eigenvalues.*

5.1 Characteristic Polynomials

Theorem 5.8: Characterstic Polynomial

Let V be a finite-dimensional vector space over a field $\mathbb{k}$. Let $F : V \longrightarrow V$ be an endomorphism. Then $\lambda \in \mathbb{k}$ is an eigenvalue of F if and only if

$$\chi_F(\lambda) = \det(\lambda \cdot \mathrm{id}_V - F) = 0_{\mathbb{k}}$$

We call χ_F the ***characteristic polynomial of F***.

Proof. We know that $\lambda \in \Bbbk$ is an Eigenvalue of F if and only if

$$\ker(F - \lambda \mathrm{id}_V) \neq \{0_V\},$$

i.e. if and only if $F - \lambda \mathrm{id}_V$ isn't injective.

We also know that $F - \lambda \mathrm{id}_V$ isn't injective if and only if it isn't surjective, i.e. if and only if it isn't invertible, i.e. if and only if

$$\det(\lambda.\mathrm{id}_V - F) = \chi_F(\lambda) = 0_{\Bbbk}.$$

□

Exercise 5.9. *Show that every endomorphism $F : V \to V$ of a finite-dimensional complex vector space V has at least one eigenvalue.*

Exercise 5.10. *Show that the constant term of a characteristic polynomial $\chi_F(\lambda)$ is given by* $(-1)^n \cdot (\det F)$.

(...)

5.2 Algebras and the Cayley-Hamilton Theorem

Theorem 5.11: Evaluation map

Let R be a commutative unital ring, and let A be a commutative unital algebra over R. Then, for every $a \in A$, there exists a unique algebra homomorphism $\mathrm{ev}_a : R[X] \to A$ such that:

1. $\mathrm{ev}_a(r) = 1_A.r$ for all constant polynomials $r \in R[X]$,
2. $\mathrm{ev}_A(X) = X$.

We call ev_a the ***evaluation map of $R[X]$ at a***.

Informally, this means that we can take elements of an algebra A over R and plug them into polynomials that were originally only defined on R, and we won't run into issues. In particular, we can plug matrices into polynomials over an underlying ring! Since ev_a is unique, we will generally just write $P(a) := \mathrm{ev}_a(P)$ from now on.

Theorem 5.12: Cayley-Hamilton

Let R be a commutative unital ring. Let $A \in R^{n\times n}$. Then $\chi_A(A) = 0$.

5.3 Minimal Polynomials

Theorem 5.13: Minimal polynomial

et F be an endomorphism of a vector space V over a field $\mathbb{k}$. Then there exists a unique polynomial $\mu_F \in \mathbb{k}[X]$ such that:

1. μ_F is either normalized or equal to $0_{\mathbb{k}[X]}$
2. μ_F is a divisor of every $P \in \mathbb{k}[X]$ with $P(F) = 0_{\mathrm{End}_{\mathbb{k}}(V)}$

We call μ_F the ***minimal polynomial of $\boldsymbol{F}$***.

Chapter 6

Normal Forms of Endomorphisms

6.1 Decomposing Vector Spaces using Polynomials

Exercise 6.1. *Let G be an abelian group. Show that G becomes a $\mathbb{Z}$-module when equipped with the operation* $n.g = \underbrace{g + \ldots + g}_{n \text{ times}}$.

Exercise 6.2. *Let V be a vector space over a field $\Bbbk$. Show that V becomes an* $\mathrm{End}(V)$-*Module when equipped with the operation $F.v = F(v)$.*

Theorem 6.3. *Any $\Bbbk[X]$-module is the same thing as a $\Bbbk$-vector space V equipped with a specified endomorphism F, and vice versa.*

Proof.

1. Let V be a $\Bbbk$-vector space. We know from the previous exercise that V is an $\mathrm{End}(V)$-module. Thus, the polynomial evaluation map $\mathrm{ev}_F : \Bbbk[X] \to \mathrm{End}(V)$ turns V into $\Bbbk$-module via

$$P.v = \mathrm{ev}_F(P)(v) = P(F)(v)$$

2. Now let M be a $\Bbbk[X]$-module. Then M becomes a $\Bbbk$-module, i.e. a $\Bbbk$-vector space, if we restrict scalar multiplication to only accept constant polynomials. Then it can easily be seen that $G(v) = X.v$ is a linear transformation.

Now, note that if we start with a vector space (V, F), then use our construction to get a module M, and then turn it back into a vector space, we will obtain our original F again, since

$$G(v) = X._M v = X(F)(v) = F(v)$$

Conversely, if we start with a module M, then turn it into a vector space (V, F), and then back into a module M, we reobtain our original scalar multiplication, since

$$P._V v = P(G)(v) = P(X._M)(v) = P._M v$$

Therefore, the two directions of our construction are inverse to each other, i.e. we actually have a bijection. □

Thus, given a $\Bbbk[X]$-module M, we can write $M = (V, F)$

Exercise 6.4. *Let $M = (V, F)$ and $N = (W, G)$ be $\Bbbk[X]$-Modules. Then a $\Bbbk[X]$-linear map $H : M \to N$ is $\Bbbk$-linear and additionally fulfills $H \circ F = G \circ H$, and any $\Bbbk$-linear map H fulfilling these properties is $\Bbbk[X]$-linear.*

Let $M = (V, F)$ be a $\Bbbk[X]$-Module. Then, by definition, a subspace $U \subseteq V$ is a $\Bbbk[X]$-submodule of M if and only if U is closed under scalar multiplication with elements of $\Bbbk[X]$.

Exercise 6.5. *Show that U is closed under scalar multiplication with elements of $\Bbbk[X]$ if and only if $Xu \in U$, which itself holds if and only if $F(u) \in U$ holds for all $u \in U$, i.e. U is* ***F*-invariant**.

Exercise 6.6.

1. *Show that the scalar multiplication map $P. = P(F)$ is linear.*
2. *Conclude that $\ker(P(F))$ and $\operatorname{im}(P(F))$ are submodules of M.*
3. *Conclude that $\ker(P(F))$ and $\operatorname{im}(P(F))$ are F-invariant.*

Theorem 6.7: Structure theorem for modules over a Euclidean ring

Let M be a module over a euclidean domain R. Let $a = bc \in R$ such that b and c share no divisors and $a.m = 0$ for all $m \in M$. Then we have:

1. $\ker(b.) = \operatorname{im}(c.)$,
2. $\operatorname{im}(b.) = \ker(c.)$,
3. $M = \ker(b.) \oplus \ker(c.)$.

Proof.

1. Let $c.m \in \operatorname{im}(c.)$ for $m \in M$. Then we have $b.(c.m) = a.m = 0$, i.e. $\operatorname{im}(c.) \subseteq \ker(b.)$. Now, applying the Euclidean algorithm gives us $1 = k_1 b + k_2 c$. If we now let $m \in \ker(b.)$, we get:

$$\begin{aligned} m &= 1m \\ &= (k_1 b + k_2 c)m \\ &= k_1 \underbrace{(bm)}_{=0} + k_2(cm) \\ &= c(k_2 m) \in \operatorname{im}(c.) \end{aligned}$$

   Thus, we also have $\ker(b.) \subseteq \operatorname{im}(c.)$, i.e. $\ker(b.) = \operatorname{im}(c.)$.

2. $\ker(c.) = \operatorname{im}(b.)$ follows from the same argument by swapping the roles of b and c.

3. Now, let $m \in M$. Then, applying our identity from before, we get

$$\begin{aligned} m &= (k_1 b + k_2 c)m \\ &= b(k_1 m) + c(k_2 m) \\ &\in \operatorname{im}(b.) + \operatorname{im}(c.) \\ &= \ker(b.) + \ker(c.), \end{aligned}$$

which gives us $M = \ker(b.) + \ker(c.)$ as desired.

If we have $m \in \ker(b.) \cap \ker(c.)$, we get

$$m = k_1(b(m)) + k_2(c(m)) = 0,$$

so $\ker(b.) \cap \ker(c.) = \{0\}$, so our sum is direct.

□

Corollary 6.8. *Let G be an abelian group. Let $n = mk \in \mathbb{Z}$ such that m and k share no divisors and $n \cdot g = 0$ for all $g \in G$. Then we have*

$$G = \ker(m\cdot) \oplus \ker(k\cdot).$$

Corollary 6.9: Chinese Remainder Theorem

Corollary 6.10. *Let V be a vector space over a field $\Bbbk$ and $F : V \to V$ be an Endomorphism. Let $P = QR \in \Bbbk[X]$ such that Q and R share no divisors and $P.v = P(F)(v) = 0$ for all $v \in V$. Then we have*

$$V = \ker(Q(F)) \oplus \ker(R(F)).$$

Corollary 6.11: P-Generalized Eigenspace Decomposition

Let V be a vector space over a field $\Bbbk$ and $F : V \to V$ be an Endomorphism. Let $P \in \Bbbk[X]$ be a normed polynomial and let $P = P_1^{n_1} \cdot \ldots \cdot P_k^{n_k}$ be its prime factorization. Then we have

$$V = \bigoplus_{i=1}^{k} \ker(P_i^{n_i}(F))$$

6.2 The Jordan Normal Form

Corollary 6.12: Generalized Eigenspace Decomposition

Let V be a vector space over a field $\Bbbk$ and $F : V \to V$ be an Endomorphism. Assume that its minimal polynomial μ_F splits into linear factors over $\Bbbk$, i.e.

$$\mu_F(x) = \prod_{i=1}^{k}(x - \lambda_i)^{n_i}$$

with $\lambda_i \in \Bbbk$. Then we have

$$V = \bigoplus_{i=1}^{k} \ker((F - \lambda_i \mathrm{id}_V)^{n_i})$$

We call $\ker((F - \lambda_i \mathrm{id}_V)^{n_i})$ the ***generalized λ_i-Eigenspace of F*** and denote it $H(F, \lambda_i)$ (from the German "Hauptraum").

Exercise 6.13. *Show that $H(F, \lambda_i)$ contains all eigenvectors of F with eigenvalue λ_i.*

Exercise 6.14. *Show that $H(F, \lambda_i)$ is an F-invariant subspace of V, i.e. a subspace such that $F(H(F, \lambda_i)) \subseteq H(F, \lambda_i)$*

Exercise 6.15. *Let $V = V_1 \oplus V_2$ be a decomposition of V into F-invariant subspaces. Let B_1 be a basis of V_1, and let B_2 be a basis of V_2. Show that:*

1. *$B = B_1 \cup B_2$ is a basis of V*
2. *If the restriction $F|_{V_1}$ of F to V_1 is represented in basis B_1 as a matrix A_1, and the restriction $F|_{V_2}$ of F to V_2 is represented in basis B_2 as a matrix A_2, then F is represented in basis B as*

$$A = \begin{pmatrix} A_1 & 0 \\ 0 & A_2 \end{pmatrix}$$

Exercise 6.16. *Let $F : V \to W$ be a linear map. Let A be a basis of* $\ker F$*, and let B be a tuple of vectors such that $F(B)$ is a basis of* $\operatorname{im} F$*. Then $A \cup B$ is a basis of V.*

We now want to make use of this to construct a Jordan basis B of a nilpotent Endomorphism N:

Corollary 6.17. *Let $N : V \to V$ be a nilpotent endomorphism. Then there exists a basis B of V such that:*

1. *$B \cup \{0\}$ is F-invariant, i.e. if $b \in B$, then we have $F(b) \in B$ or $F(b) = 0$.*
2. *For every $b \in B$, there exists at most one element $b' \in B$ with $F(b') = b$.*

We call such a basis a **Jordan basis***.*

> *Proof.* We prove this theorem in three steps: We construct a basis B, then show it is F-invariant, then show that each element of B has at most one preimage in B.

1. We construct our basis by induction over $n = \dim V$.

   If $n = 0$, then the empty set is a basis (since a sum over the empty set equals the additive identity, i.e. the zero vector) which (vacuously) fulfills our desired properties.

   Now, since the kernel of N is non-trivial, we have $\dim \text{im } N < \dim V$. Therefore, by induction, we can assume we already have a Jordan basis A of im N. We split this basis A into two parts:

   (a) $A_0 := \{a \in A \mid N(a) = 0\}$,

   (b) $A_{\neq 0} := \{a \in A \mid N(a) \neq 0\}$.

   A_0 is by definition a set of linearly independent vectors contained in $\ker A$. By the Steinitz exchange lemma, we can add a set C of Linearly independent vectors such that $A_0 \cup C$ is a basis of $\ker A$.

   $A_{\neq}0$ is the set of all A such that $N(A_{\neq 0}) \subseteq A$. We add a set D containing preimages of all the remaining elements of A, which must exist since $A \subseteq$ im N. By construction, we now have $F(A_{\neq 0} \cup D) = A$.

   Putting everything together, $A_0 \cup C$ is a basis of $\ker\ N$, and $F(A_{\neq 0} \cup D) = A$, which is a basis of im N. Therefore, by exercise 6.16, $B = A \cup C \cup D$ is a basis of V.

2. Now, B must be F-invariant since A is F-invariant by definition, C is contained in the kernel of F and thus $F(B) = \{0\}$, and D consists of preimages of A, i.e. $F(D) = A \subseteq B$.

3. Finally, the preimages of A in B are contained in either $A_{\neq 0}$ or D, which are disjoint by construction.

□

Corollary 6.18. *Let $F : V \longrightarrow V$ be a matrix whose characteristic polynomial has $n = \dim(V)$ roots. Then there exists a basis B of V such that ${}_BF_B$ is a block diagonal matrix consisting of Jordan blocks.*

Proof. Generalized Eigenspace decomposition 6.2 shows us that V is the direct sum of the generalized Eigenspaces of F. We know by exercise 6.14 that the generalized Eigenspaces of F are F-invariant. Therefore, exercise 6.15 tells us that if we find a basis B_i for each generalized Eigenspace $H(F, \lambda_i)$ such that $F|_{H(F,\lambda_i)}$ is represented by a matrix J_i in Jordan normal form, and we combine our bases into a basis $B = \bigcup_{i=1}^{n} B_i$ of V as a whole, then the matrix representation of F in basis B will be a block diagonal matrix consisting of these J_i, and thus F will also be in Jordan normal form.

Now, let F_i be the restriction of $(F - \lambda_i \mathrm{id}_V)$ to $H(F, \lambda_i)$. By definition of the generalized Eigenspace, we have

$$H(F, \lambda_i) = \ker((F - \lambda_i \mathrm{id}_V)^{n_i})$$

therefore, by construction, F_i is nilpotent. Therefore, by theorem **??**, there exists a basis B_i such that ${}_{B_i}F_{i\,B_i}$ is in Jordan normal form, with zeroes on the diagonal. But then, since we have

$$F|_{H(F,\lambda_i)} = F_i + \lambda \mathrm{id}_{H(F,\lambda_i)}$$

the matrix representation of $F|_{H(F,\lambda_i)}$ is also in Jordan form, since we simply have to add λ to the diagonal of ${}_{B_i}F_{i\,B_i}$.

Therefore, the restriction to F to any of its generalized Eigenspaces is in Jordan normal form in basis B_i. As discussed at the beginning of this proof, this means F is in Jordan normal form in basis $B = \bigcup_{i=1}^{n} B_i$. □

Exercise 6.19. *Let F be an endomorphism, and let J be its Jordan normal form. Then:*

1. *The algebraic multiplicity of each eigenvalue of F corresponds to the number of times that eigenvalue appears on the diagonal of J.*
2. *The geometric multiplicity of any eigenvalue λ of F corresponds to the number of λ-Jordan blocks with that eigenvalue in J.*
3. *The exponent of $(X - \lambda)$ in the minimal polynomial of F corresponds to the size the of largest λ-Jordan block in J.*

6.3 The Weierstrass Normal Form

Chapter 7

Inner Product Spaces

7.1 Inner Products

Definition 7.1. Let $\mathbb{k} = \mathbb{R}$ or $\mathbb{C}$. Let V and W be vector spaces over $\mathbb{k}$. Then a map $\varphi : V \to W$ is called ($\mathbb{k}$-) ***semilinear*** or ***antilinear***, if we have

1. ***Addivity:*** $\forall u, v \in V : \varphi(u + v = \varphi(u) + \varphi(v)$
2. ***Antihomogeneity:*** $\forall v \in V, \lambda \in \mathbb{k} : \varphi(\lambda.v) = \overline{\lambda}.\varphi(v)$.

Notice that for $\mathbb{k} = \mathbb{R}$, there is no difference between linearity and antilinearity, since complex conjugation leaves real numbers unchanged.

Definition 7.2. Let $\mathbb{k} = \mathbb{R}$ or $\mathbb{C}$. Let V be a $\mathbb{k}$-vector space. Then a map $s : V \times V \to \mathbb{k}$ is called a ***sesquilinear form*** if it is antilinear in the first component and linear in the second component,i.e. the map $s_w : v \to s(v, w)$ is antilinear and the map $s_v : w \to s(v, w)$ is linear.

Exercise 7.3. *Let A be a matrix with entries in $\mathbb{k} = \mathbb{R}$ or $\mathbb{C}$. Show that the map*

$$s_A(v, w) := v^*Aw$$

is sesquilinear. (Recall that v^* stands for the conjugated transpose of v.)

The latin prefix "sesqui-" means "one and a half" - a sesquilinear is a map that is linear in one component, and "sort of linear", i.e. "one-half-linear" in the other. Some sources require that a sesquilinear form is linear in the first component and antilinear in the second.

Definition 7.4. A sesquilinear form $s : V \times V \to \mathbb{k}$ is called a ***Hermitian form*** if it is ***conjugate symmetric***, i.e. $s(v, w) = \overline{s(w, v)}$.

Once again, notice that for $\mathbb{k} = \mathbb{R}$, this just means $s(v, w) = s(w, v)$.

Exercise 7.5. *Show that a matrix is Hermitian (i.e. fulfills $A = A^*$) if and only if its associated sesquilinear form $s_A(v, w) = v^*Aw$ is Hermitian.*

Definition 7.6. A sesquilinear form $s : V \times V \to \mathbb{k}$ is called ***positive semidefinite*** if $s(v, v) \geq 0$ for all $v \in V$. It is called ***positive definite*** if it additionally fulfills $s(v, v) = 0$ if and only if $v = \vec{0}$.

Positive definiteness is the primary reason we are working with sesquilinear forms instead of bilinear forms - we need to be able to compare $s(v, v)$ to 0, and there is no way to order all of $\mathbb{C}$ that is compatible with its field structure. However, if we have a sesquilinear Hermitian map, we get $s(v, v) = \overline{s(v, v)}$, i.e. $s(v, v) \in \mathbb{R}$, which means we can start comparing things again.

Definition 7.7. A ***scalar product*** or ***inner product*** on a vector space V over $\mathbb{R}$ or $\mathbb{C}$ is a positive definite Hermitian form. We call a real vector space equipped with a scalar product ***Euclidean***, and a complex vector space equipped with a scalar product ***unitary***.

Definition 7.8. The standard scalar product on $\mathbb{C}^n$ is given by

$$\langle v, w\rangle_{\mathbb{C}} = v^* w$$

Exercise 7.9. *Show that $\langle v, w\rangle_{\mathbb{C}}$ is actually a scalar product.*

Exercise 7.10. *Let $V = C^\infty([0, 1], \mathbb{R})$ be the room of infinitely differentiable functions $[0, 1] \to \mathbb{R}$. Show that the $\boldsymbol{L^2}$ scalar product*

$$\langle f, g\rangle_{L^2} = \int_0^1 f(t)g(t)\, dt$$

is actually a scalar product.

Definition 7.11. Let V be a vector space over $\mathbb{k} = \mathbb{R}$ or $\mathbb{C}$. Then a ***Norm*** is a map $\|\cdot\| : V \to \mathbb{k}$ fulfilling the following three axioms:

1. ***Positive definiteness:*** $\|v\| \geq 0$ and $\|v\| = 0$ iff. $v = \vec{0}$
2. ***Homogeneity:*** $\|\lambda.v\| = |\lambda|.\|v\|$
3. ***Triangle inequality:*** $\|v + w\| \leq \|v\| + \|w\|$

A norm is a way of assigning lengths to vectors. Once lengths of vectors are defined, one can easily define distances between vectors too:

Exercise 7.12. *Let $\|\cdot\|$ be a norm on a vector space V over a field $\mathbb{k} = \mathbb{R}$ or $\mathbb{C}$. Show that $d(v, w) := \|v - w\|$ is a metric on V.*

Theorem 7.13: Cauchy-Schwartz Inequality

Let s be a scalar product on a vector space V over a field $\mathbb{k}$. Then we have

$$|\langle v, w\rangle| \leq \sqrt{\langle v, v\rangle} \cdot \sqrt{\langle w, w\rangle}$$

One important property of scalar products is that they induce norms. Notice that the standard norm on $\mathbb{R}^n$ is just the square root of the standard scalar product on $\mathbb{R}^n$. This property generalizes to all scalar products:

Exercise 7.14. *Let $s : V \times V \to \mathbb{k}$ be a scalar product on a vector space V over a field $\mathbb{k}$. Let $\|v\|_s := \sqrt{s(v, v)}$. Show that:*

1. *$\|\cdot\|_s$ is a norm on V.*
2. *We have the* **parallelogram identity**

$$\|v + w\|_s^2 + \|v - w\|_s^2 = 2\|v\|_s^2 + 2\|w\|_s^2$$

3. *If $\mathbb{k} = \mathbb{R}$, we have the* **real polarisation formula**

$$s(v, w) = \frac{1}{4}\left(\|v + w\|_s^2 - \|v - w\|_s^2\right)$$

4. *If $\mathbb{k} = \mathbb{C}$, we have the* **complex polarisation formula**

$$s(v, w) = \frac{1}{4}\left(\|v + w\|_s^2 - \|v - w\|_s^2 - i\|v + i.w\|_s^2 + i\|v - i.w\|_s^2\right)$$

Exercise 7.15. *Find a norm on a vector space that cannot be constructed as the square root of a scalar product.*

Theorem 7.16. *Let V be a vector space over $\mathbb{k} = \mathbb{R}$ or $\mathbb{C}$. Let $\|\cdot\|$ be a norm on V fulfilling the parallelogram identity. Then there exists a corresponding scalar product s such that $\|v\| = \sqrt{s(v, v)}$*

7.2 Orthogonality

Definition 7.17. Let (V, s) be a vector space with scalar product over a field $\mathbb{k} = \mathbb{R}$ or $\mathbb{C}$. Then we call two vectors v, w ***orthogonal*** iff.

$$s(v, w) = 0$$

Exercise 7.18. *Let $B = (v_1, \dots, v_k)$ be a tuple of orthogonal vectors not containing the zero vector. Then B is linearly independent.*

Definition 7.19. A basis consisting of pairwise orthogonal vectors is called an ***orthogonal basis***. If the basis vectors are also normalized, i.e. have length 1, we call the basis an ***orthonormal basis***.

Exercise 7.20. *Let (V, s) be a vector space with scalar product. Show that $B = (b_1, \dots, b_n) \subset V$ is an orthonormal basis iff.*

$$s(b_i, b_j) = \delta_{ij}$$

(Recall that δ_{ij} is defined as the function that is 1 if $i = j$ and 0 otherwise).

Definition 7.21. Let (V, s) be a vector space with scalar product over a field $\mathbb{k} = \mathbb{R}$ or $\mathbb{C}$. Let $U \subseteq V$ be a subspace. Let $B_U := \{b_1, \dots, b_m\}$ be a basis of U. Then the **orthogonal projection of V onto U** is the map

$$\begin{aligned} p : V &\to U \\ v &\mapsto \sum_{i=1}^{m} g(b_i, v).b_i \end{aligned}$$

Exercise 7.22. 1. *Show that the orthogonal projection of V onto U leaves vectors in u unchanged.*

2. *Show that for each $v \in V$ and $u \in U$, u is orthogonal to $v - p(v)$.*

3. *Show that for each $v \in V$, $p(v)$ is the unique point in U for which $\|v - p(v)\|_s$ is minimal.*

Theorem 7.23. *Let (V, s) be a vector space with scalar product over a field $\mathbb{k} = \mathbb{R}$ or $\mathbb{C}$. Let $(v_1, \dots, v_n)$ be a basis of V. Then there are unique vectors $(b_1, \dots, b_n) \subset V$ such that for all $p \in (1, \dots, n)$, we have:*

1. $(b_1, \dots, b_p)$ *is an s-orthonormal Basis of* $\langle v_1, \dots, v_p \rangle \subseteq V$

2. $s(b_p, v_p) \in \mathbb{R}_{>0}$

These vectors b_p *can be constructed inductively via* $b_1 = \frac{v_1}{\|v_1\|}$ *followed by projection onto the already constructed basis vectors, i.e.*

$$w_p = v_p - \sum_{q=1}^{p-1} g(b_q, v_p).b_q$$
$$\dot{t}b_p = \frac{w_p}{\|w_p\|}$$

Corollary 7.24. *Every vector space with scalar product has an orthonormal basis.*

7.3 Matrix Representations of Inner Products

Definition 7.25. Let s be a sesquilinear form on a finite dimensional vector space Vover a field $\Bbbk = \mathbb{R}$ or $\mathbb{C}$. Let $B = (b_1, \dots, b_n)$ be a basis of V. Then the ***Gram matrix of B by g*** is defined as the matrix with entries

$$G_{ij} = s(b_i, b_j) \in \Bbbk^{n \times n}$$

Exercise 7.26. *Show that the Gram matrix of the standard basis of* $\mathbb{R}^n$ *by the scalar product is the identity matrix.*

Definition 7.27. A matrix $A \in \Bbbk^{n \times n}$ is called ***positive (semi-)definite*** if $s_A(v, w) = v^* A w$ is positive (semi-)definite.

Exercise 7.28. *Let s be a sesquilinear form on a finite-dimensional vector space V over a field* $\Bbbk = \mathbb{R}$ *or* $\mathbb{C}$*. Let G be its Gram matrix. Then we have* $s(v, w) = v^* G y$ *in basis B. In particular, s is a scalar product if and only if its Gram matrix is Hermitian and positive definite.*

Theorem 7.29. *Let* $A \in \Bbbk^{n \times n}$*, where* $\Bbbk = \mathbb{R}$ *or* $\mathbb{C}$*. Then the following are equivalent:*

1. *A is hermitian and psitive definite.*

2. **Cholesky Decomposition:** *There exists an upper triangular Matrix B with positive real diagonal entries such that* $A = B^* B$*.*

3. *There exists an invertible Matrix B such that* $A = B^* B$*.*

4. **Sylvester-Hurwitz Criterion:** *A is Hermitian and and all of its square upper left submatrices* A_r*, known as its* **leading principal minors**, *have strictly positive determinant.*

7.4 Dual Spaces

Definition 7.30: Dual Space

Let V be a vector space over a field $\Bbbk$. Then we define $V^* = \mathrm{Hom}_{\Bbbk}(V, \Bbbk)$. Elements of V^*, i.e. linear maps $V \to \Bbbk$, are known as ***linear forms***.

Theorem 7.31: Dual Basis

iven a basis $B = (b_1, \ldots, b_n)$ of a vector space V, the set of linear forms $E = (\varepsilon_1, \ldots, \varepsilon_n)$ given by

$$\varepsilon_p(e_q) = \delta_{pq}$$

is a basis of V^*, known as the ***dual basis*** of B.

Exercise 7.32. *Let $C^n(\mathbb{k}, \mathbb{k})$, $n \in 0, 1, \ldots, \infty$ be the set of n-times continuously differentiable functions $\mathbb{k} \to \mathbb{k}$. Then for any $x_0 \in \mathbb{k}$, the map*

$$\cdot(x_0) : f \to f^{(n)}(x_0)$$

is a linear form on $C^n(\mathbb{k}, \mathbb{k})$ (where $f^{(n)}$ denotes the n-th derivative of f).

Exercise 7.33. *The map mapping an integrable function f to $\int_0^1 f(x)\, dx$ is a linear form. More generally, for any integrable $g : [0, 1] \to \mathbb{k}$, the map taking f to $\int_0^1 \overline{g(x)} \cdot f(x)\, dx$ is a linear form.*

Theorem 7.34. *Let (V, g) be a $\mathbb{k}$-vector space with scalar product. Then g induces an injective antilinear map $g_1 : V \to V^*$ via*

$$g_1(v) = g(v, \cdot)$$

A linear form $\alpha \in V^*$ is called **representable** *by g if there exists a $v \in V$ such that $\alpha = g(v, \cdot)$.*

Definition 7.35. Let V be a $\mathbb{k}$-vector space. Then a ***antilinear form*** is an antilinear forms $\gamma : V \to \mathbb{k}$. The ***antidual space*** $\overline{V}^*$ of V is the space of antilinear maps $V \to \mathbb{k}$.

Theorem 7.36. *Let (V, g) be a finite-dimensional vector space over a field $\mathbb{k}$. Then the maps $g : V \to V^*$ and $\overline{g} : V \to V^*$ are bijective.*

Corollary 7.37. *Let (V, g) be a finite-dimensional vector space. Then all linear and antilinear forms on V are representable by g.*

Definition 7.38. Let (V, g) and (W, h) be $\mathbb{k}$-vector spaces with scalar products. Then a linear map $F : V \to W$ is called ***adjointable*** if there exists a map $G : W \to V$ such that

$$g(G(w), v) = h(w, F(v)$$

for all $v \in V$ and $w \in W$. Such a map G is known as the ***adjoint of F***, and is written $G = F^*$. If $F = F^*$, we call F ***self-adjoint***.

Proposition 7.39. *If it exists, the adjoint of a given map is unique.*

Proposition 7.40. *Adjoint maps are linear.*

Proposition 7.41. *If G is the adjoint of F, then F is the adjoint of G, i.e. we have $(F^*)^* = F$.*

Proposition 7.42. *If A is a matrix representation of F by two orthonormal bases, then the conjugate transpose A^* is a matrix representation of F^*.*

Definition 7.43. Let V and W be $\Bbbk$-vector spaces, with $\Bbbk$ a field. Let $F : V \to W$ be a linear map. Then the ***dual*** of F is the map

$$\begin{aligned} F^\vee : W^* &\to V^* \\ \beta &\mapsto \beta \circ F \end{aligned}$$

i.e.

$$F^\vee(\beta)(\vec{v}) = \beta(F(\vec{v}))$$

for $\beta \in W^*$, $\vec{v} \in V$.

We can construct a dual of any linear map this way, while linear maps involving infinite-dimensional vector spaces are generally not adjoinable.

Exercise 7.44. *Verify the following properties of the dual map:*

1. $(\mathrm{id}_V)^\vee = \mathrm{id}_{V^*}$
2. *Let $F : V \to W$ and $G : U \to V$ be linear, then $(F \circ G)^\vee = G^\vee \circ F^\vee$*
3. *The dual of a linear map is linear.*
4. *If $F : V \to W$ is given by $F(\vec{v}) = A \cdot \vec{v}$ for some matrix A, then $F^\vee(\beta) = \beta \cdot A$, where elements $\beta \in W^*$ are represented as row vectors.*

Theorem 7.45. *Let $U \subseteq V$ be vector spaces. Let $\iota : U \to V$ be the inclusion map. Let $q : V \to V/U$ be the quotient map. Then $(V/U)^* \cong \mathrm{im}\ q^\vee = \ker\ \iota^\vee$.*

Proof. Since q maps V to V/U, we have $q^\vee : (V/U)^* \to \mathrm{im}\ q^* \subseteq V^*$. Since q is linear, so is $q^\vee$. By definition, any map surjectively maps onto its image. Therefore, all we need to show to obtain $(V/U)^* \cong \mathrm{im}\ q^\vee$ is injectivity.

So, let $\beta \in \ker\ q^\vee \subseteq (V/U)^*$, i.e. let $q^\vee(\beta) = \beta(q)$ be the zero map. Since q is surjective, $\beta(q(v)) = 0$ for all $v \in V$ already implies $\beta(w) = 0$ for all $w \in V/U$, so β must have already been the zero map.

Finally, we want to show $\mathrm{im}\ q^\vee = \ker\ \iota^\vee$.

First off, recall that $\iota^\vee : V^* \to U^*$. By the universal property of the quotient map, for every $\beta \in \ker \iota^\vee \subseteq V^*$, there exists $\overline{\beta} : V/U \to \Bbbk$ (i.e. $\overline{\beta} \in (V/U)^*$) with $\beta = \overline{\beta} \circ q = \overline{\beta}(q) \in \mathrm{im}\ q^\vee$, so we have $\ker \iota^\vee \subseteq \mathrm{im}\ q^\vee$.

Conversely, for any $\beta \in (V/U)^*$ and $v \in (V/U)$, we have

$$q^\vee(\beta)(v) = \beta(q(v)) = \beta(0) = 0,$$

so we also have $\mathrm{im} q^\vee \subseteq \ker \iota^\vee$. □

Theorem 7.46

Let (V, g) and (W, h) be $\Bbbk$-vector spaces with scalar products. Let $h_1 : W \longrightarrow W^*$ and $g_1 : V \longrightarrow V^*$ be the antilinear maps induced by g and h, i.e.

$$h_1(\vec{w}) = h(\vec{w}, \cdot)$$
$$g_1(\vec{v}) = g(\vec{v}, \cdot)$$

Let $F : V \longrightarrow W$ be linear. Then $g_1 \circ F^* = F^\vee \circ h_1$.

Proof. Let $v \in V, w \in W$, then we have:

$$\begin{aligned} g_1(F^*(w))(v) &= g(F^*(w), v) \\ &= h(w, F(v)) \\ &= h_1(w)(F(v)) \\ &= F^\vee(h_1(w))(v) \end{aligned}$$

□

Corollary 7.47. *If V is finite-dimensional, $F : V \longrightarrow W$ is adjointable.*

Proof. If V is finite-dimensional, g_1 is invertible. We get $F^* = g_1^{-1} \circ F^\vee \circ h_1$. □

Definition 7.48. Let V and W be $\Bbbk$-vector spaces. Let $F : V \longrightarrow W$ be linear, and let $s : W \times W \longrightarrow \Bbbk$ be a sesquilinear form. Then we define the ***pullback of s by F*** to be the map $F^*s : V \times V \longrightarrow \Bbbk$ given by

$$(F^*s)(u, v) = s(F(u), F(v))$$

Exercise 7.49. *Verify that:*

1. *F^*s is a sesquilinear form.*
2. *If s is Hermitian, F^*s is Hermitian.*
3. *If s is positive semidefinite, F^*s is positive semidefinite.*
4. *If s is positive definite and F is injective, F^*s positive definite.*
5. *If we fix bases of V and W such that F is represented by a matrix A and s is represented by a Gram matrix G, then F^*s is represented by A^*GA.*
6. *Given $F : V \longrightarrow W$ antilinear, we can achieve the same results by defining $F^*s(u, v) = s(F(v), F(u))$.*

Chapter 8

Normal Endomorphisms

Definition 8.1

Let (V, g) be a $\mathbb{k}$-vector space with scalar product. Let $F : V \to V$ be an adjointable endomorphism. Then we call F ***normal*** if $F^* \circ F = F \circ F^*$.

Exercise 8.2.

1. *Show that self-adjoint maps are normal.*
2. *A linear map with $F^* = -F$ is called* **skew**. *Show that skew maps are normal.*
3. *A linear map represented in a given basis by a matric of the form*

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$$

is normal.

Definition 8.3

Let (V, g) and (W, h) be $\mathbb{k}$-vector spaces with scalar products. Then a linear map $F : V \to W$ is called an ***isometric embedding*** if $g(v, w) = h(F(v), F(w))$ for all $v, w \in V$.

Exercise 8.4. *Show that:*

1. *Isometric embeddings preserve lengths and distances.*
2. *Isometric embeddings preserve angles.*
3. *Isometric embeddings are injective.*
4. *If orthonormal bases are chosen for V and W, then the matrix representation A of F fulfills $A^*A = E_n$.*
5. *If* $\dim V = \dim W$, *F is invertible with $F^{-1} = F^*$. An invertible isometric embedding is called an* **isometry**.
6. *Isometries are normal.*

Theorem 8.5: Fundamental Theorem of Normal Endomorphisms

Let (V, g) be a finite-dimensional complex vector space with scalar product. Let $F : V \to V$ be an endomorphism. Then there exists an orthonormal basis of (V, g) consisting of eigenvectors of F if and only if F is normal.

Note that this only works for complex numbers, not for reals:

Exercise 8.6. *Recall that any endomorphism represented by a matrix of the form*

$$A = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$$

is normal. Show that, if $b \neq 0$, A has no real eigenvalues.

Corollary 8.7: A Normal Form for Complex Normal Endomorphisms

Let (V, g) be a finite-dimensional real vector space with scalar product. Let $F : V \to V$ be an endomorphism. Then there exists an orthonormal basis of (V, g) in which F is represented by a block diagonal matrix consisting of 1×1 blocks and 2×2 blocks of the form

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$$

with $b > 0$ if and only if F is normal. If such a representation exists, it is unique up to permutation of the blocks.

Corollary 8.8: Spectral Theorem for Self-Adjoint Maps

Let (V, g) be a finite dimensional real or complex vector space with scalar product. Let $F : V \to V$ be an endomorphism. Then there exists an orthonormal basis of V in which F is represented by a diagonal matrix with real entries if and only if F is self-adjoint.

Exercise 8.9. *Derive an analogous spectral theorem for skew maps.*

Corollary 8.10. *Let (V, g) be a finite-dimensional real or complex vector space with scalar product. Let s be a sesquilinear form on V. Then there exists a g-orthonormal Basis of V in which s is represented by a diagonal matrix with real entries if and only s is hermitian.*

Corollary 8.11: Singular Value Decomposition

Let (V, g) and (W, h) be finite-dimensional real or complex vector spaces with scalar products. Let $F : V \to W$ be linear. Then there exist orthonormal bases of V and W in which F is represented by a diagonal matrix with uniquely determined real entries a_n such that $a_1 \geq a_2 \geq \dots$ and $a_n = 0$ if $n \geq \operatorname{rk} F$. These entries are known as the ***singular values*** of F.

Singular value decomposition is fundamental to the JPG compression algorithm, which represents an image by a matrix, then calculates the singular value decomposition of that matrix, and then sets all of the singular values below a given threshold to zero, leading to less data needing to be stored while still maintaining the singular values that 'noticably contribute' to the resulting image.

8.1 Linear Isometries

Corollary 8.12: Normal Form for Real Isometries

Let (V, g) be a finite-dimensional real vector space with scalar product g. Let $F : V \to V$ be a linear isometry. Then there exists an orthonormal basis of (V, g) in which F is represented by a block diagonal matrix consisting of 1×1 blocks and 2×2 blocks of the form

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$$

with $a^2 + b^2 = 1$.

Corollary 8.13: Normal Form for Complex Isometries

Let (V, g) be a finite-dimensional complex vector space with scalar product g. Let $F : V \to V$ be a linear isometry. Then there exists an orthonormal basis of (V, g) in which F is represented by a diagonal matrix whose entries have absolute value 1.

Corollary 8.14

Let (V, g) be a finite-dimensional complex vector space with scalar product g. Let $F : V \to V$ be a linear isometry. Then V can be decomposed into a direct sum of subspaces U of dimension one or two, such that F acts on the one-dimensional subspaces as the identity or reflection, and on the two-dimensional subspaces as a rotation around the origin.

Exercise 8.15. *A linear isometry is orientation-preserving if and only if the number of invariant subspaces on which it acts as a reflection is even.*

8.2 Affine Isometries

Theorem 8.16. *Let $(A, \vec{A}, s))$, $(B, \vec{B}, g)$ be finite-dimensional normed affine spaces of the same dimension over vector spaces with scalar products. Then there exists an affine isometry $F : A \to B$.*

Theorem 8.17. *Let $(A, \vec{A}, g)$ be a finite-dimensional normed $\mathbb{k}$-affine space over a vector space with scalar product. Let $F : A \to A$ be an affine sometry. Then there exists a point $o \in A$ and a vector $x \in \mathrm{Eig}(\vec{F}, 1)$ such that*

$$F(a) = o + x + \vec{F}(a - o)$$

x is uniquely determined by A, while o is an arbitrary point chosen from an affine-F-invariant subspace $(B, \mathrm{Eig}(\vec{F}, 1)) \subseteq (A, \vec{A})$.

Definition 8.18. We call $(B, \mathrm{Eig}(\vec{F}, 1))$ the ***axis set*** of F, and call lines of the form

$$\{b + \lambda.x \mid \lambda \in \mathbb{k}\} \subseteq B$$

with $b \in B$ the ***axes*** of F.

Our theorem thus states that any affine isometry can be represented as a linear map composed with a translation along the axis set of F.

Chapter 9

Revisiting Synthetic Geometry

9.1 Incidence Geometries

Definition 9.1: Incidence Geometries

An ***incidence geometry*** is a set X equipped with a set $G \subseteq \mathcal{P}(X)$ of subsets of X known as ***lines***, such that:

1. Each line contains at least two points,
2. For every pair of points, there exists exactly one line containing both.

Instead of "x is contained in $g \in G$", we often say "x lies on $g \in G$".

Exercise 9.2. *Let R be a division ring. Then $X := R^n$ can be turned into an incidence geometry by defining lines to be all sets of the form*

$$g = \{p + \lambda v : \lambda \in R\},$$

where $p, v \in R^n$.

Moving forward, this will be the canonical choice of incidence geometry on R^n.

Definition 9.3. A set of points lying on the same line is called ***colinear***.

Definition 9.4. A set of lines containing a common point is called ***confluent***.

Definition 9.5. A set of three non-colinear points is known as a ***triangle***.

Definition 9.6: Betweenness

A ***betweenness relation*** on an incidence geometry (X, G) is a subset $Z \subseteq X^3$ such that the following properties hold:

1. ***Induced Orderings on Lines:*** On every line $g \in G$, there exists an ordering $\leq$ such that $(x, y, z) \in Z$ if and only if $x \leq y \leq z$ or $x \geq y \geq z$.

2. ***Pasch's Axiom:*** Let $p, q, r \in X$. Let

$$[p, q] := \{x \in X : (p, x, q) \in Z\}$$

denote the ***line segment between p and q***. Then every line $g \in G$ either intersects none of the line segments $[p, q]$, $[q, r]$, $[r, p]$, or at least two.

An incidence geometry equipped with a betweenness relation is known as an ***ordered incidence geometry***.

Exercise 9.7. *Let K be an ordered field. Then:*

1. *The set of all tuples (x, y, z), where $x, y, z \in K^n$ fulfill $y = \lambda x + \mu z$ with $\lambda, \mu \in [0, 1] \subset K$ and $\lambda + \mu = 1$, gives a betweenness relation on K^n.*

2. *There exists an affine isomorphism $\varphi : K \cong (g \in G)$ between any line in G and K.*

3. *The ordering on K can be recovered from the betweenness relation on K^n by taking the unique ordering $\leq$ on $g \in G$ that fulfills $\varphi(0) \leq \varphi(1)$.*

Moving forward, this will be the canonical choice of betweenness relation on K^n.

Definition 9.8. Given an ordered incidence geometry (X, G, Z), a ***ray*** in X is a set $A \subseteq X$ of the form

$$A = \{x \in g \mid x \leq p\},$$

where $g \in G$ is a line and $p \in g$ is a point on that line.

Definition 9.9: Congruence Groups

Given an ordered incidence geometry (X, G, Z), a ***congruence group*** of (X, G, Z) is a group K of automorphism of (X, G, Z) such that every pair of rays in $A, B \subseteq X$ has exactly two elements $h, k \in K$ such that $h(A) = k(A) = B$.

Intuitively, one of these two elements will be a rotation and the other a rotation composed with reflection along an axis.

Exercise 9.10. *Let K be an ordered field. Let s be the standard scalar product in K^2. Then the set of s-invariant affine automorphisms of K^2 forms a congruence group of K^2.*

Moving forward, this will be the canonical choice of congruence group on K^2.

Definition 9.11. We say that an ordered incidence geometry has the ***least upper bound property*** if every nonempty subset $A \subset g \in G$ that is bounded from above has a least upper bound.

Definition 9.12. We call an ordered incidence geometry (X, G, Z) with congruence group K ***near-euclidean*** if (X, G, Z) fulfills the supremum property and G is nonempty.

Definition 9.13. We say that an incidence geometry (X, G) fulfills the ***parallel postulate*** iff. for every line $g \in G$ and every point $p \notin G$, there exists at most one line $h \in G$ such that $p \in h$ and $h \cap g = \emptyset$.

Exercise 9.14. *Let $K \subsetneq \mathbb{R}$ be a proper subfield of $\mathbb{R}$ containing square roots of every positive element. Then K^2 fulfills the parallel postulate.*

The primary goal of this chapter will be proving the following result:

Theorem 9.15: Classification of near-euclidean geometries

Let (X, G, Z) be a near-euclidean incidence geometry. Then:

1. If (X, G, Z) fulfills the parallel postulate, it is isomorphic to $\mathbb{R}^2$.
2. If (X, G, Z) doesn't fulfill the perallel postulate, it is isomorphic to the ***hyperbolic plane*** H.

9.2 Projection Maps

Definition 9.16. Let V be a vector space. Then a ***linear hyperplane*** is a vector subspace of W of codimension 1, i.e.

$$\operatorname{codim}(W \subseteq V) := \dim(V/W) = 1$$

Equivalently, $W \neq V$ such that V is generated by W and any vector $v \in V \setminus V$. If V is finite-dimensional, we have $\dim(V/W) = \dim V - \dim W$, and thus a linear hyperplane is a subspace W whose dimension is one lower than the dimension of V. An ***affine Hyperplane*** is defined entirely analogously as an affine subspace of codimension one.

Exercise 9.17.

1. *In any affine space, the intersection of an affine line with another affine subspace is either empty, a single point, or the entire line.*
2. *For any affine hyperplane H and any line g with directional vector $\vec{v} \notin \vec{H}$, the intersection of g and H consists of a single point.*

Definition 9.18. Given an affine space E, an affine hyperplane $H \subset E$, and a vector $\vec{v} \in \vec{E} \setminus \vec{H}$, the ***parallel projection of E onto H in direction $\vec{v}$*** is the map

$$\pi = \pi_{\vec{v}} = \pi_{\vec{v},H} : E \to H$$

assigning to each point e the intersection of the affine line $(e + \langle \vec{v} \rangle)$ with H.

Definition 9.19. Given an affine space E, an affine hyperplane $H \subset E$, and a point $q \in E \setminus H$, the ***central projection of E onto H centered at q*** is the map

$$\zeta = \zeta_q = \zeta_{q,H} : E \setminus (q + \vec{H}) \to H$$

assigning to each point $e \in E$ the intersection of the line though e and q with H.

Exercise 9.20. *The image of a line g under central projection is either:*

1. *A line,*
2. *a pointed line, i.e. a line with a single missing point (the* **vanishing point***),*
3. *a single point,*
4. *or the empty set.*

9.3 Projectivization

Definition 9.21: Projectivization

Let $\mathbb{k}$ be a field. Let V be a $\mathbb{k}$-vector space. Then the set of all lines through the origin in V is called the the ***projectivization of V*** and denoted by $\mathbb{P}V$.

In the finite-dimensional case, we have $\dim \mathbb{P}V = \dim V - 1$. It is therefore common to write

$$\mathbb{P}^n\mathbb{k} := \mathbb{P}(\mathbb{k}^{n+1})$$

Beware, this means $\mathbb{P}\mathbb{k} \neq \mathbb{P}^1\mathbb{k}$! The set $\mathbb{P}^1\mathbb{k}$ is called the ***projective line over*** $\mathbb{k}$, the set $\mathbb{P}^2\mathbb{k}$ is called the ***projective plane over*** $\mathbb{k}$.

Definition 9.22. Given a vector space V, we call a subset $g \subseteq \mathbb{P}V$ a ***projective line*** or $\mathbb{P}$***-line*** if there exists a vector subspace $g' \subseteq V$ such that $\dim g' = 2$ and $g = \mathcal{P}g'$.

Lemma 9.23. *For any vector space V, the set $\mathbb{P}V$ forms an incidence geometry with $\mathbb{P}$-lines as the lines.*

Lemma 9.24. *In a projective plane over any field, any two different $\mathbb{P}$-lines intersect in exactly one point.*

Definition 9.25. An ***embedding of incidence geometries*** is an injective map $X \hookrightarrow Y$ such that points in X are colinear if and only if their images are colinear. A bijective embedding of incidence geometries is known as a ***colineation***.

Lemma 9.26. *Any injective vector space homormophism $W \hookrightarrow V$ induces an embedding $\mathbb{P}W \hookrightarrow \mathbb{P}V$.*

Definition 9.27

Let E be an affine space. Then we can equip the disjoint union

$$\mathbb{V}E := E \sqcup \mathbb{P}\vec{E}$$

with an incidence structure by designating lines to be:

1. Sets $\mathbb{V}g := g \sqcup \{\vec{g}\}$ for any affine line $g \in E$,
2. and $\mathbb{P}$-lines $l = \mathbb{P}l'$.

We call this structure the ***projective completion of E***. We call points in $\mathbb{P}\vec{E}$ ***points at infinity***.

Exercise 9.28. *If E is a two-dimensional affine space, then its projective completion $\mathbb{V}E$ contains exactly one line containing only infinitely distant points. We call this line the* **line at infinity**.

Exercise 9.29. *Given an affine space E, the trivial embeddings $E \hookrightarrow \mathbb{V}E$ and $\mathbb{P}\vec{E} \hookrightarrow \mathbb{V}E$ are embeddings of incidence geometries. Any injective affine map $F \hookrightarrow E$ of affine spaces is an embedding of incidence geometries and induces an embedding $\mathbb{V}E \hookrightarrow \mathbb{V}F$ of their projective completions.*

Theorem 9.30

Let V be a vector space. Let E be an affine hyperplane in V not containing the origin. Then

$$\mathbb{V}E \cong \mathbb{P}V$$

Corollary 9.31: Projective lines are projective completions

Let F be a field. Then we have $\mathbb{P}^1 F \cong \mathbb{V}F = F \sqcup \{\infty\}$.

Exercise 9.32.

1. *For any field $\mathbb{F}$, we have*

$$\mathbb{P}^n\mathbb{F} \cong \bigsqcup_{i=0}^{n} \mathbb{F}^i$$

2. *For a finite Field $\mathbb{F}_q$, we have*

$$\left|\mathbb{P}^n\mathbb{F}_q\right| = \sum_{i=0}^{n} q^i$$

Chapter 10

Hyperbolic Geometry

10.1 Circle Inversions

Definition 10.1. Let E be a Euclidean plane. Then the corresponding ***extended Euclidean plane*** is the set

$$\hat{E} = E \cup \{\infty\},$$

adding a "point at infinity" to E.

Definition 10.2. We call a subset $L \subseteq \hat{E}$ a ***generalized circle*** if it is either a circle in E or the union of a line in E with $\{\infty\}$. We call the former ***real circles*** and the latter ***extended lines***.

Definition 10.3. Let E be a euclidean plane. Then we assign to each generalized circle $L \subseteq \hat{E}$ the ***circle inversion map***

$$s_L : \hat{E} \to \hat{E},$$

which is given by the ordinary reflection at L with the additional rule $s_L(\infty) = \infty$ if L is an extended line, and by

$$c + \vec{v} \mapsto \begin{cases} c + \left(\frac{r}{\|\vec{v}\|}\right)^2 \vec{v} & \vec{v} \neq \vec{0} \\ \infty & \vec{v} = \vec{0} \end{cases}$$

with $\infty \mapsto c$ if L is a circle with center c and radius r.

Exercise 10.4. *s_L is an involution, i.e. we have $s_L^2 = \mathrm{id}$.*

Exercise 10.5. *For any two points $p, q \in \hat{E}$, there exists a circle inversion s_L with $s_L(p) = q$ and $s_L(q) = p$*

Exercise 10.6. *Every circle inversion maps generalized circles to generalized circles.*

10.2 Möbius Transformations

Definition 10.7. A composition of circle inversions is known as a ***Möbius transformation***.

Exercise 10.8. *The set* $\text{Möb}(\hat{E})$ *of Möbius transformations of a given extended Euclidean plane* $\hat{E}$ *forms a group under composition.*

Exercise 10.9. *Any two generalized circles in a generalized Euclidean plane can be mapped onto each other by a Möbius transformation.*

Theorem 10.10. *Given an extended Euclidean plane* E*, its Möbius transformations are precisely the bijections* $\varphi : \hat{E} \xrightarrow{\sim} \hat{E}$ *mapping generalized circles to generalized circles.*

Theorem 10.11. *Any Möbius transformation that leaves a given generalized circle* L *pointwise unchanged is either the circle inversion* s_L *at that circle or the identity map.*

Exercise 10.12. *Given a generalized Euclidean plane* $\hat{E}$*, a Möbius transformation* $\varphi : \hat{E} \to \hat{E}$ *and a generalized circle* $L \subseteq \hat{E}$*, we have* $\varphi \circ s_L \circ \varphi^{-1} = s_{\varphi(L)}$

Exercise 10.13. *Given a generalized Euclidean plane* $\hat{E}$*, a Möbius transformation* $\varphi : \hat{E} \to \hat{E}$*, and a generalized circle* $L \subseteq \hat{E}$*, we have* $\varphi(L) = L$ *if and only if* $\varphi \circ s_L = s_L \circ \varphi$.

Definition 10.14. Let $\hat{E}$ be a generalized Euclidean plane. Given a pair of generalized circles $L, M \subset \hat{E}$ we call L ***orthogonal to M***, which we write $L \perp M$, if $L \neq M$ and $s_L(M) = M$.

Theorem 10.15. *Let* $L, M \subset \hat{E}$ *be generalized circles in a generalized Euclidean plane* $\hat{E}$ *such that* $L \perp M$*. Let* $\varphi : \hat{E} \to \hat{E}$ *be a Möbius transformation. Then* $\varphi(L) \perp \varphi(M)$.

Proposition 10.16. *Let* E *be a Euclidean plane,* $g \subset E$ *a line,* $H \subset E \setminus g$ *an open Half-plane with* g *as its boundary, and*

$$K_H = \text{Möb}_H(\hat{E}) := \left\{\varphi \in \text{Möb}(\hat{E}) \,\middle|\, \varphi(H) = H\right\}$$

the set of Möbius transformations leaving H *invariant. Then:*

1. *Any Möbius transformation that leaves* H *invariant also leaves the extended line* $\hat{g}$ *invariant.*

2. $\text{Möb}_H(\hat{E})$ *operates transitively on* H*, i.e. any point of* H *can be mapped onto any other by a Möbius transformation* $\varphi \in \text{Möb}_H(\hat{E})$.

3. $\text{Möb}_H(\hat{E})$ *is generated by the set of circle inversions* s_L *with* $L \perp \hat{g}$.

4. *Different elements of* K_H *are still different when restricted to* H.

10.3 Complex Möbius Geometry

Definition 10.17. We denote by γ the "double complex conjugation map"

$$\begin{aligned} \gamma : \mathbb{C}^2 &\to \mathbb{C}^2 \\ (w, z) &\mapsto (\overline{w}, \overline{z}) \end{aligned}$$

Theorem 10.18. *The set of all antilinear automorphisms of $\mathbb{C}^2$ is given by*

$$\mathrm{GL}(\mathbb{C}^2)\gamma := \left\{ F \circ \gamma \,\middle|\, F \in \mathrm{GL}(\mathbb{C}^2) \right\}$$

Definition 10.19. We denote by

$$\mathrm{GL}(\mathbb{C}^2)\left\langle \gamma \right\rangle := \mathrm{GL}(\mathbb{C}^2) \sqcup \mathrm{GL}(\mathbb{C}^2)\gamma$$

the set of all automorphisms of $\mathbb{C}^2$ that are either linear or antilinear.

Since $\mathbb{P}^1\mathbb{C}$ consists of subsets of $\mathbb{C}^2$, every map $\mathbb{C}^2 \to \mathbb{C}^2$ mapping lines to lines acts on $\mathbb{P}^1\mathbb{C}$ in a trivial way. By theorem 9.31 $\mathbb{P}^1\mathbb{C} \cong \hat{\mathbb{C}}$, the map

$$\begin{aligned} \iota_{\mathbb{C}} : \hat{\mathbb{C}} \xrightarrow{\sim} \mathbb{P}^1\mathbb{C} &\hookrightarrow \mathbb{C}^2 \\ z &\mapsto (z, 1) \\ \infty &\mapsto (1, 0) \end{aligned}$$

is a bijection, which means we also get an action on $\hat{\mathbb{C}}$.

Theorem 10.20: Action of $\mathrm{GL}(\mathbb{C}^2)\left\langle \gamma \right\rangle$ on $\hat{\mathbb{C}}$

The canonical action gives us a group homomorphism

$$\begin{aligned} \varphi : \mathrm{GL}(2; \mathbb{C})\left\langle \gamma \right\rangle &\to \mathrm{Ens}^\times(\hat{\mathbb{C}}) \\ \gamma &\mapsto (z \mapsto \overline{z}) \\ M := \begin{pmatrix} a & b \\ c & d \end{pmatrix} &\mapsto \left(z \mapsto \begin{cases} \frac{az+b}{cz+d} & z \neq \infty \wedge cz + d \neq 0 \\ \frac{a}{c} & z = \infty \wedge c \neq 0 \\ \infty & z \neq \infty \wedge cz + d = 0 \\ \infty & z = \infty \wedge c = 0 \end{cases} \right) \end{aligned}$$

Proof.

1. Let $z \in \mathbb{C}$. Translating directly from $\mathbb{C}$ to gives $\mathbb{C}^2$ $\iota_{\mathbb{C}}(z) = \binom{z}{1}$, which means $M(\iota_{\mathbb{C}}(z)) = \binom{az+b}{cz+d}$.
   (a) If $cz+d$, then this point lies on the same line through the origin as $\binom{\frac{az+b}{cz+d}}{1}$, which is then translated back from $\mathbb{P}^1\mathbb{C}$ to $\hat{\mathbb{C}}$ as the point $\frac{az+b}{cz+d}$.
   (b) If $cz + d = 0$, then this point lies on the same line through the origin as $\binom{1}{0}$, which is then translated back from $\mathbb{P}^1\mathbb{C}$ to $\hat{\mathbb{C}}$ as the point ∞.
2. If $z = \infty$, then $\iota_{\mathbb{C}}(\infty) = \binom{1}{0}$, which gets mapped to $\binom{a}{c}$, from which we can proceed as before.

□

Theorem 10.21. *Our canonical group homomorphism* $\varphi : \mathrm{GL}(\mathbb{C}^2)\langle y\rangle \to \mathrm{Ens}^{\times}(\hat{\mathbb{C}})$ *is a surjective group homomorphism*

$$\varphi : \mathrm{GL}(\mathbb{C}^2)\langle \gamma\rangle \twoheadrightarrow \mathrm{M\ddot{o}b}(\hat{\mathbb{C}})$$

whose kernel is the group consisting of all scalar multiples of the identity.

Proof. 1. Since $\mathrm{GL}(\mathbb{C}^2)$ is generated by the diagonal matrices together with the matrices $S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $\mathrm{GL}(2,\mathbb{C})\langle\gamma\rangle$ is generated by these matrices along with γ. It therefore suffices to show that the images of these matrices are Möbius transformations. Now, according to our previous analysis of the maps generated by φ, these images are as follows:

   (a) A diagonal matrix with entries λ and μ gets mapped to $\frac{\lambda}{\mu} \cdot z$, which corresponds to a scaled rotation.

   (b) S gets mapped to $z \mapsto \frac{1}{z}$, which is the circle inversion at the unit circle composed with a reflection along the real axis.

   (c) T gets mapped to the translation $z \mapsto z + 1$.

   (d) γ gets mapped to the conjugation $z \mapsto \overline{z}$, which is just the reflection along the real axis.

   All of these maps are known Möbius transformations.

2. We now know that the image of φ contains inversion at the unit circle, all translations, and all scalings. Therefore, it contains all inversions along proper circles. It also contains reflection along the extended real axes and all rotations at the origin, so it also contains reflections along extended lines. Thus´, the image contains the generators of $\mathrm{M\ddot{o}b}(\hat{\mathbb{C}})$, and therefore the entire group.

3. An antilinear automorphism cannot stabilize every one-dimensional subspace. Thus, antilinear automorphisms cannot be mapped the identity, i.e. they cannot be contained in the kernel.

   Meanwhile, the only linear automorphisms stabilizing every one-dimensional subspace are scalings, i.e. multiples of the identity.

□

Definition 10.22: Projective Linear Groups

Given a vector space V, the ***projective linear group*** $\mathrm{PGL}(V)$ is defined as the quotient group of $\mathrm{GL}(V)$ by the group $Z(V)$ of all scalar multiples of the identity. The ***projective special linear group*** $\mathrm{PSL}(V)$ is analogously defined as the quotient group of $\mathrm{SL}(V)$ by the group $SZ(V)$ of all multiples of the identity by a unit scalar.

Theorem 10.23: Möbius group as a projective linear group

$$\text{Möb}^+(\hat{\mathbb{C}}) \cong \text{PGL}\big(\mathbb{C}^2\big)$$

Definition 10.24. We denote by τ reflection along the axis $\langle e_1 \rangle$, i.e.

$$\begin{aligned} \tau : \mathbb{R}^2 &\to \mathbb{R}^2 \\ \begin{pmatrix} a \\ b \end{pmatrix} &\mapsto \begin{pmatrix} a \\ -b \end{pmatrix} \end{aligned}$$

Theorem 10.25: Action of $\text{SL}(\mathbb{C}^2)$ on $\hat{\mathbb{C}}$

Our surjective group homomorphism $\varphi : \text{GL}(\mathbb{C}^2)\,\langle\gamma\rangle \twoheadrightarrow \text{Möb}\big(\mathbb{C}^2\big)$ induces a surjective group homomorphism

$$\text{SL}(\mathbb{R}^2)\,\langle\tau\gamma\rangle \twoheadrightarrow \text{Möb}_{\mathbb{H}}(\hat{\mathbb{C}})$$

whose kernel is $\pm\text{Id}_{\mathbb{R}^2}$.

Theorem 10.26: Conjugations of $\mathbb{H}$ as a projective special linear group

$$\text{Möb}^+_{\mathbb{H}}(\hat{\mathbb{C}}) \cong \text{PSL}\big(\mathbb{R}^2\big)$$

Part V

Elementary Differential Geometry

Chapter 1

Differentiation in $\mathbb{R}^n$

1.1 Differentials and Partial Derivatives

1.2 Rules of Differentiation

Theorem 1.1: Chain Rule in $\mathbb{R}^n$

Let $f : U \to \mathbb{R}^m$ and $g : V \to \mathbb{R}^r$ with $U \subset \mathbb{R}^n$, $V \subset \mathbb{R}^m$ open and $f(U) \subseteq V$. Let f be differentiable at a point $p \in U$ and let g be differentiable at $f(p)$. Then

$$D(g \circ f)(p) = Dg(f(p)) \circ Df(p)$$

Exercise 1.2. *Verify that this version of the chain rule agrees with the familiar chain rule for $f, g : \mathbb{R} \to \mathbb{R}$.*

Proof. By definition of the differential, we have:

$$\begin{aligned} f(x) &= f(p) \\ &\quad + Df(p)(x-p) \\ &\quad + \varepsilon_f(x)\|x-p\| \\ g(f(x)) &= g(f(p)) \\ &\quad + Dg(f(p))(f(x)-f(p)) \\ &\quad + \varepsilon_g(f(x))\|f(x)-f(p)\| \end{aligned}$$

For some functions $\varepsilon_f : U \to \mathbb{R}^m$, $\varepsilon_g : U \to \mathbb{R}^r$ which are continuous in p and $f(p)$ and fulfill $\varepsilon_f(p) = 0$ and $\varepsilon_g(f(p)) = 0$.
The second equation tells us:

$$\begin{aligned} g(f(x)) &= g(f(p)) \\ &\quad + Dg(f(p))(f(x)-f(p)) \\ &\quad + \varepsilon_g(f(x))\|f(x)-f(p)\| \end{aligned}$$

Plugging in our equation for $f(x)$ now lets us regroup our terms into an equation of the same form as the definition of the differential:

$$
\begin{aligned}
g(f(x)) = g(f(p))& \\
&+ Dg(f(p))\left[\overbrace{f(p) + Df(p)(x-p) + \varepsilon_f(x)\|x-p\|}^{f(x)} - f(p)\right] \\
&+ \varepsilon_g(f(x))\left\|\overbrace{f(p) + Df(p)(x-p) + \varepsilon_f(x)\|x-p\|}^{f(x)} - f(p)\right\| \\
&= g(f(p)) \\
&+ Dg(f(p))\left[Df(p)(x-p) + \varepsilon_f(x)\|x-p\|\right] \\
&+ \varepsilon_g(f(x))\left\|Df(p)(x-p) + \varepsilon_f(x)\|x-p\|\right\| \\
&= g(f(p)) \\
&+ Dg(f(p))Df(p)(x-p) \\
&+ Dg(f(p))\varepsilon_f(x)\|x-p\| \\
&+ \varepsilon_g(f(x))\left\|Df(p)(x-p) + \varepsilon_f(x)\|x-p\|\right\| \\
&= g(f(p)) \\
&+ Dg(f(p))Df(p)(x-p) \\
&+ \|x-p\| \cdot \underbrace{\left(\varepsilon_f(x)Dg(f(p)) + \varepsilon_g(f(x))\left\|Df(p)\left(\frac{x-p}{\|x-p\|}\right) + \varepsilon_f(x)\right\|\right)}_{:=\varepsilon_{g\circ f}(x)} \\
&= g(f(p)) \\
&+ (Dg(f(p)) \circ Df(p))(x-p) \\
&+ \varepsilon_{g\circ f}(x)\|x-p\|
\end{aligned}
$$

All that remains is to show that $\varepsilon_{g\circ f}(x)$ is continuous if we set

$$\varepsilon_{g\circ f}(p) = 0.$$

Since we have $\varepsilon_f(p) = 0$ and $\varepsilon_g(f(p)) = 0$, we have our desired result if we can show that

$$\left\|Df(p)\left(\frac{x-p}{\|x-p\|}\right)\right\| = \frac{\|Df(p)(x-p)\|}{\|x-p\|}.$$

is bounded. Since $\mathbb{R}^n$ is finite-dimensional, we can find a constant c such that

$$\|Df(p)(x-p)\| \leq c \cdot \|x-p\|,$$

which means that our quotient is indeed bounded, completing the proof.

□

Theorem 1.3: Product Rule in $\mathbb{R}^n$

Let $U \subseteq \mathbb{R}^n$. Let $f, g : U \to \mathbb{R}$ at a point $p \in U$. Then

$$D(f \cdot g)(p) = Df(p) \cdot g(p) + f(p) \cdot Dg(p).$$

Exercise 1.4. *Verify that this version of the product rule agrees with the familiar product rule for $f, g : \mathbb{R} \to \mathbb{R}$.*

Proof. We define the multiplication function $m(x, y) := x \cdot y$. The partial derivatives are $\partial_1 m(x, y) = y$ and $\partial_2 m(x, y) = x$, i.e.

$$Dm(x, y) = \begin{pmatrix} y & x \end{pmatrix}$$

Applying the chain rule now gives

$$\begin{aligned}
D(f \cdot g)(p) &= D(m \circ (f, g))(p) \\
&= Dm((f, g)(p)) \circ D(f, g)(p) \\
&= Dm(f(p), g(p)) \circ (Df(p), Dg(p)) \\
&= Dm(f(p), g(p)) \cdot (Df(p), Dg(p)) \\
&= \begin{pmatrix} g(p) & f(p) \end{pmatrix} \cdot \begin{pmatrix} Df(p) \\ Dg(p) \end{pmatrix} \\
&= Df(p) \cdot g(p) + f(p) \cdot Dg(p).
\end{aligned}$$

□

Chapter 2

The Analysis II Twins

2.1 The Inverse Function Theorem

Theorem 2.1: Inverse Function Theorem

et $U \subseteq \mathbb{R}^n$ be open. Let $f : U \to \mathbb{R}^n$ be continuously differentiable. Let $x \in U$ be a point such that $Df(x) : \mathbb{R}^n \to \mathbb{R}^n$ is invertible. Then there exists an open neighbourhood V of x such that:

1. $f(V)$ is an open neighbourhood of $f(x)$.
2. $f : U \to f(H)$ is a C^1-diffeomorphism, i.e. invertible with continuously differentiable inverse.

Lemma 2.2. *Let $U \subset \mathbb{R}^n$ be open. Let $f : U \to \mathbb{R}^k$ be continuously differentiable. Let $p, q \in U$ be points whose connecting line is contained in U, i.e.*

$$[p, q] = \{p + \lambda(p - q) \mid \lambda \in [0, 1]\} \subseteq U.$$

Let

$$L = \max_{r \in [p,q]} \|Df(r)\|_{\text{sup}}.$$

Then

$$d_2(f(p), f(q)) \leq L \cdot d_2(p, q)$$

Proof. Let $h = q - p$. Let $g(\lambda) := f(p + \lambda h)$. Since g is a composition of continuously differentiable functions, it is continuously differentiable. By definition, we have $g(0) = f(p)$ and $g(1) = f(q)$. Applying the fundamental theorem of integration gives

$$\begin{aligned} f(q) - f(p) &= g(1) - g(0) \\ &= \int_0^1 g'(t)\, dt. \end{aligned}$$

Applying the chain rule gives

$$\int_0^1 g'(t)\, dt = \int_0^1 Df(p + th)(h)\, dt$$

$\square$

2.2 The Implicit Function Theorem

Chapter 3

Integration in $\mathbb{R}^n$

Chapter 4

Curves

Chapter 5

Surfaces

Part VI

Numerical Mathematics

Chapter 1

Numerical Linear Algebra

1.1 Operator Norms and Conditioning

Definition 1.1. Let $x \in V$ be an element of a normed vector space. Let $\widetilde{x} \in V$ be an approximation of x. Then the ***relative error*** of $\widetilde{x}$ is

$$\varepsilon(x, \widetilde{x}) := \frac{\|x - \widetilde{x}\|}{\|x\|}.$$

Definition 1.2. A function $\Phi(x) : V \to W$ is called ***poorly conditioned*** (at $x \in X$) if a small pertubation in the input can lead to a large pertubation in the output, i.e. there exists $\widetilde{x}$ such that

$$\varepsilon_\Phi := \varepsilon(\Phi(x), \Phi(\widetilde{x})) \gg \varepsilon(x, \widetilde{x}) =: \varepsilon_x.$$

$\Phi(x)$ is called ***well-conditioned*** if there exists a 'reasonable' constant c such that

$$\varepsilon_\Phi \leq c \cdot \varepsilon_x$$

for all $x, \widetilde{x}$.

Lemma 1.3. *Multiplication $\Phi(x, y) := x \cdot y$ of two numbers is well conditioned, with the approximation*

$$\varepsilon_\Phi \leq \varepsilon_x \varepsilon_y + \varepsilon_x + \varepsilon_y$$

Proof.

$$
\begin{aligned}
\varepsilon_\Phi &= \frac{\|\widetilde{x}\widetilde{y} - xy\|}{\|xy\|} \\
&= \frac{\|\widetilde{x}\widetilde{y} - x\widetilde{y} + x\widetilde{y} - xy\|}{\|xy\|} \\
&= \frac{\|(\widetilde{x} - x)\widetilde{y} + x(\widetilde{y} - y)\|}{\|xy\|} \\
&\leq \frac{\|(\widetilde{x} - x)\widetilde{y}\|}{\|xy\|} + \frac{\|x(\widetilde{y} - y)\|}{\|xy\|} \\
&= \frac{\|\widetilde{x} - x\|}{\|x\|}\frac{\|\widetilde{y}\|}{\|y\|} + \frac{\|\widetilde{y} - y\|}{\|y\|} \\
&= \frac{\|\widetilde{x} - x\|}{\|x\|}\frac{\|\widetilde{y} - y + y\|}{\|y\|} + \frac{\|\widetilde{y} - y\|}{\|y\|} \\
&\leq \frac{\|\widetilde{x} - x\|}{\|x\|}\frac{\|\widetilde{y} - y\|}{\|y\|} + \frac{\|\widetilde{x} - x\|}{\|x\|}\frac{\|y\|}{\|y\|} + \frac{\|\widetilde{y} - y\|}{\|y\|} \\
&= \frac{\|\widetilde{x} - x\|}{\|x\|}\frac{\|\widetilde{y} - y\|}{\|y\|} + \frac{\|\widetilde{x} - x\|}{\|x\|} + \frac{\|\widetilde{y} - y\|}{\|y\|} \\
&= \varepsilon_x \varepsilon_y + \varepsilon_x + \varepsilon_y
\end{aligned}
$$

□

Definition 1.4. An algorithm is called ***unstable*** if rounding errors significantly increase the relative error beyond what the conditioning of the problem would imply, i.e. if the function Φ is also being approximated by a function $\widetilde{\Phi}$ and

$$\varepsilon_{\widetilde{\Phi}} := \varepsilon(\widetilde{\Phi}(\widetilde{x}), \Phi(x)) \gg \varepsilon_\Phi$$

A function is stable if and only if every individual step of the calculation is well-conditioned.

Theorem 1.5. *Subtraction of approximately equal real numbers x and $y := (1 + \varepsilon_y)x$ is poorly conditioned.*

Proof. Let $\widetilde{x} := (1 + \varepsilon_x)x$. Then

$$\varepsilon_\Phi := \frac{\|(\widetilde{x} - y) - (x - y)\|}{\|x - y\|} = \frac{\|\varepsilon_x x\|}{\|-\varepsilon_y x\|} = \frac{\varepsilon_x}{\varepsilon_y}$$

If ε_y's order of magnitude is equal or smaller than that of ε_y, this relative error will be significantly larger than 1. □

Corollary 1.6. *The function*

$$\widetilde{\Phi}_1(x) := \left(\frac{1}{x}\right) - \left(\frac{1}{x+1}\right)$$

is unstable, since it is poorly conditioned even though the mathematically equivalent function

$$\widetilde{\Phi}_2(x) := \frac{1}{x(x+1)}$$

is well-conditioned.

Definition 1.7: Induced Operator Norm

Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be normed vector spaces. Then the operator norm on $\mathrm{Hom}(V, W)$ is defined by

$$\|F\|_{V,W} = \sup \{\|F(x)\|_W \mid x \in V : \|x\|_V = 1\}$$

The operator norm of F thus tells us the largest norm that any normalized vector ends up with after being transformed.

Equivalently, it tells us the radius of the smallest W-circle around the image of the normalized V-circle.

Theorem 1.8. $\|\cdot\|_{V,W}$ *is a norm.*

Exercise 1.9. *Show that* $\|\mathrm{id}_V\|_{V,V} = 1$ *for any norm* $\|\cdot\|_V$ *on a vector space* V*.*

Exercise 1.10. *Show that, given any eigenvalue* λ *of* $F \in \mathrm{Hom}(V, V)$ *and any norm* $\|\cdot\|_V$ *on* V*,*

$$\|F\|_{V,V} \geq |\lambda|$$

Theorem 1.11. *Let* V, W *be* $\mathbb{F}$*-vector spaces. Then*

$$\|F(x)\|_{V,W} = \inf \{c \in \mathbb{F} \mid \forall x \in V : \|F(x)\|_W \leq c \cdot \|x\|_V\}$$

> *Proof Sketch.* The largest factor by which $\|x\|_V$ can increase is equivalently the smallest upper bound on such factors. "□"

Note that this means $\|F(x)\|_W \leq \|F\|_{V,W} \cdot \|x\|_V$ for all $x \in V$.

Corollary 1.12. *Let* $F \in \mathrm{Hom}(V, W)$ *and* $G \in \mathrm{Hom}(W, U)$*. Then*

$$\|G \circ F\|_{V,U} \leq \|G\|_{W,U} \cdot \|F\|_{V,W}.$$

Lemma 1.13. *The infimum is a minimum, i.e. there exists an* x *attaining it.*

Lemma 1.14. *Let* $F \neq 0$*. Let* $x \in V$ *with* $\|x\|_V \leq 1$*. Then* $\|F(x)\|_W = \|F\|_{V,W}$ *implies* $\|x\| = 1$*.*

Proposition 1.15. *The* l^1*-norm*

$$\|x\|_1 = \sum_{i=1}^{n} |x_i|$$

on $\mathbb{R}^n$ *and* $\mathbb{R}^m$ *induces the* **column sum norm**

$$\|A\|_1 = \max_{j=1,\dots,n} \sum_{i=1}^{m} |a_{ij}|$$

on $\mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^m)$*.*

Proposition 1.16. *The l^∞-norm*

$$\|x\|_\infty = \max\{|x_i| \mid i = 1, \ldots, n\}$$

on $\mathbb{R}^n$ and $\mathbb{R}^m$ induces the **row sum norm**

$$\|A\|_\infty = \max_{j=1,\ldots,m} \sum_{j=1}^{n} \left|a_{ij}\right|$$

on $\mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^m)$.

Definition 1.17

The ***spectral radius*** $\rho(F)$ of an endomorphism $F \in \mathrm{Hom}(V, V)$ of a finite-dimensional vector space is the absolute value of the largest eigenvalue of F.

Proposition 1.18. *The l^2-norm*

$$\|x\|_2 = \sqrt{\sum_{i=1}^{n} |x_i|^2}$$

on $\mathbb{R}^n$ and $\mathbb{R}^m$ induces the **spectral norm**

$$\|A\|_2 = \sqrt{\rho(A^\top A)}$$

on $\mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^m)$.

Definition 1.19. The ***Frobenius norm*** on $\mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^m)$ is the natural norm given by $\mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^m) \cong \mathbb{R}^{mn}$:

$$\|A\|_F := \sqrt{\sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij}^2}$$

Exercise 1.20. *Show that the Frobenius norm is not an operator norm.*

Theorem 1.21. *Let $\|\cdot\|$ be an operator norm on* $\mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^n)$. *Let $A \in \mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^n)$ be invertible. Let $x, \widetilde{x}, b, \widetilde{b} \in \mathbb{R}^n \setminus \{0\}$ such that $Ax = b$ and $A\widetilde{x} = \widetilde{b}$. Then*

$$\varepsilon_x \leq \|A\| \left\|A^{-1}\right\| \varepsilon_b.$$

This means that the product $\|A\|\left\|A^{-1}\right\|$ tells us how much our accumulated relative error increases when we solve a system of linear equations. We want to give this product a special name:

Definition 1.22

Let $\|\cdot\|$ be an operator norm on $\mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^n)$. Let $A \in \mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^n)$ be invertible. Then we define the ***condition number*** of A to be

$$\mathrm{cond}(A, \|\cdot\|) = \|A\| \left\|A^{-1}\right\|$$

If $\|\cdot\|_p$ is a p-norm, we write $\mathrm{cond}(A, \|\cdot\|_p) =: \mathrm{cond}_p(A)$.

Exercise 1.23. *Show that* $\mathrm{cond}(A, \|\cdot\|) \geq 1$ *for all $A \in \mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^m)$ and any operator norm $\|\cdot\|$.*

Exercise 1.24. *Show that if A is symmetric with eigenvalues $\lambda_1, \ldots, \lambda_n$, then*

$$\operatorname{cond}_2(A) = \frac{\max |\lambda_i|}{\min |\lambda_j|}$$

Exercise 1.25. *Show that the linear map given by*

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 1+\varepsilon \end{pmatrix}$$

is poorly conditioned by approximating $\operatorname{cond}_2(A)$.

Definition 1.26: Generalized Condition Number

The condition number of an arbitrary function $\Phi : X \to Y$ between normed real vector spaces is defined as

$$\kappa_{\Phi(x)} = \inf \{\kappa \in \mathbb{R}_{\geq 0} \mid \varepsilon_\Phi \leq \kappa \cdot \varepsilon_x\}$$

Exercise 1.27. *Show that* $\operatorname{cond}(A) = \sup \{\kappa_{Ax} \mid x \in V\}$.

Exercise 1.28. *Show that if Φ is differentiable, we have $\kappa_\Phi(x) = \frac{\|D\Phi(x)\|\|x\|}{\|\Phi(x)\|}$*

Definition 1.29: Stability Indicator

The ***stability indicator*** of a numerical algorithm $\widetilde{\Phi}_\varepsilon$ with rounding accuracy ε is

$$\inf \left\{\sigma \geq 0 \,\middle|\, \exists \delta > 0 : \forall \varepsilon \leq \delta : \varepsilon_{\widetilde{\Phi}_\varepsilon} \leq \sigma \kappa_{\Phi(x)} \varepsilon \right\} \cdot \varepsilon$$

1.2 Triangular Decompositions

Solving a system of linear equations corresponding to a triangular matrix is trivial. Thus, if we can write a given matrix as a product of two triangular matrices, further computations become significantly easier.

Exercise 1.30. *Let U, V be upper triangular matrices. Then $U \cdot V$ is an upper triangular matrix.*

Exercise 1.31. *Let U be an upper triangular matrix. Then if U is invertible, U^{-1} is an upper triangular matrix whose diagonal entries are the inverses of the diagonal entries of U.*

Theorem 1.32. *Let $A \in \operatorname{Hom}(\mathbb{R}^n, \mathbb{R}^n)$ be an upper triangular matrix such that all upper left submatrices are invertible. Then there exists a unique decomposition*

$$A = LU$$

such that U is an invertible upper triangular matrix and L is an invertible lower triangular matrix with normalized diagonal entries.

Proof. For $n = 1$, we simply set $L = 1$ and $U = A$.
A decomposition $A_{n-1} = L_{n-1}U_{n-1}$ for the upper left block of size $n-1$ in A can be extended to a decomposition $A = LU$ by solving a system of linear equations, whose solvability is guaranteed by the invertibility of L_{n-1} and U_{n-1}:

$$\begin{aligned} & A = LU \\ \iff & \begin{pmatrix} A_{n-1} & a_U \\ a_L^\top & a_{nn} \end{pmatrix} = \begin{pmatrix} L_{n-1} & 0 \\ l_L & 1 \end{pmatrix} \cdot \begin{pmatrix} U_{n-1} & u_U \\ 0 & u_{nn} \end{pmatrix} \\ & = \begin{pmatrix} L_{n-1} \cdot U_{n-1} & L_{n-1}u_U \\ (U_{n-1}^\top l_L)^\top & l_L^\top u_U + u_{nn} \end{pmatrix} \end{aligned}$$

Since we've already solved the upper left block $A_{n-1} = L_{n-1}U_{n-1}$, we simply need to solve the equations

$$\begin{aligned} a_U &= L_{n-1}u_U \\ a_L &= U_{n-1}l_L \\ a_{nn} &= l_L^\top u_U + u_{nn} \end{aligned}$$

Since L_{n-1} and U_{n-1} are invertible, the first two equations are uniquely solvable (and obtaining these solutions is trivial, since both matrices are triangular). The solutions u_U and l_L then give a unique solution for u_{nn} as well. □

Exercise 1.33. *If A is positive definite, it has an LU-decomposition.*

Definition 1.34. We call a matrix $A \in \mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^n)$ ***diagonally dominant*** if the diagonal entries of A are at least as large as the sum over the other entries in a given row of A:

$$\forall i \in [1..n] : \sum_{j=1, j \neq i}^{n} |a_{ij}| \leq |a_{ii}|$$

If this equality is strict, we call A ***strictly diagonally dominant***.

Exercise 1.35. *Show that diagonally dominant matrices are invertible, and thus have an LU-decomposition.*

Note that $\mathrm{cond}(A) \leq \mathrm{cond}(L)\mathrm{cond}(U)$, meaning that LU-decomposition can lead to a decrease in numerical stability. However, in practical applications, this decrease is generally not noticable.

If A is symmetrical, then A is uniquely defined by the entries on the diagonal and below - thus, a natural followup question would be to ask if we can find a decomposition $A = LL^\top$ of A into the product of a lower diagonal matrix with its transpose.

Exercise 1.36. *Show that such a decomposition $A = LL^\top$ can only exist if A is positive definite.*

Exercise 1.37. *Let A be symmetric and positive definite. Then there exists a unique decomposition $A = LL^\top$, known as the* **Cholesky decomposition** *of A.*

1.3 Linear Regression

Many practical problems lead to systems of linear equations with more equations than variables - for example, we might try to find a line that approximates messy data linearly.

In such tasks, an exact solution is generally not achievable. Instead, we seek to find an x that minimizes the error $\|Ax - b\|$.

Theorem 1.38: Gauß Normal Equation

1. The minimizers of $\|Ax - b\|$ are exactly the solutions of

$$A^\top Ax = A^\top b.$$

2. The solution set is non-empty.

3. For any two solutions x, y, we have $Ax = Ay$.

The Gauß normal equation is incredibly useful for establishing theoretical results, but we have $\mathrm{cond}_2(A^\top A) = \mathrm{cond}_2(A)^2$, so solving a linear regression problem by numerically solving the normal equation is generally impractical.

Exercise 1.39. *$A^\top A$ is invertible if* $\ker A = \{0\}$.

Since orthogonal matrices preserve the euclidean norm, we try to construct an orthogonal matrix Q such that QA is a (possibly non-square) upper triangular matrix, in which case we have

$$\|QAx - Qb\|_2 = \|Ax - b\|_2.$$

Since orthogonal matrices preserve norms, we have $\mathrm{cond}_2(QA) = \mathrm{cond}_2(A)$. This will end up leading us to a more numerically stable solution to linear regression.

Theorem 1.40

Given a normalized vector $n \in \mathbb{R}^n$, reflection along the hyperplane $x^\top n = 0$ is given by

$$P_n = \mathrm{id}_{\mathbb{R}^n} - 2nn^\top.$$

Proof. We have $P_n(x) = x - 2nn^\top x = x - 2\langle n, x\rangle .x$, i.e. P_n orthogonally projects x onto our hyperplane, and then applies the same translation another time, mapping x onto its reflection. □

Lemma 1.41. *Let $x \in \mathbb{R}^n \setminus \langle e_i \rangle$. Define*

$$v = x + (\mathrm{sgn}(x_1)\|x\|_2).e_1,$$

declaring $\mathrm{sgn}(0) = 1$. *Let $w = v/\|v\|_2$. Then*

$$P_w(x) = (-\mathrm{sgn}(x_1)\|x\|_2).e_i.$$

Theorem 1.42: QR-decomposition

Let $A \in \mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^m)$ with $m \geq n$ and $\ker A = \{0\}$. Then there exist $Q \in \mathrm{O}(m)$ and an invertible upper triangular matrix $R \in \mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^m)$ such that $A = QR$.

Proof Sketch. Using our Lemma, we can bring A into upper triangular form R by repeated application of hyperplane reflections to its lower right submatrices, at which point Q is given by the product of these reflections. "□"

QR-decomposition is $\mathcal{O}(n^3)$, which is asymptotically the same as LU-decomposition. However, the actual runtime of QR-decomposition is generally about twice as long.

Theorem 1.43

Let $A \in \mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^m)$ with $m \geq n$ and $\ker A = \{0\}$. Then QR-decomposition of A gives a solution x of the linear regression problem $\|Ax - b\|_2$ via

$$R_U x = b_U,$$

where

$$\begin{pmatrix} b_U \\ b_L \end{pmatrix} = Q^\top b, \quad \begin{pmatrix} R_U \\ 0 \end{pmatrix} = R = Q^\top A$$

Theorem 1.44: Singular Value Decomposition

Let $A \in \mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^m)$. Then there exist orthonormal bases B_n, B_m of $\mathbb{R}^n$ and $\mathbb{R}^m$ such that the diagonal of ${}_{B_m}A_{B_n}$ contains the square roots of the non-zero eigenvalues of $A^\top A$, known as the ***singular values*** of A, ordered by size and followed by some amount of zeroes.

Definition 1.45. Let $A = U\Sigma V^\top$ be a singular value decomposition of A. Let Σ^+ be constructed from Σ by inverting all of the singular values. Then $A^+ := V\Sigma^+ U^\top$ is known as the ***pseudoinverse*** or ***Moore-Penrose-inverse*** of A.

Exercise 1.46. *Show that $A^+ = A^{-1}$ if A is square.*

Proposition 1.47. *A^+ is the unique matrix fulfilling all of the following equations:*

1. $AA^+A = A$,
2. $A^+AA^+ = A^+$,
3. $(AA^+)^\top = AA^+$,
4. $(A^+A)^\top = A^+A$.

Theorem 1.48. *A^+b is a minimal solution of the linear regression problem $Ax \approxeq b$.*

1.4 Eigenvalue Problems

1.4.1 Localization

Theorem 1.49. *Every eigenvalue of a matrix $A \in \mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^n)$ is contained in a set of the form*

$$K_i := \left\{ z \in \mathbb{C} \;\middle|\; |z - a_{ii}| \le \sum_{j \neq i} |a_{ij}| \right\}.$$

These sets are known as **Gerschgorin circles***.*

Notice that K_i is a circle of radius $\sum_{j\neq i} |a_{ij}|$ centered on a_{ii}. Importantly, this tells us that if the off-diagonal entires of A are small, the diagonal entries will always give us a good approximation of the eigenvalues.

Proof. Let $x \in \mathbb{C}^n \setminus \{0\}$ be an eigenvector of A with eigenvalue λ. Let x_i be the entry of x with the largest absolute value. Then we have

$$\lambda x_i = (Ax)_i = \sum_{j=1}^{n} a_{ij} x_j.$$

Dividing by $x_i \neq 0$ gives

$$\lambda = (Ax)_i = \sum_{j=1}^{n} a_{ij} \frac{x_j}{x_i},$$

subtraction of a_{ii} gives

$$\lambda - a_{ii} = (Ax)_i = \sum_{j\neq i}^{n} a_{ij} \frac{x_j}{x_i}.$$

Applying the triangle equality gives

$$|\lambda - a_{ii}| \le (Ax)_i = \sum_{j\neq i}^{n} \left| a_{ij} \frac{x_j}{x_i} \right|,$$

and since x_i is maximal, we have $|x_j|/|x_i| \le 1$, i.e.

$$|\lambda - a_{ii}| \le (Ax)_i = \sum_{j\neq i}^{n} |a_{ij}|.$$

□

Theorem 1.50. *Let $A \in \mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^n)$ be symmetric. Then the maximal and minimal eigenvalues of A are the maximum and minimum of the* **Rayleigh quotient**

$$\begin{aligned} R(x) : \mathbb{R}^n \setminus \{0\} &\to \mathbb{R} \\ x &\mapsto \frac{x^\top A x}{\|x\|_2^2} \end{aligned}$$

Proof Sketch. By the spectral theorem, we can choose an orthonormal basis of $\mathbb{R}^n$ consisting of eigenvectors of A. In this basis, we find that

$$x^\top A x = \sum_{i=1}^{n} \lambda_i x_i^2$$

We can bound this expression from above and below as

$$\lambda_{\min} \sum_{i=1}^{n} x_i^2 \leq x^\top A x \leq \lambda_{\max} \sum_{i=1}^{n} x_i^2$$

which gives us our desired result, since

$$\lambda_{\min} \sum_{i=1}^{n} x_i^2 = \|x\|_2^2.$$

"□"

Theorem 1.51. *Let $A \in \mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^n)$ be diagonalizable with $A = VDV^{-1}$. Let $\lambda_B \in \mathbb{C}$ be an eigenvalue of $A + B$ for any $B \in \mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^n)$. Then there exists an eigenvalue $\lambda_A \in \mathbb{C}$ of A such that*

$$|\lambda_B - \lambda_A| \leq \mathrm{cond}_2(V)\|B\|_2.$$

Corollary 1.52. *If A is normal, we get*

$$|\lambda_B - \lambda_A| \leq \|B\|_2.$$

This means that finding eigenvalues of normal matrices is a well-conditioned problem - adding a small amount of noise, represented by another matrix B, to a matrix A won't change the eigenvalues much. This doesn't hold for arbitrary matrices - for example, the ***Frobenius matrix***

$$A = \begin{pmatrix} 0 & & & & -a_0 \\ 1 & 0 & & & -a_1 \\ & \ddots & \ddots & & \vdots \\ & & 1 & 0 & -a_{n-2} \\ & & & 1 & -a_{n-1} \end{pmatrix}$$

has characteristic polynomial $p(t) = (-1)^n \sum_{i=0}^{n} a_i t^i$, with $a_n = 1$. In general, adding a small constant to an arbitrary polynomial can change the roots quite significantly, so finding the eigenvalues of A is generally not well-conditioned.

1.4.2 Power Iteration

Theorem 1.53: Von Mises iteration

Let $A \in \mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^n)$. Let $x_0 \in \mathbb{R}^n \setminus \{0\}$. Then the sequence

$$x_{n+1} = \frac{Ax_n}{\|Ax_n\|_2}$$

converges to a normalized eigenvector of A corresponding to the eigenvalue with the largest absolute value, as long as x_0 is not orthogonal to that eigenvector.
More specifically, let v_i be normed eigenvalues of A corresponding to eigenvalues λ_i, ordered by absolute value. Let $x_0 = \sum_{i=1}^{n} \alpha_i v_i$ be the representation of x_0 by our eigenvalues. Then there exists a constant c such that

$$|\|Ax_K\|_2 - |\lambda_1|| \leq 4c \cdot \|A\|_2 \left|\frac{\lambda_2}{\lambda_1}\right|^k$$

Proposition 1.54.

$$x_n = \frac{A^k x_0}{\left\|A^k x\right\|_2}$$

Proposition 1.55. *Under "suitable conditions", applying von Mises iteration to* $(A - \mu I_n)^{-1}$ *gives the eigenvalue closest to* μ*.*

Theorem 1.56: QR Iteration

Let A_0 be diagonalizable. Then the sequence

$$\begin{aligned} A_k &= Q_k R_k \\ A_{k+1} &= R_k Q_k, \end{aligned}$$

obtained by repeatedly finding the QR decomposition of A_k and then swapping the factors, converges to a diagonal matrix containing the eigenvalues of A_0, ordered by absolute value.

In praxis, this iteration generally converges even if A_0 is not diagonalizable. Each step of this iteration is $O(n^3)$.

This can be decreased to a one-time cost of $O(n^3)$ followed by $O(n^2)$ for every subsequent step by transforming A into *Hessenberg form* (an upper triangular matrix $\hat{A}$ with one extra diagonal below the main diagonal), whose QR decomposition is then cheaper to calculate.

1.4.3 The Jacobi Eigenvalue Algorithm

The Gerschogorin circle theorem tells us that we can find the eigenvalues of A by making A 'almost diagonal', i.e. by transforming A so that the off-diagonal entries become negligible. This is the idea behind the ***Jacobi eigenvalue algorithm*** (Not to be confused with the *Jacobi Method*, which is an algorithm for approximating the solution to a system of linear equations that we will meet later).

Definition 1.57. Let $A \in \mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^n)$. We define $N(A)$ to be the Euclidean norm of the off-diagonal entries of A, i.e.

$$N(A) = \sum_{i \neq j} a_{ij}^2$$

Proposition 1.58. *For every diagonal entry a_{ii}, there exists an eigenvalue λ such that*

$$|\lambda - a_{ii}| \leq \sqrt{N(A)}.$$

Lemma 1.59. *Let $A \in \mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^n)$ be symmetrical. Let $R_{ij}(\theta)$ be a rotation of θ degrees in the plane spanned by the basis vectors b_i and b_j. Then $B = R_{ij}(\theta)^\top A R_{ij}(\theta)$ fulfills*

$$\begin{aligned} N(B) &= N(A) - 2(a_{ij}^2 - b_{ij}^2), \\ b_{ij} = b_{ji} &= \cos(2\theta) a_{ij} + \frac{1}{2}\sin(2\theta)(a_{ii} - a_{jj}) \end{aligned}$$

Lemma 1.60. *Let $a_{ij} \neq 0$. Let*

$$\theta = \frac{1}{2}\arctan\left(\frac{2a_{ij}}{a_{jj} - a_{ii}}\right)$$

Then $B = R_{ij}(\theta)^\top A R_{ij}(\theta)$ fulfills $b_{ij} = b_{ji} = 0$.

> *Proof.* We saw in the last lemma that
>
> $$b_{ij} = \cos(2\theta) a_{ij} + \frac{1}{2}\sin(2\theta)(a_{ii} - a_{jj})$$
>
> Setting $b_{ij} = 0$ gives
>
> $$\begin{aligned} 0 &= \cos(2\theta) a_{ij} + \frac{1}{2}\sin(2\theta)(a_{ii} - a_{jj}) \\ \iff \tan(2\theta) &= \frac{\sin(2\theta)}{\cos(2\theta)} = \frac{2a_{ij}}{a_{jj} - a_{ii}} \end{aligned}$$
>
> □

Note that performing this trick multiple times may return entries we previously transformed to zero back to being non-zero. However, if we whack the off-diagonal entries in descending order of absolute value, we do see convergence:

Theorem 1.61. *If we choose a_{ij} to be the off-diagonal entry of A with the largest absolute value and define $R_{ij}(\theta)$ and B as in the previous lemma, we get*

$$N(B) \leq \left(1 - \frac{2}{n(n-1)}\right) N(A)$$

> *Proof.* Since a_{ij} is the element with the largest absolute value, we have
>
> $$N(A) \leq n(n-1) a_{ij}^2,$$

giving us

$$N(B) = N(A) - 2a_{ij}^2 \leq \left(1 - \frac{2}{n(n-1)}\right) N(A)$$

□

Repeated iterations of this procedure is known as the ***Jacobi eigenvalue algorithm***. The algorithm requires $O(n^2 \log(1/\varepsilon))$ iterations to guarantee $N(A_{k+1}) \leq \varepsilon$. Searching for the largest off-diagonal entries can be impractical, so instead one often simply iterates over all off-diagonal entries.

1.5 Contraction Methods

Theorem 1.62. *Let $A \in \mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^n)$. Then the spectral radius $\rho(A)$ is the greatest lower bound of all operator norms of A.*

Corollary 1.63. *If $\rho(A) < 1$, any affine map $\Phi : x \mapsto Ax + b$ is a contraction.*

We can decompose any matrix A into a sum

$$A = L + D + U$$

of a strictly lower triangular matrix, a diagonal matrix, and a strictly upper triagonal matrix. Since D, $D + U$, and $D + L$ are easy to invert (as long as they *are* invertible), this approach leads to an entire family of iterative algorithms for solving systems of linear equations.

The ***Jacobi method*** iterates via

$$\begin{aligned} Lx^k + Dx^{k+1} + Ux^k &= b \\ \Leftrightarrow x^{k+1} &= -D^{-1}(A - D)x^k + D^{-1}b \\ &:= M^J + c^J. \end{aligned}$$

The ***Gauß-Seidel method*** iterates via

$$\begin{aligned} Lx^{k+1} + Dx^{k+1} + Ux^k &= b \\ \Leftrightarrow x^{k+1} &= -(L + D)^{-1}Ux^k + (L + D)^{-1}b \\ &:= M^{GS}x^k + c^{GS}. \end{aligned}$$

We want to find sufficient conditions that A needs to fulfill in order for these iterations to be contractions, and thus to converge.

Exercise 1.64. *Show that both methods converge if A is strictly diagonally dominant.*

Theorem 1.65. *If A is symmetric and positive definite, the Gauß-Seidel method iterates over a contraction, and thus converges.*

Definition 1.66. We call a matrix $A \in \mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^n)$ ***reducible*** if there exists a decomposition of $[1, ..n]$ into disjoint index sets I, J such that $a_{ij} = 0$ for all $(i, j) \in I \times J$.

Theorem 1.67. *If A is reducible, we can decompose the system of linear equations $Ax = b$ into two systems $A_{II}x_I = b_I$ and $A_{JJ}x_J = b_J - A_{JI}x_I$.*

Theorem 1.68. *If A is reducible and diagonally dominant, it is invertible with non-zero diagonal entries.*

Theorem 1.69. *If A is irreducible and diagonally dominant, the Jacobi and Gauß-Seidel methods iterate over contractions, and thus converge.*

Chapter 2

Numerical Analyis

2.1 Polynomial Interpolation

Definition 2.1. The ***Lagrange interpolation*** problem consists of finding a polynomial $p \in \mathcal{P}_n$ such that $p(x_i) = y_i$ for some $a \leq x_0 < \ldots < x_n \leq b$ and $y_0, \ldots, y_n$.

A convenient solution to this problem would be finding a set of polynomials L_i such that $L_i(x_j) = \delta_{ij}$, since those would let us solve any interpolation problem using the right linear combination. A first idea is

$$L_i(x) := \prod_{i \neq j} (x - x_j)$$

which gives us $L_i(x_j) = 0$ if $i \neq j$ and $L_i(x_i) \neq 0$. Now, we only need to normalize these polynomials such that $L_i(x_i) = 1$. This can be done by setting

$$L_i(x) := \prod_{i \neq j} \frac{x - x_j}{x_i - x_j}$$

Theorem 2.2

Given some $x_0 < x_1 \ldots x_n$, the polynomials

$$L_i(x) := \prod_{i \neq j} (x - x_j)$$

fulfill $L_i(x_j) = \delta_{ij}$. They are known as ***Lagrange polynomials***.

Corollary 2.3. *The polynomial*

$$p(x) = \sum_{i=0}^{n} y_i \cdot L_i(x)$$

is the unique solution of the Lagrange interpolation problem.

Exercise 2.4. *Show that* $\sum_{i=0}^{n} L_i(x) = 1$ *for all* $x \in [a, b]$.

Theorem 2.5. *Let* $f \in C^{n+1}([a, b])$ *such that* $f(x_i) = y_i$. *Let* p *be the solution of the corresponding Lagrange polynomial interpolation problem. Then for every* $x \in [a, b]$, *there exists a* $\xi \in [a, b]$ *such that*

$$f(x) - p(x) = \frac{1}{(n+1)!} \cdot f^{(n+1)}(\xi) \cdot \prod_{j=0}^{n} (x - x_j).$$

Corollary 2.6.

$$\|f-p\|_{\sup} \leq \frac{\left\|f^{(n+1)}\right\|_{\sup}}{(n+1)!} \cdot (b-a)^{n+1}$$

Corollary 2.7. *Increasing the number of nodes x_i or decreasing $b-a$ leads to uniform convergence of the interpolation polynomials to f, as long as the derivatives of f don't grow rapidly.*

In practice, this leads to uniform convergence for most f. Given uniformly spaced nodes, one counterexample is given by ***Runge's function*** $f(x) = \frac{1}{1+25x^2}$.

However, evaluating the interpolation polynomial can be difficult and unstable in practice. We now focus our attention on deriving a scheme that allows a stable evaluation of $p(x)$ in $O(n^2)$ time.

Definition 2.8. Given nodes $x_0, \ldots, x_n$ and values $y_0, \ldots, y_n$ and $j \in [0..n]$, we define $p_{i,j}$ as the Lagrange interpolation polynomial fulfilling $p_{i,j}(x_k) = y_k$ for all $k \in [i..i+j]$

Note that $p_{i,0}(x) = y_i$ and $p_{0,n}(x) = p(x)$.

Theorem 2.9: Neville Scheme

The polynomials $p_{i,j}$ are given by $p_{i,0}(x) = y_i$ and

$$p_{i,j}(x) = \frac{(x-x_i) \cdot p_{i+1,j-1}(x) - (x-x_{i+j})p_{i,j-1}(x)}{x_{i+j}-x_i}$$

Recursive implementations of the Neville scheme are highly inefficient. A simple successive "forwards" evaluation scheme leads to a runtime in $O(n^2)$.

Definition 2.10. Given $x_0, \ldots, x_{n-1}$, the ***Newton basis*** $q_0, \ldots, q_n$ of $\mathcal{P}_n$ is given by

$$q_j(x) = \prod_{k=0}^{j-1}(x-x_k)$$

Theorem 2.11: Divided Differences

Given $p \in \mathcal{P}_n$ and $x_0, \ldots, x_n \in [a,b]$, we can find the representation $p(x) = \sum_{j=0}^{n} \lambda_j q_j(x)$ of p in the corresponding newton basis by setting $\lambda_{i,0} = y_i$ and iterating

$$\lambda_{i,j} = \frac{\lambda_{i+1,j-1} - \lambda_{i,j-1}}{x_{i+j}-x_i}$$

p can then be evaluated easily using the Horner Scheme

$$p(x) = \lambda_0 + (x-x_0) \cdot (\lambda_1 + (x-x_1) \cdot (\lambda_2 + \ldots (\lambda_{n-1} + (x - x_{n-1}\lambda_n))))$$

This approach has the advantage of allowing us to easily add more nodes later.

Now, we shift our attention to the problem of lowering interpolation error. One way of doing this is by cleverly choosing our nodes x_i.

Definition 2.12. The ***Chebyshev-Polynomials*** on $t \in [-1, 1]$ are given by

$$T_n(t) = \cos(n \cdot \arccos(t)).$$

The roots of a Chebyshev polynomial are known as ***Chebyshev nodes***.

Lemma 2.13. *We have* $|T_n(t)| \leq 1$ *for all* $t \in [-1, 1]$.

Lemma 2.14. *The Chebyshev polynomials are given by the recurrence relation*

$$\begin{aligned} T_0(t) &= 1 \\ T_1(t) &= t \\ T_{n+1}(t) &= 2tT_n(t) - T_{n-1}(t) \end{aligned}$$

Corollary 2.15. *We have*

$$T_{n+1}(x) = 2^n \prod_{j=0}^{n} (x - t_j)$$

Lemma 2.16. *The roots of* T_n *are given by*

$$t_j = \cos\left(\frac{(j + \frac{1}{2})\pi}{n}\right)$$

for $j \in [0..n-1]$.

Lemma 2.17. *The extreme values of* T_n *are given by*

$$s_j = \cos\left(\frac{j\pi}{n}\right)$$

for $j \in [0..n]$.

Theorem 2.18. *Among all possible choices of nodes for polynomial interpolation, Chebyshev nodes minimize the supremum norm of the interpolation polynomial. If Chebyshev nodes are used, the interpolation error becomes*

$$\|f - p\|_{\sup} \leq 2^{-n} \frac{\left\|f^{(n+1)}\right\|_{\sup}}{(n+1)!}$$

Definition 2.19. The ***Hermite interpolation problem*** consists of finding a polynomial $p \in \mathcal{P}_N$ such that $p^{(k_i)}(x_i) = y_i^{(k_i)}$, where $N + 1 = \sum_{i=0}^{n}(k_i + 1)$.

Proposition 2.20. *Hermite interpolation has a unique solution. Given only a single node, the solution is the corresponding taylor polynomial.*

2.2 Spline Interpolation

Definition 2.21

A function $s : [a, b] \to \mathbb{R}$ is known as a ***Spline of degree m and order k*** if $s \in C^k([a, b])$ is a piecewise combination of polynomials of degree m over some partition I_n of $[a, b]$ into n nodes. The space of m, k-splines on P is written $\mathcal{S}^{m,k}(I_n)$. If no order is given, it is generally assumed that $k = m - 1$.

Proposition 2.22. *For $k \geq m$, we have $\mathcal{P}_m|_{[a,b]} = \mathcal{S}^{m,k}(I_n)$.*

Theorem 2.23. *Linear spline interpolation is solved by $s = \sum_{i=0}^{n} y_i \varphi_i$, where φ_i are the* **hat functions***, given by linearly interpolating $\varphi_i(x_j) = \delta_j$*

Theorem 2.24.

$$\dim \mathcal{S}^{m,m-1}(I_n) = n + m$$

Definition 2.25. Given a cubic spline interpolation problems, we generally want to fulfill one of the following boundary conditions:

1. ***Natural boundary conditions***, given by
$$s''(a) = s''(b) = 0$$
2. ***Complete boundary conditions*** or ***Hermite boundary conditions***, given by
$$s'(a) = \alpha,$$
$$s'(b) = \beta$$
for some $\alpha, \beta \in \mathbb{R}$.
3. ***Periodic boundary conditions*** $s'(a) = s'(b)$, $s''(a) = s''(b)$

If a thin, elastic board has its position fixed at certain points, the shape that the board takes is exactly the solution of the corresponding cubic spline interpolation problem.

Theorem 2.26. *Given a spline s and any other function g interpolating the same points and fulfilling the same boundary condtions, the curvature of s is a lower bound to the curvature of g.*

Theorem 2.27. *Let s_1 be the linear spline interpolating $f \in C^2([a,b])$. Then*

$$\|f - s_1\|_{L^2} \leq \frac{h^2}{2} \|f''\|_{L^2}$$

with $h = \max\{x_i - x_{i-1} \mid i \in [1..n]\}$.

Theorem 2.28. *Let s_3 be a cubic spline interpolating $f \in C^2([a,b])$. Then*

$$\|f - s_3\|_{L^2} \leq \frac{h^4}{4} \left\|f^{(4)}\right\|_{L^2}$$

with $h = \max\{x_i - x_{i-1} \mid i \in [1..n]\}$.

2.3 Discrete Fourier Transform

Definition 2.29: Real Trigonometric Interpolation

Given an even number of evenly spaces nodes $x_0, \ldots, x_{n-1}$ with $n = 2m$ even, the ***real trigonometric ineterpolation problem*** consists of finding $a_k, b_k \in \mathbb{R}$, $k \in [1..m-1]$ and $a_0, a_m \in \mathbb{R}$ such that

$$T(x) = \frac{a_0}{2} + \sum_{k=1}^{m-1} (a_k \cos(kx) + b_k \sin(kx)) + \frac{a_m}{2} \cos(mx)$$

fulfills $T\left(\frac{2\pi j}{n}\right) = y_j$ for all j.

Trigonometric interpolation can be written much more compactly in the complex case:

Definition 2.30: Real Trigonometric Interpolation

Given an even number of evenly spaces nodes $x_0, \ldots, x_{n-1}$ with $n = 2m$ even, the ***real trigonometric ineterpolation problem*** consists of finding $c_k \in \mathbb{C}$, $k \in [1..n-1]$ such that

$$T(x) = \sum_{k=0}^{n-1} c_k e^{ikx}$$

fulfills $T\left(\frac{2\pi j}{n}\right) = y_j$ for all j.

Theorem 2.31. *A set of coefficients c_k solves the complex trigonometric interpolation problem iff. the coefficients a_k, b_k defined by*

$$\begin{aligned} a_0 &= 2c_0 \\ a_k &= c_k + c_{2m-k} \\ b_k &= i(c_k - c_{2m-k}) \\ a_m &= 2c_m \end{aligned}$$

solve the corresponding real trigonometric interpolation problem.

If we write $y = (y_0, \ldots, y_{n-1})^\top$ and $\vec{\omega}_n^k = (\omega_n^{jk})_{j \in [1..n]}$, the complex trigonometric interpolation problem is given by the system of linear equations

$$y = \sum_{k=0}^{n-1} \beta_k \vec{\omega}_n^k$$

Theorem 2.32

The family $(\vec{\omega}_n^k)_{k \in [1..n]}$ is an orthogonal basis of $\mathbb{C}^n$ fulfilling

$$\left\langle \vec{\omega}_n^k, \vec{\omega}_n^l \right\rangle = n\delta_{kl}$$

known as the ***Fourier basis***.

Complex trigonometric interpolation thus comes down to performing a change of basis from the standard basis into the Fourier basis. We know from linear algebra that the matrix $T_n = (\vec{\omega}_n^0, \ldots, \vec{\omega}_n^n)$ is the basis change matrix from the Fourier basis into the standard basis. Thus, its inverse must be the basis change matrix from the standard basis into the Fourier basis.

Exercise 2.33. *Show* $T_n^{-1} = \frac{1}{n} T_n^*$

Thus, we get:

Theorem 2.34: Discrete Fourier Transform

1. Change of basis from the standard basis of $\mathbb{C}^n$ into the Fourier basis, known as the ***discrete Fourier transform***, is given by

$$\beta = \frac{1}{n} T_n^* y$$

2. Change of basis from the Fourier basis of $\mathbb{C}^n$ into the standard basis, known as the ***inverse discrete Fourier transform*** or as ***Fourier synthesis***, is given by

$$y = T_n \beta$$

2.4 Numerical Integration

The goal of numerical integration, also known as ***quadrature***[3] is numerical approximation of the integral operator

$$I(f) = \int_a^b f(x)\, dx.$$

[3] Based on the ancient Greek practice of finding the area of a shape by constructing a square of equal area.

Definition 2.35. A ***quadrature formula*** on an interval $[a, b]$ is a linear map $Q : C^0([a, b]) \to \mathbb{R}$ of the form

$$Q(f) = \sum_{i=0}^{n} w_i . f(x_i)$$

Definition 2.36. The number

$$\|Q\| = \frac{1}{b-a} \sum_{i=0}^{n} |w_i|$$

is known as the ***stability indicator*** of Q.

The Riemann integral is defined as the limit of the quadrature formula

$$Q(f) = \sum_{k=0}^{n-1} (x_{i+1} - x_i) . f(x_i)$$

as the distance between the nodes approaches zero.

Definition 2.37. A quadrature formula Q is called ***exact of degree d*** if $Q(p) = I(p)$ for all $p \in \mathcal{P}_d$.

Exercise 2.38. *A quadrature formula is exact of degree* 0 *if and only if* $\sum_{i=0}^{n} w_i = b - a$.

Lemma 2.39. *Given $n = 2m$ equidistant nodes, a quadrature formula on these n nodes that is exact of degree $2d$ is also exact of degree $2d + 1$*

Theorem 2.40

The ***Newton-Cotes-Formula***

$$Q(f) = \sum_{i=0}^{n} \left(\int_a^b L_i(x)\, dx \right) . f(x_i)$$

is exact of degree n. The case $n = 0$ is known as the ***middle point rule***, the case $n = 1$ is known as the ***trapeze rule***, and the case $n = 2$ is known as the ***Simpson rule***.

Theorem 2.41. *The trapeze rule is given in closed form by*

$$T(f) = \frac{b-a}{2}(f(a) + f(b))$$

Theorem 2.42. *The Simpson rule is given in closed form by*

$$T(f) = \frac{b-a}{6}\left(f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right).$$

since we chose equidistant nodes x_0, x_1, x_2, it is exact of degree 3.

We can increase the accuracy of a quadrature formula by partitioning our interval $[a, b]$ into smaller intervals, and applying a quadrature formula to each one separately. We write this as

$$Q^n(f) = \sum_{k=1}^{n} Q_k(f|_{(x_{j-1}, x_k)})$$

A summed quadrature formula is called exact of degree d if

$$|Q^n(f) - I(f)| = O\left(\left(\frac{b-a}{n} \right)^d \right)$$

Proposition 2.43. *The summed trapeze rule is exact of degree 2. The summed Simpson rule is exact of degree 4.*

Lemma 2.44. *A quadrature formula with $n + 1$ weights has exactness degree at most $2n + 1$.*

Proof. Let $Q(f) = \sum_{i=0}^{n} w_i \cdot f(x_i)$ be our quadrature formula. Let $p(x) = \prod_{i=0}^{n}(x - x_i)^2$. Then we have $Q(p) = 0$, but $I(p) > 0$. Since $p \in \mathcal{P}_{2n+2}$, Q cannot have exactness $2n + 2$. □

Lemma 2.45. *A quadrature formula with $n + 1$ weights is exact of degree n if and only if*

$$w_i = \int_a^b L_i(x)\, dx$$

Proof. 1. If Q is exact of degree n, it must integrate the lagrange polynomials exactly. We get:

$$\begin{aligned}\int_a^b L_i(x)\,dx &= \sum_{j=0}^{n} w_j L_i(x_j)\\ &= \sum_{j=0}^{n} w_j \delta_{ij}\\ &= w_i\end{aligned}$$

2. Any quadrature formula Q that integrates the Lagrange polynomials exactly must also integrate any linear combination of Lagrange polynomials exactly. Since the Lagrange polynomials at $n+1$ nodes form a basis of $\mathcal{P}_n$, it follows that Q integrates any polynomial of degree n exactly.

□

Lemma 2.46. *If a quadrature formula is exact of even degree $n = 2m$, we have $w_i > 0$ for all i.*

Proof Sketch.

$$w_i = \int_a^b L_i(x)^2 > 0$$

"□"

We will now devote our attention to deriving the ***Gauß quadrature***, which chooses $n + 1$ quadrature points such that the maximum possible exactness degree is achieved.

Proposition 2.47. *Let $\omega \in C^0([a,b])$ with $\omega(x) > 0$. Then $\langle f, g\rangle_\omega = \int_a^b f(x)g(x)\omega(x)\,dx$ is a scalar product on $C^0([a,b])$.*

Using Gram-Schmidt starting with the standard monomial basis, we can construct an ω-orthonormal base $(\pi_i)_{i\in[0..n]}$ of $\mathcal{P}_n$ such that $\pi_i \in \mathcal{P}_i$ for all i.

Proposition 2.48. *Using $\omega(x) = (1-x^2)^{-\frac{1}{2}}$ on $(-1,1)$ gives the Chebyshev polynomials.*

Proposition 2.49. *Using $\omega(x) = 1$ on $[-1,1]$ gives the Legendre polynomials.*

Lemma 2.50. *Every orthonormal polynomial π_i has i roots of order 1 lying in the given interval (a,b).*

Theorem 2.51

Let x_i be the roots of π_{n+1} for some $\omega \in C^0([a,b])$. Then the quadrature formula with roots x_i and weights

$$w_i = \int_a^b L_i(x)\omega(x)\,dx$$

is exact of degree $2n+1$.

If the error of a summed quadrature formula $T(h)$ is of order h^γ for a $\gamma \in \mathbb{N}$, we have

$$T(h) = T(0) + \varphi(h)$$

with $\varphi(0) = 0$ and $|\varphi(h)| \leq ch^\gamma$ for some $c \in \mathbb{R}$. Taylor expansion of φ at 0 gives

$$\varphi(h) = c_1 h^\gamma + c_2 h^{\gamma+1} + o(h^{\gamma+1})$$

for some $c_1, c_2 \in \mathbb{R}$ and, i.e.

$$T(h) = T(0) + c_1 h^\gamma + c_2 h^{\gamma+1} + o(h^{\gamma+1})$$

If we evaluate at $h/2$, we get

$$T(h/2) = T(0) + \frac{c_1}{2^\gamma} h^\gamma + \frac{c_2}{2^{\gamma+1}} h^{\gamma+1} + o\left(\frac{h^{\gamma+1}}{2^{\gamma+1}}\right)$$

This lets us eliminate the $c_1 h^\gamma$-term by setting

$$T^*(h) := \frac{T(h) - 2^\gamma T(h/2)}{1 - 2^\gamma} = T(0) + c_2 \frac{1 - 2^{-1}}{1 - 2^\gamma} \cdot h^{\gamma+1} + o(h^{\gamma+1}),$$

which is a new summed quadrature formula with an error of $h^{\gamma+1}$, instead of h^γ, which can be a significant improvement for $\gamma \ll 1$. Given a non-polynomial function $f \in C^k([a,b])$ whose exact integral $I(f)$ is known, we can investigate the errors

$$e_h = \left|I(f) - Q^N(f)\right|$$

for various $h > 0$. Since $e_h \approx c_1 h^\gamma$, we can evauate at two different step sizes $h, H > 0$ and get

$$c_1 \approx \frac{e_h}{h^\gamma} \approx \frac{e_H}{H^\gamma}$$

and thus

$$\gamma \approx \frac{\log(e_h/e_H)}{\log(h/H)} = \frac{\log(e_h) - \log e_H}{\log(h) - \log(H)}$$

We can now try out various pairs h, H, and know that we have obtained the correct $\gamma \in h$ if these different pairs all give nearly identical answers.

2.5 Nonlinear Optimization

Let $U \subseteq \mathbb{R}^n$. Then typical problems include:

1. Find $x^* \in U$ such that $f(x^*) = 0$ for some $f : U \to \mathbb{R}^m$.

2. Find $x^* \in U$ such that $g(x^*) = \min_{x \in U} g(x)$ for some $g : U \to \mathbb{R}$.

In general, neither of these problems have closed form solutions. Instead, approximate solutions are found by iterative algorithms, i.e. we find $(x_k)_{k\in\mathbb{N}}$ with $x_k \to x^*$.

Definition 2.52. A numerical algorithm which returns a sequence $(x_k)_{k\in\mathbb{N}}$ approximating a solution x^* given some initial vector $x_0 \in U$ is known as:

1. ***Globally convergent***, if $x_k \to x^*$ for all $x_0 \in U$.

2. ***Locally convergent***, if there exists some $\varepsilon > 0$ such that $x_k \to x^*$ for all $x_0 \in U$ with $d(x_0, x^*) < \varepsilon$.

Definition 2.53. If $x_k \neq x^*$ for all k, we can define the ***rate of convergence*** of x_k by saying that a locally convergent algorithm has ***convergence order*** $\boldsymbol{\alpha \geq 1}$ if there exists a $q \in \mathbb{R}$ such that the errors $\delta_k = d(x^* - x_k)$ fulfill

$$\limsup_{k\to\infty} \frac{\delta_{k+1}}{\delta_k^\alpha} = q.$$

If $\alpha = 1$, we additionally demand $q \leq 1$.

Proposition 2.54. *If $\Phi : \mathbb{R}^n \to \mathbb{R}^n$ is a contraction, the basic fixed point iteration $x_{k+1} = \Phi(x_k)$ is globally linearly convergent.*

Proposition 2.55. *If $\Phi \in C^1(\mathbb{R}, \mathbb{R})$ has a fixed point x^*, then the fixed point iteration $x_{k+1} = \Phi(x_k)$ is locally convergent if $|\Phi'(x^*)| < 1$ and divergent if $|\Phi'(x^*)| > 1$.*

2.6 Computing Roots Numerically

Theorem 2.56. Bisection Algorithm: *Let $f \in C^0([a,b], \mathbb{R})$ with $f(a)f(b) \leq 0$. Set $\varepsilon > 0$. Consider the following algorithm:*

1. *Set $(a_0, b_0) = (a, b)$ and $k = 0$.*

2. *Set $c_k = \frac{a_k + b_k}{2}$ and*

$$(a_{k+1}, b_{k+1}) = \begin{cases} (a_k, c_k) & \text{if} f(a_k)f(c_k) \leq 0 \\ (c_k, b_k) & \text{otherwise} \end{cases}$$

3. *Return b_{k+1} if $d(b_{k+1}, a_{k+1}) \leq \varepsilon$. Otherwise set $k \leftarrow k+1$ and return to step 2.*

Then iteration stops after $J \leq 1 + \log_2\left(\frac{b-1}{\varepsilon}\right)$ steps, the interval $[a_J, b_J]$ contains a root, and if $\varepsilon = 0$, then the sequence $(c_k)_{k\in\mathbb{N}}$ converges to a root linearly with $q = \frac{1}{2}$.

Proposition 2.57. *The bisection algorithm converges linearly.*

Theorem 2.58. Secant Method: *Let $f \in C^0([a,b] \to \mathbb{R})$. Then the following algorithm converges to a root of f, if one exists:*

1. *Let $\varepsilon > 0$, $x_0 = a$, $x_1 = b$, and $k = 1$.*

2. *If $f(x_k) \neq f(x_{k-1})$, set*

$$x_{k+1} = x_k - \frac{x_k - x_{k-1}}{f(x_k) - f(x_{k-1})} f(x_k)$$

3. *Return x_{k+1} if $d(x_{k+1}, x_k) \leq \varepsilon$. Otherwise set $k \leftarrow k+1$ and return to step 2.*

Proposition 2.59. *The secant method converges superlinearly, but subquadratically, with convergence order $\varphi = \frac{1+\sqrt{5}}{2}$ (the golden ratio!)*

If $f \in C^1(U \subseteq \mathbb{R}^n, \mathbb{R}^n)$, Taylor approximation gives

$$0 = f(x^*) = f(x) + Df(x)(x^* - x) + \varphi(d(x^*, x)) \approx f(x) + Df(x)(x^* - x),$$

from which we can set $x := x_k$ to get a new iteration

$$x_{k+1} = x_k - Df(x_k)^{-1} f(x_k) \approx x^*$$

as long as $Df(x_k)$ is regular. This is the (multidimensional analogue of) the Newton method, and is generally much more efficient than the secant method.

Theorem 2.60. *Let $f \in C^2(U, \mathbb{R}^n)$. Let $x^* \in U$ be a root of f such that $Df(x^*)$ is invertible. Then there exists an $\varepsilon > 0$ such that the Newton method converges for any $x_0 \in U$ with $d(x^*, x_0) < \varepsilon$. Additionally, there exists a constant $c \geq 0$ such that the iterates $(x_k)_{k \in \mathbb{N}}$ fulfill*

$$\|x^* - x_{k+1}\| \leq c \|x^* - x_k\|^2,$$

i.e. the Newton method converges quadratically.

If $d(x^*, x_0) \geq \varepsilon$, the Newton method might diverge. In this case, introducing a damping factor $\omega \in (0, 1)$ and iterating

$$x_{k+1} = x_k - \omega Df(x_k)^{-1}$$

can lead to convergence, but makes convergence asymptotically slower (no longer quadratic in general).

Part VII

Complex Analysis

Part VIII

Point-Set Topology

Chapter 1

Topological Spaces

Chapter 2

Filters and Convergence

Theorem 2.1. *Let X be a topological space. Let $U \subseteq X$. Then U is open iff. U is contained in every ultrafilter converging to some point in U.*

Proof. $\Rightarrow$: By definition, if U is an open neighborhood of x, any filter convering to x must contain U.

$\Leftarrow$: Let $U \subseteq X$ and assume $U \in \mathcal{U}$ for every ultrafilter converging to a point $u \in U$. Assume U is not open. Then there exists a point $u \in U$ such that every open neighborhood of u contains at least one point in $X \setminus U$. But then any two members of the family

$$\{X \setminus U\} \cup \{U(u)\}$$

have nonempty intersection. Therefore, this family can be extended to an Ultrafilter $\mathcal{U}_u$. Since $\mathcal{U}_u$ contains all open neighborhoods of u by construction, it converges to u, and must thus contain U. Since it also contains $X \setminus U$, we arrive at a contradiction.

□

Definition 2.2. Let $\mathcal{F}$ be a filter on X. Let $f : X \to Y$. Then the ***pushforward Filter of f on $\mathcal{F}$*** is the set

$$f_*(\mathcal{F}) := \left\{A \subseteq Y \,\middle|\, p_i^{-1}[A] \in \mathcal{F}\right\},$$

Exercise 2.3. *Verify that the pushforward of a filter is a filter, and that the pushforward of an ultrafilter is an ultrafilter.*

Theorem 2.4: Characterization of continuity in terms of Ultrafilters

Let $f : X \to Y$. Then f is continuous if and only if $f_*(\mathcal{U})$ converges to $f(x)$ for every ultrafilter $\mathcal{U}$ that converges to $x \in X$.

Proof. 1. Assume f is continuous. Let $\mathcal{U}$ be an ultrafilter on X converging to $x \in X$.

Let $U \subseteq Y$ be an open neighbourhood of $f(x)$. Then, since f is continuous, $f^{-1}(U)$ is an open neighbourhood of x. We thus have $f^{-1}(U) \in \mathcal{U}$, which gives us $U \in f_*(\mathcal{U})$ by the definition of the pushforward filter.

Thus, every open neighbourhood of $f(x)$ is contained in $f_*(\mathcal{U})$, so $f_*(\mathcal{U})$ converges to $f(x)$.

2. Assume that the pushforward of any ultrafilter converging to x converges to $f(x)$. Let $U \subseteq Y$ be open.

   Let $\mathcal{U}$ be an ultrafilter on X converging to some point $x \in f^{-1}(U)$. Then, by our hypothesis, $f_*(\mathcal{U})$ converges to $f(x)$. Since U is an open neighborhood of $f(x)$, we have $U \in f_*(\mathcal{U})$, and thus $f^{-1}(U) \in \mathcal{U}$. Thus, by theorem 2.1, $f^{-1}(U)$ is open, so f is continuous.

□

Theorem 2.5. *Let $(X_i, \mathcal{O}_i)_{i\in I}$ be a family of topological spaces. Let $\mathcal{F}$ be a filter on $\prod_{i\in I} X_i$. Then $\mathcal{F}$ converges to a point $x \in \prod_{i\in I} X_i$ if and only if each pushforward filter $(p_i)_*(\mathcal{F})$ converges to $p_i(x) = x_i$.*

Proof. $\Rightarrow$: Assume that $\mathcal{F}$ converges to $x \in \prod_{i\in I} X_i$. Since all projections p_i are continuous, each $(p_i)_*(\mathcal{F})$ converges to $p_i(x) = x_i$ by theorem 2.4.

$\Leftarrow$: Assume that each $(p_i)_*(\mathcal{F})$ converges to $p_i(x) = x_i$. By definition of the product topology, the sets $S_i := p_i^{-1}(U_i)$ for open $U_i \in X_i$ form a subbase of $\prod_{i\in I} X_i$. Therefore, any open neighborhood $U \in \mathcal{U}(x)$ contains an open neighborhood $B = \bigcap_{i\in J\subseteq I} p_i^{-1}(U_i)$ for some open neighborhoods $U_i \in \mathcal{U}(x_i)$.

Since each $(p_i)_*(\mathcal{F})$ converges to x_i, we have $U_i \in (p_i)_*(\mathcal{F})$ for each i. Therefore, by definition of the pushforward filter, we have $p_i^{-1}(U_i) \in \mathcal{F}$ for all i. Since filters are closed under intersection, we get $B \in \mathcal{F}$, and since $B \subseteq U$ we also get $U \in \mathcal{F}$ as desired.

□

Chapter 3

Separation Axioms

A very common axiom is the Hausdorff separation axiom, which says that any two distinct points can be housed orf from each other by disjoint open sets.

"Topology via Logic", Steven Vickers

Definition 3.1. Let $(X, \mathcal{O})$ be a topological space. Let $x, y \in X$. Then we call x and y ***topologically distinguishable*** iff. $\mathcal{U}(x) \neq \mathcal{U}(y)$, i.e. one of the points has a neighborhood not containing the other.

Definition 3.2: $T0$ **Space**

Let $(X, \mathcal{O})$ be a topological space. Then we call $(X, \mathcal{O})$ ***T0*** if every pair of points is topologically distinguishable.

Definition 3.3: $T1$ **Space**

Let $(X, \mathcal{O})$ be a topological space. Then we call $(X, \mathcal{O})$ ***T1*** if for every pair of points, both of them have neighbourhoods not containing the other:

$$\begin{aligned}&\forall x, y \in X :\\&\exists U_x \in \mathcal{U}(x), U_y \in \mathcal{U}(y) :\\&\quad x \notin U_y, y \notin U_x.\end{aligned}$$

Exercise 3.4. *Show that T1 implies T0.*

Exercise 3.5. *Find a space that is T0, but not T1.*

Exercise 3.6. *Show that $(X, \mathcal{O})$ is T1 if and only if every singleton set is closed.*

Exercise 3.7. *Show that $(X, \mathcal{O})$ is T1 if and only if every subset of X is the intersection of all of its neighborhoods:*

$$\forall Y \subseteq X : Y = \bigcap \{U \mid U \in \mathcal{U}(Y)\}$$

Definition 3.8: Hausdorff Space

Let $(X, \mathcal{O})$ be a topological space. Then we call $(X, \mathcal{O})$ ***T2*** or ***Hausdorff*** if every pair of points can be housed orf from each other, i.e. they have disjoint neighbourhoods:

$$\begin{aligned} \forall x \neq y \in X : \\ \exists U_x \in \mathcal{U}(x), U_y \in \mathcal{U}(y) : \\ U_x \cap U_Y = \emptyset \end{aligned}$$

Exercise 3.9. *Show that every Hausdorff space is T1.*

Exercise 3.10. *Let $(X, \mathcal{O})$ be a topological space. Then $(X, \mathcal{O})$ is Hausdorff if and only if every convergent filter has a unique limit.*

Theorem 3.11: Characterization of Hausdorff spaces by diagonals

Let $(X, \mathcal{O})$ be a topological space. Then $(X, \mathcal{O})$ is Hausdorff iff. the diagonal

$$\Delta := \{(x, x) \mid x \in X\}$$

is closed in the product topology.

Proof. $\Rightarrow$ Assume $(X, \mathcal{O})$ is Hausdorff. Let $(x, y) \notin \Delta$. Then $x \neq y$ by definition of Δ. Therefore, there exist disjoint neighbourhoods U_x, U_y of x and y. Since these neighborhoods are disjoint, we have

$$\Delta \cap (U_x \times U_y) = \emptyset$$

. Therefore, every point $(x, y) \notin \Delta$ has an open neighborhood $(U_x \times U_y)$ that doesn't intersect Δ, i.e. Δ is closed.

$\Leftarrow$ Let $x \neq y$. Assume that Δ is closed. Then $(x, y) \notin \Delta$ has a basic open neighborhood $U_x \times U_y$ that doesn't intersect Δ. Since $U_x \times U_y$ doesn't intersect Δ, U_x and U_y must be disjoint.

□

Theorem 3.12. *Let $(X, \mathcal{O})$ be a topological space. Then $(X, \mathcal{O})$ is Hausdorff iff. every point is the intersection of the closures of all of its neighbourhoods:*

$$\forall x \in X : \{x\} = \bigcap \left\{ \overline{U} \;\middle|\; U \in \mathcal{U}(x) \right\}$$

Proof. $\Rightarrow$ Let $(X, \mathcal{O})$ be Hausdorff. Let $x, y \in X$ and $y \notin \{x\}$, i.e. $y \neq x$. Then, by Hausdorff, there exist disjoint neighborhoods U_x and U_y of x and y. Since U_x and U_y are disjoint, we have $U_x \subseteq X \setminus U_y$. But then, by monotonicity of the closure operator, we get

$$\overline{U}_x \subseteq \overline{X \setminus U_y} = X \setminus U_y,$$

i.e. $y \notin \overline{U}_x$, i.e.

$$y \notin \bigcap \left\{ \overline{U} \,\middle|\, U \in \mathcal{U}(x) \right\}.$$

$\Leftarrow$ Let $x \neq y$. Let $\{x\} = \bigcap \left\{ \overline{U} \,\middle|\, U \in \mathcal{U}(x) \right\}$. Then there must be an open neighborhood U_x of x whose closure doesn't contain y. But then the complement $U_y := X \setminus \overline{U}_x$ of that closure is an open neighborhood of y that is disjoint from U_x.

□

Definition 3.13: Regular Space

Let (X, O) be a topological space. Then we call (X, O) ***regular*** if every point can be housed orf from any closed set not containing it:

$$\begin{aligned} &\forall X \setminus C \in O : \\ &\forall x \notin C : \\ &\exists U_x \in \mathcal{U}(x), U_C \in \mathcal{U}(C) : \\ &\quad U_x \cap U_C = \emptyset \end{aligned}$$

We call (X, O) ***T3*** or ***regular Hausdorff***, if it is regular and Hausdorff.

Exercise 3.14. *Find a topological space that is regular, but not T0.*

Exercise 3.15. *Show that a topological space is regular Hausdorff (T3) if and only if it is regular and T1.*

Definition 3.16: Completely Regular Space

Let (X, O) be a topological space. Then we call (X, O) ***completely regular*** if every point can be separated from any closed set not containing it by a continuous function:

$$\begin{aligned} &\forall (X \setminus C) \in O : \\ &\forall x \notin C : \\ &\exists f \in C(X, [0,1]_{\mathbb{R}}) : \\ &\quad f(x) = 1,\ f[A] \subseteq \{0\} \end{aligned}$$

We call (X, O) ***T3a***, ***T3$\frac{1}{2}$***, or ***completely regular Hausdorff***, if it is completely regular and Hausdorff.

Exercise 3.17. *Find a space that is completely regular, but not T0.*

Exercise 3.18. *Show that a space if completely regular Hausdorff if and only if it is completely regular and T1.*

Theorem 3.19. *Every zero-dimensional space (X, O) is completely regular.*

Proof. Since X is zero-dimensional, it has a basis consisting of clopen sets. Given any clopen set U, the function

$$f_U : x \mapsto \begin{cases} 1 & x \in U \\ 0 & x \in X \setminus U \end{cases}$$

is continuous. Now, let C be a closed set not containing x. Then $X \setminus C$ is an open neighbourhood of x. We can now choose a basic (i.e. clopen) neighbourhood $U \subseteq X \setminus C$ of x, at which point f_U is our desired function separating x and C. □

Theorem 3.20. *Let $(X, \mathcal{O})$ be completely regular. Then the set*

$$\left\{ f^{-1}[U] \,\middle|\, f \in C((X, \mathcal{O}), [0,1]_{\mathbb{R}}), U \in \mathcal{O}_{\mathbb{R}} \right\}$$

is a basis of X.

Proof.

1. Since the f are continuous, every basis set is open.
2. Let $x \in X$. Let U be an open neighbourhood of x. Since X is completely regular, there exists a continuous function f separating $\{x\}$ and $X \setminus U$. Then the basic set $f^{-1}[(0,1]]$ must be a subset of U, since every element outside of U gets mapped to 0. Thus, every open neighhourhood contains a basic neighbourhood.

□

Theorem 3.21: Completely regular cubes

Every completely regular Hausdorff space $(X, \mathcal{O})$ can be embedded into the cube

$$\prod_{f \in C(X,[0,1])} [0,1]$$

Proof. We choose the trivial embedding $\iota : x \mapsto (f(x))_{f \in C(X,[0,1])}$.

1. ι is continuous, since the projections of ι are exactly the elements of $C(X, [0,1])$, i.e. continuous.
2. Let $x \neq y$. By T1, all singletons are closed. Therefore, by complete regularity, there exists a function f with $f(x) = 1$ and $f(y) = 0$. Therefore, we have $\iota(x)_f \neq \iota(y)_f$, i.e. $\iota(x) \neq \iota(y)$.
3. By 3.20, we only need to show openness for the basis sets

$$\left\{ f^{-1}[U] \,\middle|\, f \in C((X, \mathcal{O}), [0,1]_{\mathbb{R}}), U \in \mathcal{O}_{\mathbb{R}} \right\}.$$

For these sets, we get:

$$\begin{aligned}
\iota\big[f^{-1}[U]\big] &= \iota[(p_f \circ \iota)^{-1}[U]] \\
&= \iota[\iota^{-1}(p_f^{-1}[U])] \\
&= \iota\left[\iota^{-1}\left[U \times \prod_{g \neq f}[0,1]\right]\right] \\
&= \left(U \times \prod_{g \neq f}[0,1]\right) \cap \iota[X]
\end{aligned}$$

Where $\left(U \times \prod_{g \neq f}[0,1]\right)$ is open, since every component besides the f-component is open.

□

Definition 3.22: Normal Space

Let $(X, \mathcal{O})$ be a topological space. Then we call $(X, \mathcal{O})$ ***normal*** if every pair of disjoint closed sets can be housed orf from each other by disjoint neighbourhoods:

$$\begin{aligned}
&\forall X \setminus C_1, X \setminus C_2 \in \mathcal{O} : \\
&\quad C_1 \cap C_2 = \emptyset \implies \\
&\exists U_1 \in \mathcal{U}(C_1), U_2 \in \mathcal{U}(C_2) : \\
&\quad U_1 \cap U_2 = \emptyset
\end{aligned}$$

We call $(X, \mathcal{O})$ ***T4***, or ***normal Hausdorff***, if it is normal and Hausdorff.

Exercise 3.23. *Find a topological space that is normal, but not T0.*

Showing that T4 implies T3 is not an easy task - this statement is *Urysohn's Lemma*, which we will prove later in this chapter after a lot of preparation.

Chapter 4

Compactness

The familiar notions of open converings and compactness generalize directly from metric spaces to topological spaces. The most major difference is that sequential compactness is generally weaker than "covering compactness", and has to be replaced with a notion of compactness based on ultrafilter convergence.

Definition 4.1. Let $(X, \mathcal{O})$ be a topological space. Then we call a set $\mathcal{U} \subseteq \mathcal{O}$ of open subsets an ***open covering*** of X if every point of X is contained in the union of all elements of $\mathcal{U}$, i.e. if

$$\bigcup_{U_i \in \mathcal{U}} U_i = X$$

.

Definition 4.2: Compactness

Let $(X, \mathcal{O})$ be a topological space. Then we call $(X, \mathcal{O})$ ***compact*** if every open covering of X contains a finite open covering as a subset. We call a subset $K \subseteq X$ compact if it is compact as a topological space equipped with the subspace topology.

Exercise 4.3. *Every closed subset of a compact space is compact.*

Exercise 4.4. *Let $(X, \mathcal{O})$ be compact and $f : X \to Y$ be continuous. Then $f[X]$ is compact.*

Exercise 4.5. *Let $(X, \mathcal{O})$ be a nonempty topological space. Then the following are equivalent:*

1. *$(X, \mathcal{O})$ is compact, i.e. every open covering has a finite subcovering.*
2. *Every set $\mathcal{A}$ of closed sets whose intersection is the empty set has a finite subset whose intersection is the empty set.*
3. *If $\mathcal{A}$ is a set of closed sets such that the intersection over any finite subset of $\mathcal{A}$ is non-empty, the intersection of $\mathcal{A}$ is non-empty.*
4. *Every filter on $(X, \mathcal{O})$ has a limit point.*
5. *Every ultrafilter on $(X, \mathcal{O})$ is convergent.*

Theorem 4.6. (Alexander Subbase Theorem) *Let $(X, \mathcal{O})$ be a topological space. Let $\mathcal{S}$ be a subbasis. Then X is compact if every open covering of X by sets of $\mathcal{S}$ has a finite subcovering.*

Proof. Assume that X is not compact. Then there exists a non-convergent ultrafilter $\mathcal{F}$, i.e. we have

$$\begin{aligned}&\forall x \in X :\\ &\exists U_x \in \mathcal{U}(x) :\\ &\ (X \setminus U_x) \in \mathcal{F}\end{aligned}$$

By definition of subbases, every $U_x \in \mathcal{U}(x)$ contains an open subset O_x of the form

$$O_x = \bigcap_{k=0}^{n} S_{k,x}$$

for some $S_{k,x} \in \mathcal{S}$ and $n \in \mathbb{N}$. We therefore have

$$\begin{aligned}X \setminus U_x &\subseteq X \setminus O_x\\ &= X \setminus \bigcap_{k=0}^{n} S_{k,x}\end{aligned}$$

which implies $X \setminus \bigcap_{k=0}^{n} S_{k,x} \in \mathcal{F}$ by the filter axioms. Since the intersection of this set with $\bigcap_{k=0}^{n} S_{k,x} \in \mathcal{F}$ is empty, and filters are closed under finite intersection, $\mathcal{F}$ cannot contain every $S_{k,x}$. Therefore, since $\mathcal{F}$ is an ultrafilter, it must contain at least one set $X \setminus S_{k,x}$ for every x. We will refer to these $S_{k,x}$ with $X \setminus S_{k,x} \in X$ as S_x.

Since every S_x is an open neighborhood of x, and thus contains x, the set of all S_x is an open convering of X. By compactness, there must be a finite subcovering

$$\bigcup_{x \in X'} S_x = X.$$

Now, since $\mathcal{F}$ contains every $X \setminus S_x$, it also contains

$$\bigcap_{x \in X'} (X \setminus S_x) = X \setminus \bigcup_{x \in X'} S_x = X \setminus X = \emptyset.$$

which contradicts $\mathcal{F}$ being a filter. □

Lemma 4.7. *Let $(X, \mathcal{O})$ be a Hausdorff space. Let $K \subseteq X$ be compact. Then K can be separated from any point $x \in X \setminus K$ by open neighborhoods, i.e:*

$$\begin{aligned}&\forall x \in X :\\ &\exists U_x \in \mathcal{U}(x), U_K \in \mathcal{U}(K) :\\ &\ U_x \cap U_K = \emptyset\end{aligned}$$

Exercise 4.8. *Let $(X, \mathcal{O})$ be compact and Hausdorff. Let $A, B \subseteq X$ be closed. Then A*

and B can be separated from each other by disjoint open neigborhoods.

Exercise 4.9. *Every compact subset of a Hausdorff space is closed.*

Exercise 4.10. *Every continuous function from a compact space into a Hausdorff space is closed.*

Exercise 4.11. *Every continuous injective function from a compact space into a Hausdorff space is an embedding.*

Theorem 4.12: Tychonoff's Theorem

Let $(X_i, \mathcal{O}_i)_{i\in I}$ be a family of topological spaces. Then if all X_i are compact, $\prod_{i\in I} X$ is compact.

Proof.

1. If there exists an $i \in I$ such that $X_i = \emptyset$, then $\prod X_i = \emptyset$ is compact.
2. Now, assume $\prod_{i\in I} X_i \neq \emptyset$. Then there exists an ultrafilter $\mathcal{U}$ on $\prod_{i\in I} X_i$.

   Since each X_i is compact, each pushforward $(p_i)_*(\mathcal{U})$ converges to an $x_i \in X_i$. Thus, by theorem 2.5, $\mathcal{U}$ converges to $(x_i)_{i\in I}$ and $\prod_{i\in I} X_i$ is compact.

□

Chapter 5

Topologies on Algebraic Structures

5.1 Topological Groups

Definition 5.1

Let $(G, \circ)$ be a group. Let $\mathcal{O}$ be a topology on G. Then we call $(G, \circ, \mathcal{O})$ a ***topological group*** iff. the maps $^{-1} : g \to g^{-1}$ and $\circ : G \times G \to G$ are continuous.

Exercise 5.2. *Show that $(G, \circ, \mathcal{O})$ is a topological group if and only if the map*

$$\varphi : (g, h) \mapsto g \circ h^{-1}$$

is continuous.

Exercise 5.3. *Let $(G, \circ, \mathcal{O})$ be a topological groups. Then for any $g \in G$, the left and right group operation maps $l_g : h \mapsto g \circ h$ and $r_g : h \mapsto h \circ g$ are homeomorphisms.*

Exercise 5.4. *Let $(G, \circ, \mathcal{O}_G)$ be a topological group. Let $(H, \circ, \mathcal{O}_H)$ be a normal subgroup equipped with the supspace topology, and let $(G/H, \circ, \mathcal{O}_q)$ be the quotient group, equipped with the quotient topology. Then the quotient map $p : G \to G/H$ is continuous and open.*

Corollary 5.5. *If H is closed in G and G is Hausdorff, then G/H is Hausdorff. If G/H is T1, then H is closed.*

Part IX

Algebraic Topology

Chapter 1

Homotopy

Definition 1.1

Let X, Y be topological spaces. Let $f, g : X \to Y$ be continuous. Then f is called ***homotopic*** to g via a map h iff. h is a "continous interpolation between f and g, i.e:

1. h is a continous map $X \times [0,1] \to Y$,
2. For all $x \in X$, we have $h(x, 0) = f(x)$,
3. For all $x \in X$, we have $h(x, 1) = g(x)$.

We call h a ***free homotopy***, or simply a ***homotopy***, between f and g. We write $f \sim_h g$ if f is a homotopy between f and g. We write $f \sim g$ if there exists a homotopy between f and g.

Exercise 1.2. *Show that the property of being homotopic is an equivalence relation.*

Definition 1.3. We say that f and g are ***homotopic relative to a subset $A \subseteq X$*** and write $f \sim_{\text{rel } A}$ or $f \sim_{h,\text{rel } A}$ iff:

1. $f \sim_h g$,
2. For all $t \in [0,1]$, we have $h(t, \cdot)|_A = f|_A = g|_A$

I personally find the designation "homotopic relative to A" quite unintuitive, given that the property of being homotopic relative to a given subset is stronger than the property of being homotopic.

Exercise 1.4. *Show that $f \sim_h g$ is equivalent to $f \sim_{h,\text{rel } \emptyset} g$.*

Exercise 1.5. *Let $f_0, f_1 : X \to Y$ and $g_0, g_1 : Y \to Z$ be continuous. Let h_0 be a homotopy between f_0 and f_1 and let h_1 be a homotopy between f_1 and g_1. Then $H : (x, t) \mapsto h_1(h_0(x, t), t)$ is a homotopy between $g_0 \circ f_0$ and $g_1 \circ f_1$.*

Definition 1.6

Let $f : X \to Y$ be continuous. Then f is called a ***homotopy equivalence*** between X and Y iff. there exists a map $g : Y \to X$ such that $g \circ f$ is homotopic to id_X and $f \circ g$ is homotopic to id_Y. We write $X \simeq_{f,g} Y$, $X \simeq_f Y$, or simply $X \simeq Y$.

Exercise 1.7. *Show that the property of being homotopy equivalent is an equivalence relation.*

Exercise 1.8. *Show that homeomorphism implies homotopy equivalence.*

Definition 1.9. A topological space X is called ***contractible*** iff. X is homotopy equivalent to a one-point space.

Exercise 1.10.

1. *Show that*

$$h : (x, t) \mapsto x - tx$$

is a homotopy between $\mathrm{id}_{\mathbb{R}^n}$ *and* $f : \mathbb{R}^n \mapsto \{0\}$.

2. *Conclude that* $\mathbb{R}^n$ *is contractible.*

3. *Conclude that homotopy equivalence does not imply homeomorphism.*

Exercise 1.11. *A topological space X is contractible if and only if there exists a point $x_0 \in X$ such that* id_X *is homotopic to the constant map* $X \longrightarrow x_0$.

Definition 1.12. Let X be a subset of a topological vector space. Then we call X ***star-shaped*** iff. there exists a point $x_0 \in X$ such that X contains the connecting line between x_0 and any other point $x \in X$, i.e.

$$\begin{aligned} \forall x \in X &: \\ \forall t \in [0, 1] &: \\ (1 - t)x_0 + tx &\in X \end{aligned}$$

Exercise 1.13. *Show that any star-shaped set is contractible. Conclude that any convex set is contractible.*

Lemma 1.14. *Let X be a topological space. Let $A \subseteq X$. Then A and X are homotopy equivalent if there exists a homotopy between* id_X *and* $\mathrm{id}_X|_A$, *i.e. a continuous map $h : X \times [0, 1] \longrightarrow X$ such that for all $x \in X$:*

1. $h(x, 0) = x$,
2. $h(x, 1) \in A$,
3. $x \in A \implies h(x, 1) = x$

1.1 The Fundamental Group

Definition 1.15. Let X be a topological space. Then a continuous map $\gamma[0,1] \to X$ with $\gamma(0) = \gamma(1)$ is known as a ***loop***. We denote the set of all loops in X with $\gamma(0) = \gamma(1) = x_0$ as $\Omega(X, x_0)$.

Lemma 1.16. *Let $\gamma : [0,1] \to X$ be a loop. Let $\varphi : [0,1] \to [0,1]$ be a continuous map such that $\gamma(0) = 0$ and $\gamma(1) = 1$. Then $\gamma \circ \varphi$ and γ are homotopic relative to $\{0,1\}$.*

Proof. The map

$$h : (x,t) \mapsto \gamma((1-t)x + t\varphi(x))$$

is a homotopy between γ and $\gamma \circ \varphi$. □

Definition 1.17. Let $f, g \in \Omega(X, x_0)$. Then we define the ***concatenation of f and g*** as the map

$$(f * g)(t) = \begin{cases} f(2t) & 0 \leq t \leq 1/2, \\ g(2t-1) & 1/2 \leq t \leq 1. \end{cases}$$

Definition 1.18. Let $\gamma \in \Omega(X, x_0)$. We define the ***inverse loop*** of γ to be the loop that "moves through gamma backwards", i.e.

$$\overline{\gamma}(t) := \gamma(1-t)$$

Exercise 1.19. *Verify that $\overline{\overline{\gamma}} = \gamma$.*

Exercise 1.20. *Let $\gamma \in \Omega(X, x_0)$. Then $\gamma * \overline{\gamma}$ is homotopic to $e_{x_0} : X \to \{x_0\}$ relative to $\{0,1\}$.*

Theorem 1.21: Existence of the fundamental group

Let X be a topological space. Let $x_0 \in X$. Then the quotient space

$$\pi_1(X, x_0) = \Omega(X, x_0)/\sim_{\text{rel }(0,1)}$$

of loops modulo homotopy becomes a group, known as the ***fundamental group of (X, x_0)***, when equipped path concatenation

$$[f] * [g] = [f * g],$$

as the group operation. The neutral element of this group is $e_{x_0} := (x \mapsto x_0)$.

Theorem 1.22. *If we additionally define*

$$\begin{aligned} \pi_1(f) : \pi_1(X, x) &\mapsto \pi_1(Y, f(x)) \\ [\gamma] &\mapsto [f \circ \gamma] \end{aligned}$$

for all continuous $f : X \to Y$, the map π_1 becomes a covariant functor from the category of topological spaces into the category of groups.

Definition 1.23. Whenever we need to make the base point clear, we write $\pi_1(f,x) := \pi_1(f) : \pi_1(X,x) \mapsto \pi_1(Y, f(x))$

Theorem 1.24: Fundamental groups at different base points

Let X be a topological space. Let $x, y \in X$. Then every continuous path $\delta : [0,1] \to X$ induces a group isomorphism

$$\begin{aligned} J_\delta : \pi_1(X, \delta(0)) &\cong \pi_1(X, \delta(1)) \\ [\gamma] &\mapsto \left[\overline{\delta} * \gamma * \delta\right] \end{aligned}$$

Proof. 1. Since loop concatenation preserves homotopy, J_δ sends different representatives of the same equivalence class to the same equivalence class.

2. J_δ goes from $\pi_1(X,x)$ to $\pi_1(X,y)$, since for any $\gamma : [0,1] \to X$ with $\gamma(0) = \gamma(1) = x$, $\overline{\delta}$ traces out a path from $\delta(1)$ to $\delta(0)$, then γ traces out a loop from $\delta(0)$ to $\delta(0)$, and then δ traces out a path from $\delta(0)$ back to $\delta(1)$, i.e. $\overline{\delta} * \gamma * \delta$ is a loop from $\delta(1)$ to $\delta(1)$.

3. J_δ is a group homomorphism, since

$$\begin{aligned} J_\delta([\gamma_1 * \gamma_2]) &= \left[\overline{\delta} * \gamma_1 * \gamma_2 * \delta\right] \\ &= \left[\overline{\delta} * \gamma_1 * e_{x_0} * \gamma_2 * \delta\right] \\ &= \left[\overline{\delta} * \gamma_1 * \delta * \overline{\delta} * \gamma_2 * \delta\right] \\ &= \left[\overline{\delta} * \gamma_1 * \delta\right] * \left[\overline{\delta} * \gamma_2 * \delta\right] \\ &= J_\delta([\gamma_1]) * J_\delta([\gamma_2]) \end{aligned}$$

(Recall that any map that respects the group operation also respects the neutral element and inverses, and is thus a homomorphism.)

4. J_δ is bijective with $J_\delta^{-1} = J_{\overline{\delta}}$:

$$\begin{aligned} J_\delta(J_{\overline{\delta}}(\gamma)) &= \left[\overline{\delta} * \left(\delta * \gamma * \overline{\delta}\right) * \delta\right] = [\gamma] \\ J_{\overline{\delta}}(J_\delta(\gamma)) &= \left[\delta * \left(\overline{\delta} * \gamma * \delta\right) * \overline{\delta}\right] = [\gamma] \end{aligned}$$

□

Corollary 1.25. *If X is path connected, $\pi_1(X,x) \cong \pi_1(X,y)$ for any $x, y \in X$.*

Definition 1.26. Let $\gamma_0 \in \Omega(X,x)$ an $\gamma_1 \in \Omega(X,y)$. Then we call $h : [0,1] \times [0,1] \to X$ a ***free homotopy of loops***, or simply ***loop homotopy***, between γ_0 and γ_1 iff. h is a homotopy between γ_0 and γ_1 such that $h_t(x) := h(x,t)$ is a loop for any fixed $t \in [0,1]$.

Exercise 1.27. 1. *Verify that h is a loop homotopy iff. h is a homotopy and $h_t(0) = h_t(1)$ for any $t \in [0,1]$.*

2. *Show that any $\{0,1\}$-homotopy is a loop homotopy.*

The additional constraint that and $h(\cdot,t)$ is a loop may seem superfluous, but there are in fact homotopies between loops that aren't loop homotopies:

Exercise 1.28. 1. *Let $\gamma_0 = \gamma_1 : [0,1] \to \{0\}$ be constant loops. Let $h(x,t) := xt(1-t)$. Then h is a homotopy between γ_0 and γ_1, but fails to be a loop homotopy.*

2. *Let $\gamma_0(x) = e^{2\pi i x}$. Let $\gamma_1(x) = e^{4\pi i x}$. Then $h(x,t) := e^{2\pi i(1+t)x}$ is a homotopy between γ_0 and γ_1, but fails to be a loop homotopy.*

3. *Assume that any loop $\gamma : [0,1] \to S^1$ with 1 as its base point is homotopic a loop of the form $\gamma(x) := e^{2\pi n i x}$ for some $n \in \mathbb{Z}$ (we will prove this later). Does 2. imply that the fundamental group $\pi_1(S^1, 1)$ is trivial?*

Definition 1.29. A loop $\gamma \in \Omega(X,x)$ is called ***contractible in X*** iff. there exists an $x \in X$ and a loop homotopy h such that $\gamma \sim_h e_x$.

Lemma 1.30. *$\gamma \in \Omega(X,x)$ is contractible in X if $\gamma \sim_{\text{rel } 0,1} e_{\gamma(0)}$.*

Theorem 1.31. *Let $\gamma \in \Omega(X,x)$ be a loop. Let $S^1 \cong I/\{0,1\}$ be equipped with the quotient topology. Then γ is contractible if and only if $\gamma \circ p : S^1 \to X$ has a continuous extension to the closed unit ball $\overline{B}^2$.*

Theorem 1.32. *Let h be a homotopy between $f, g : X \to Y$. Let $x_1 \in X$. Then*

$$\begin{aligned} h(x_1,\cdot) : [0,1] &\to Y \\ t &\mapsto h(x_1,t) \end{aligned}$$

is a path from $f(x_1)$ to $g(x_1)$ which fulfills

$$\pi_1(g, x_1) = J_{h(x_1,\cdot)} \circ \pi_1(f, x_1)$$

Recall that $\pi_1(g,x_1)$ is the map $[\gamma] \mapsto [g \circ \gamma]$, and that J_δ is the map $[\gamma] \mapsto [\overline{\delta} * \gamma * \delta]$, so this theorem states that

$$[g \circ \gamma] = \left[\overline{h}(x_1,\cdot) * (f \circ \gamma) * h(x_1,\cdot)\right]$$

for any $\gamma \in \pi_1(X, x_1)$, or, equivalently, that the diagram

$$\begin{tikzcd} & \pi_1(X,x_1) \arrow[dl, "\pi_1(f,x_1)"] \arrow[dr, "\pi_1(g,x_1)"] & \\ \pi_1(Y,f(x_1)) \arrow[rr, "J_{h(x_1,\cdot)}"] & & \pi_1(Y,g(x_1)) \end{tikzcd}$$

commutes.

Proof. Define

$$H_t(x) := h(x_1, t + (1-t)x).$$

We have $(f \circ \gamma) \in \Omega(X, f(x_1)$ and $(g \circ \gamma) \in \Omega(X, g(x_1))$, since

$$\begin{aligned} f(\gamma(0)) &= f(x_1) = f(\gamma(1)) \\ g(\gamma(0)) &= g(x_1) = g(\gamma(1)) \end{aligned}$$

Then a homotopy between $g \circ \gamma$ and $\overline{h}(x_1, \cdot) * (f \circ \gamma) * h(x_1, \cdot)$ is given by:

$$k(x,t) := (\overline{H_t} * (h(\cdot, t) \circ \gamma) * H_t)(x),$$

since we get:

$$\begin{aligned} k(x,0) &= (\overline{H_0} * (h(\cdot, 0) \circ \gamma) * H_0)(x) \\ &= (\overline{H_0} * (f \circ \gamma) * H_0)(x) \\ &= (\overline{h}(x_1, \cdot) * (f \circ \gamma) * h(x_1, \cdot))(x) \end{aligned}$$

and

$$\begin{aligned} k(x,1) &= (\overline{H_1} * (h(\cdot, 1) \circ \gamma) * H_1)(x) \\ &= (h(x_1, (1-1) + (1-(1-1))(\cdot)) * (g \circ \gamma) * h(x_1, 1 + (1-1)(\cdot)))(x) \\ &= (h(x_1, \cdot) * (g \circ \gamma) * h(x_1, 1))(x) \\ &= (e_{g(x_1)} * (g \circ \gamma) * e_{g(x_1)})(x) \\ &= (g \circ \gamma)(x), \end{aligned}$$

where the second to last step follows since the loop $(g \circ \gamma)$ ends on the base point $g(x_1)$. □

Lemma 1.33. *Let $\gamma : [0,1] \to X$ be continuous. Let $f : X \to Y$. Then*

$$\pi_1(f, \gamma(1)) \circ J_\gamma = J_{f \circ \gamma} \circ \pi_1(f, \gamma(0)),$$

i.e. the diagram

$$\begin{tikzcd} \pi_1(X, \gamma(0)) \arrow[r, dashed, "\pi_1(f,\gamma(0))"] \arrow[d, "J_\gamma"] & \pi_1(Y, f(\gamma(0))) \arrow[d, "J_{f\circ\gamma}"] \\ \pi_1(X, \gamma(1)) \arrow[r, dashed, "\pi_1(f,\gamma(1))"] & \pi_1(Y, f(\gamma(1))) \end{tikzcd}$$

commutes.

Proof. Let $\sigma \in \Omega(X, \gamma(0))$. Then we have

$$\begin{aligned} \pi_1(f, \gamma(1)) \circ J_\gamma([\sigma]) &= \pi_1(f, \gamma(1))([\overline{\gamma} * \sigma * \gamma]) \\ &= [f \circ (\overline{\gamma} * \sigma * \gamma)] \\ &= [(f \circ \overline{\gamma}) * (f \circ \sigma) * (f \circ \gamma)] \\ &= \left[\overline{(f \circ \gamma)} * (f \circ \sigma) * (f \circ \gamma)\right] \\ &= J_{f \circ \gamma}([f \circ \sigma]) \\ &= J_{f \circ \gamma} \circ \pi_1(f, \gamma(0))([\sigma]) \end{aligned}$$

□

Theorem 1.34. *Let $f : X \to Y$ be a homotopy equivalence. Let $x_0 \in X$. Then the map $\pi_1(f, x_0)$ is a group isomorphism $\pi_1(X, x_0) \cong \pi_1(Y, f(x_0))$.*

Chapter 2

Fibrations and Coverings

Definition 2.1: Locally Trivial Fibration

Let $\pi : Y \to X$ be continuous. Then π is called a ***locally trivial fibration*** iff. for every $x \in X$, there exists an open neighborhood U such that there exists a homeomorphism $h_u : \pi^{-1}[U] \cong U \times \pi^{-1}[x]$ fulfilling $p_1 \circ h_U = \pi|_{\pi^{-1}[U]}$.

$\pi^{-1}[x]$ is called the ***Fiber of π at x***, and sometimes written F_x.

$$\begin{tikzcd} Y \supseteq \pi^{-1}[U] \arrow[r, "h_U"'] \arrow[d, "\pi"] & U \times \pi^{-1}[x] \arrow[dl, "p_1"] \\ X \supseteq U & \end{tikzcd}$$

Definition 2.2. Let $\pi : Y \to X$ be a locally trivial fibration. Then:

1. If all $\pi^{-1}[x]$ are homeomorphic, we call π a ***fiber bundle***.
2. If every $\pi^{-1}[x]$ is discrete, we call π a ***covering map***.
3. If we can pick $U = X$, π is called a ***trivial fibration***.

Exercise 2.3. *If we replace $\pi^{-1}[x]$ in the definition by an arbitrary topological space F_U, we get $F_U \cong \pi^{-1}[x]$. Thus, both definitions are equivalent.*

Lemma 2.4. *Let $\pi : Y \to X$ be a locally trivial fibration. Let X be connected. Then π is a trivial fibration.*

Proof. Let $x_0 \in X$. Let $V = \{x \in X \mid F_x \cong F_{x_0}\}$ be the set of all points whose fibers are homeomorphic to the fiber of x_0. Let U be a neighborhood as in the definition. Then the fibers of all $x \in U$ must be homeomorphic by definition. Thus, every point in V has a neighborhood contained in V, so V is open. We have

$$X \setminus V = \bigcup_{y \notin V} \left\{x \in X \;\middle|\; F_x \cong F_y\right\} := \bigcup_{y \notin V} V_y$$

Since every V_y is open by the same argument, $X \setminus V$ must be open. But since X is connected with $(X \setminus V) \cup V = X$ and V nonempty, $X \setminus V$ must be empty. □

Theorem 2.5. *A continuous function $\pi : Y \to X$ is a covering map if and only if every $x \in X$ has an open neighborhood U such that $\pi^{-1}[U]$ can be decomposed into disjoint open sets $V_i \subseteq Y$ such that $\pi|_{V_i}$ is a homeomorphism $V_i \cong U$.*

Proof. $\Rightarrow$: Let π be a covering map. Let $I = F_U$. Since h_U is bijective, the sets $V_i := h_U^{-1}[U \times \{i\}]$ are disjoint. Since h_U is a homeomorphism and $I = F_U$ is discrete, the V_i are also open. Since $p_1|_{U\times\{i\}}$ is a homeomorphism $U \times \{i\} \cong U$, the composition

$$\pi|_{V_i} = (p_1|_{U\times\{i\}}) \circ h|_{V_i}$$

must be a homeomorphism $V_i \mapsto U$.

$\Leftarrow$: Define

$$\begin{aligned} h_U : \pi^{-1}[U] &\cong U \times I \\ y \in V_i &\mapsto (\pi(y), i). \end{aligned}$$

Then the restriction of h_U to each V_i is by definition a homeomorphism $V_i \cong U \cong U \times \{i\}$, and so the whole of h_U is a homeomorphism $\pi^{-1}[U] = \bigsqcup_{i\in I} V_i \cong \bigsqcup_{i\in I} U \times \{i\} = U \times I$.

□

Definition 2.6: Lifting

Let $\pi : Y \to X$ be a locally trivial fibration. Let $f : Z \to X$ be continuous. Then a continuous map $\hat{f} : Z \to Y$ is called a ***lift from f to Y*** if $\pi \circ \hat{f} = f$.

Any path can be lifted:

Theorem 2.7. *Let $\pi : Y \to X$ be a locally trivial fibration. Let $\gamma : [0, 1] \to XX$ be continuous. Let $\pi(y_0) = \gamma(0)$. Then there exists a lift $\hat{\gamma} : [0, y] \to Y$ with $\hat{\gamma}(0) = y_0$ such that $\pi \circ \hat{f} = f$.*

And lifts respect fundamental groups:

Theorem 2.8. *Let $\pi : Y \to X$ be a locally trivial fibrations. Let $f : Z \to X$. Let $\hat{f} : Z \to Y$ be a lift of f. Then we have:*

$$\begin{aligned} &\forall z_0 \in Z : \\ &\exists y_0 \in Y : \\ &\quad \hat{f}(z_0) = y_0, \\ &\quad \pi_1(f)[\pi_1(Z, z_0)] \subseteq \pi_1(\pi)[\pi_1(Y, y_0)] \end{aligned}$$

Proof.

$$\begin{aligned}\pi_1(f)[\pi_1(Z, z_0)] &= (\pi_1(\pi \circ \hat{f}))[\pi_1(Z, z_0)]\\ &= (\pi_1(\pi) \circ \pi_1(\hat{f}))[\pi_1(Z, z_0)]\\ &=\end{aligned}$$

□